



```
. irf graph irf, irf(lags) name(irf2, replace)
    name: SLP_EE
    log: C:\Users\jamel\Dropbox\Latex\PROJECTS\24-03-emft-adb-private\manuscript\SI_EE\Replication_EE\SLP_EE.s
> mcl
    log type: text
opened on: 22 Jun 2025, 21:19:57

.
. // Version of Stata
.
. version 19.5

.
. // Set scheme
.
. set scheme Cleanplots

.
. graph set window fontface "Arial Narrow"

.
. // Clear consol and data
.
. cls

.
. clear

.
. // Set the paths
.
. *global Docs = "C:\Users\jamel\Dropbox\Latex\"
. *global Proj = "PROJECTS\24-03-emft-adb\estimates"
.
. *cd "${Docs}"
. *cd "${Proj}"
.
. **# Install the packages (may require to install of other related required packages)
.
. ssc install locproj, replace
checking locproj consistency and verifying not already installed...
all files already exist and are up to date.

. ssc install heatplot, replace
checking heatplot consistency and verifying not already installed...
all files already exist and are up to date.

. ssc install ridgeline, replace
checking ridgeline consistency and verifying not already installed...
all files already exist and are up to date.

. ssc install outreg2, replace
checking outreg2 consistency and verifying not already installed...
all files already exist and are up to date.

. ssc install schemepack, replace
checking schemepack consistency and verifying not already installed...
all files already exist and are up to date.

.
. **# Data preparation
```


dlngdp_bk	float	%9.0g		* BK: Potential real GDP growth (percent)
dlngdp_ucm	float	%9.0g		* UCM: Potential real GDP growth (percent)
dlngdp_bw	float	%9.0g		* BW: Potential real GDP growth (percent)
dlngdp_5exp	float	%9.0g		* Exp(WEO): Potential real GDP growth (percent)
dlngdp_pfa	float	%9.0g		PFA: Potential real GDP growth (percent)
dlng_contpfa	float	%9.0g		Contribution of capital accumulation to potential growth (percentage points)
dlngl_contpfa	float	%9.0g		Contribution of labor to potential growth (percentage points)
dlntfp_pfa	float	%9.0g		Potential TFP growth (percent)
ers	float	%9.0g		Exchange Rate Stability Index
CURGDP	double	%10.0g		GDP (current US\$)
RES	double	%10.0g		Total reserves minus gold (current US\$)
RESGDP	float	%9.0g		Total reserves minus gold (% of GDP)
IT	float	%9.0g		Inflation Targeters
vul	double	%10.0g		ND-GAIN overall vulnerability
read	double	%10.0g		ND-GAIN overall readiness
ccode	str3	%9s		Country Code
group	byte	%18.0g	group	Country group
region	byte	%31.0g	region	Geographical region
inc	byte	%19.0g	inc	Income group
ggdy	double	%10.0g		General government gross debt, % of GDP [Government debt sustainability]
pby	double	%10.0g		Primary balance, % of GDP [Government debt sustainability]
cby	double	%10.0g		Cyclically-adjusted balance, % of potential GDP [Government debt sustainability]
fby	double	%10.0g		Fiscal balance, % of GDP [Government debt sustainability]
dfggd	double	%10.0g		General government gross debt, % of average tax revenues [Government debt sustainai]
dffb	double	%10.0g		Fiscal balance, % of average tax revenues [Government debt sustainability]
ggdma	double	%10.0g		General government gross debt, % of 10-year moving average GDP [Government debt]
fbma	double	%10.0g		Fiscal balance, % of 10-year moving average GDP [Government debt sustainability]
fxsovsh	double	%10.0g		General government debt in foreign currency, % of total [Balance sheet compositi]
secnres	double	%10.0g		Debt securities held by nonresidents, % of total [Balance sheet composition]
fordebtsh	double	%10.0g		General government debt held by nonresidents, % of total [Balance sheet composi]
concggd	double	%10.0g		Concessional external debt stocks, % of external public debt [Balance sheet comp]
avglife	double	%10.0g		Sovereign debt average maturity, years [Balance sheet composition]
debtduy	double	%10.0g		Central government debt maturing in 12 months or less, % of GDP [Balance sheet c]
xtdebt	double	%10.0g		Total external debt stocks, % of GDP [External and private sector debt]
fxdebtall	double	%10.0g		External debt in foreign currency, % of total [External and private sector debt]
prdebt	double	%10.0g		Private external debt stocks, % of GDP [External and private sector debt]
pscy	double	%10.0g		Domestic credit to private sector, % of GDP [External and private sector debt]
stdebtall	double	%10.0g		Short-term external debt stocks, % of total [External and private sector debt]
stdebtres	double	%10.0g		Short-term external debt stocks, % of reserves [External and private sector debt]
xtdebtres	double	%10.0g		Total external debt stocks, % of reserves [External and private sector debt]
xtdebtrxg	double	%10.0g		Total external debt stocks, % of reserves excluding gold [External and private s]
sovrate	double	%10.0g		Foreign currency long-term sovereign debt ratings, index from 1-21 [best] [Marke
nd_gain	float	%9.0g		ND-GAIN overall score
countryname	str30	%30s		IMF country names
xrates	double	%10.0g		IMF IFS, Exchange Rate, National Currency Per USD
xrates_gw	float	%9.0g		IMF IFS, Exchange Rate, National Currency Per USD, growth rate (winsorized)
bonds1	double	%10.0g		IMF IFS, Interest Rates, Government Bonds, yield, percentage
bonds2	double	%10.0g		JPM EMBI sovereign bond index, yield, percentage
bonds3	float	%9.0g		Sovereign bonds, IMF IFS or JPM EMBI, yield, percentage
bonds_tw	float	%9.0g		Government bonds, JPM EMBI & IMF, yields %, 100*x/(x+100)(winsorized)

bills	double	%10.0g	IMF IFS, Treasury Bills, interest rates, percentage
bills_tw	float	%9.0g	IMF IFS, Treasury Bills, yields %, 100*x/(x+100)(winsorized)
floating	float	%10.0g	IMF AREAER coded float or free float
stocks_gw	float	%9.0g	MSCI country index, price, USD, growth rate (winsorized)
stocksL	double	%10.0g	MSCI country index, price, local currency units
stocksL_gw	float	%9.0g	MSCI country index, price, local currency units, growth rate (winsorized)
msciworld_gw	float	%9.0g	MSCI world index, price, USD, growth rate (winsorized)
msci_usd_eur	double	%10.0g	MSCI stock prices at country level, in USD or EUR
bondsUS_tw	float	%9.0g	US government bonds, yields %, 100*x/(x+100)(winsorized)
billsUS_tw	float	%9.0g	US government bills, yields %, 100*x/(x+100)(winsorized)
imfnew	float	%9.0g	Dummy, 1 if new program approved
unsc	float	%10.0g	Dummy, 1 for election date of non-permanent member of UNSC
unsc_startdate	float	%10.0g	Dummy, 1 for 24 months of serving as non-permanent member of UNSC
unsc24	float	%9.0g	Dummy, UNSC election month until 24 months after election date
unsc_full	float	%9.0g	Dummy, UNSC election month until end of serving time
gdp_w	double	%10.0g	IMF WEO, GDP, Constant Prices, Change y/y, annual (Jan value), winsorized
cpi	double	%10.0g	CPI % yoy, national sources
cpi_tw	float	%9.0g	CPI % yoy, national sources, 100*x/(x+100) (winsorized)
prgt	float	%8.0g	PRGT member, dummy
VIXCLS	double	%14.2f	CBOE Volatility Index: VIX
fbsusgap	double	%10.0g	Sustainability gap, fiscal balance, % of GDP [Government debt sustainability]
pbsusgap0	double	%10.0g	Sustainability gap, primary balance, country-specific conditions, % of GDP [Gove
pbsusgap1	double	%10.0g	Sustainability gap, primary balance, historical conditions, % of GDP [Government
pbsusgap2	double	%10.0g	Sustainability gap, primary balance, current conditions, % of GDP [Government de
pbsusgap3	double	%10.0g	Sustainability gap, primary balance, stressed conditions, % of GDP [Government d
pbsusgap4	double	%10.0g	Sustainability gap, primary balance, benign conditions, % of GDP [Government deb
cds5y	double	%10.0g	5-year sovereign CDS spreads, basis points [Market perception]
banking	float	%9.0g	1 = Laeven and Valencia 2020: systemic banking crisis
currency	int	%10.0g	1 = Laeven and Valencia 2020: currency crisis
debt	int	%10.0g	1 = Laeven and Valencia 2020: debt crisis

* indicated variables have notes

Sorted by: imfcode period

. sum

Variable	Obs	Mean	Std. dev.	Min	Max
country	0				
imfcode	6,567	557.5829	262.2908	111	968
period	6,567	2006	9.52263	1990	2022
CAB	5,312	-2.510943	13.70318	-240.495	311.7461
NIIP	3,600	-.1903194	1.125594	-6.823869	12.67608
EDS	3,790	62.67229	67.0741	.2386041	1111.27
GDeficit	5,487	-2.512284	13.04986	-557.499	125.135
GDebt	5,092	56.47414	45.14327	0	600.117
CPI	5,519	25.1936	369.6068	-16.85969	23773.13
CREDIT	1,383	79.63016	42.81665	8.15	278.5
REER	3,023	101.741	27.20756	33.01393	741.6998
FUELX	4,629	15.56177	26.08907	0	99.98648
FUELM	4,827	13.60473	8.507112	.0104926	67.77688
MKT	1,989	63.88655	110.3168	.0092022	1777.226
kaopen	5,482	.1991533	1.562826	-1.931087	2.299361
ka_open	5,482	.5035495	.3694233	0	1
FD	5,824	.2831532	.2195336	0	1
FI	5,824	.3683459	.2219489	0	1
FM	5,824	.1868549	.2410327	0	1
extconf	4,401	9.910577	1.608372	0	12
bureau	4,401	2.220897	1.118027	0	4
corruption	4,401	2.871279	1.269621	0	6
demoacc	4,401	3.938433	1.625584	0	6

ethnictens	4,401	4.018178	1.300544	0	6
govstab	4,401	7.742257	1.819912	.6666667	12
intconf	4,401	9.069823	2.053463	0	12
laworder	4,401	3.744376	1.367858	0	6
milpol	4,401	3.836609	1.740627	0	6
reltensions	4,401	4.588569	1.29756	0	6
socioeco	4,401	5.73132	2.317912	0	11
invprofile	4,401	7.851435	2.350494	0	12
REM	5,364	4.119802	7.835214	0	167.4331
milexgdp	4,642	.022462	.0288844	3.66e-06	1.173498
dlngdp_mv	2,125	3.102448	2.302033	-9.163499	22.28646
dlngdp_cf	2,125	3.082225	2.516415	-7.105941	21.55679
dlngdp_hp	2,125	3.125434	2.256471	-5.802664	19.1487
dlngdp_bk	2,125	3.04979	2.777852	-10.14352	26.96705
dlngdp_ucm	2,165	3.036703	1.699798	-4.382327	11.83436
dlngdp_bw	2,125	3.09847	2.198428	-10.62781	19.2406
dlngdp_5exp	5,173	4.297102	2.488258	-9.042277	65.03672
dlngdp_pfa	1,942	3.235463	1.977139	-2.466566	13.82682
dlng_contpfa	1,942	1.470878	1.209911	-2.695527	9.766214
dlng_contpfa	1,942	.7531268	.6740357	-1.012055	2.858675
dlngtfp_pfa	1,942	1.011458	1.179201	-5.27768	6.808414
ers	5,396	.6119759	.3136414	.001342	1
CURGDP	6,292	2.86e+11	1.31e+12	4.29e+07	2.54e+13
RES	5,484	3.70e+10	1.92e+11	40812.32	3.86e+12
RESGDP	5,417	17.62543	34.35094	.0079961	1653.938
IT	6,567	.1137506	.3175323	0	1
vul	4,968	.4420377	.0955	.2441626	.6961713
read	5,049	.403483	.1321845	.1148045	.8231771
ccode	0				
group	6,336	1.802083	.3984604	1	2
region	6,336	3.416667	1.852322	1	7
inc	6,336	2.802083	1.042113	1	5
ggdy	5,069	57.20957	44.91472	.002	600.117
pby	5,416	-.4817977	6.519113	-186.818	126.464
cby	4,084	-2.274798	5.002638	-52.32261	40.88641
fby	5,596	-2.184784	6.710723	-151.333	125.135
dfggd	4,995	466.5801	844.4819	.0313969	27964.48
dffb	5,488	-15.1032	264.278	-13768.27	3927.696
ggdma	4,936	86.30542	92.39212	.0033874	2004.972
fbma	5,412	-3.168582	11.07769	-118.2264	247.6643
fxsovsh	779	29.37355	30.06265	0	100
secnres	965	3.393978	5.258295	0	85.31197
fordebtsh	1,312	41.03018	21.38795	1.956478	100
concgdp	3,561	37.62886	28.9273	0	100
avglife	915	6.68283	2.899997	.3123288	20
debtduy	1,533	7.170218	8.491593	0	70.74007
xtdebt	4,688	101.1262	329.5746	.6027712	6596.141
fxdebtall	631	77.58295	26.85264	1.004835	100
prdebt	3,468	75.83949	373.9769	0	6385.398
pscy	5,669	56.4842	60.67687	.0014018	407.4
stdebtall	4,690	16.40842	16.15377	0	88.98281
stdebtres	4,213	1804.956	19067.79	0	567255.3
xtdebtres	4,245	5091.573	41221.72	2.604098	1059468
xtdebttrng	4,245	7221.107	55037.56	2.694032	1411828
sovrage	3,560	12.64293	5.229736	1	21
nd_gain	4,968	47.9819	10.44913	25.41625	76.99231
countryname	0				
xrates	4,525	1486375	9.99e+07	.0006333	6.72e+09
xrates_gw	4,518	.3884252	1.167915	-2.606477	12.41035
bonds1	1,431	6.915087	8.064942	-.49	209.6
bonds2	1,029	7.839989	6.024717	.801	97.011

bonds3	2,180	7.227482	7.511966	-.49	209.6
bonds_tw	2,180	6.350907	3.883431	.4380725	23.31288
bills	2,237	9.626031	17.24744	-1.028333	340
bills_tw	2,237	7.480569	6.708904	-.1101211	36.30573
floating	4,625	.5731892	.4946678	0	1
stocks_gw	1,489	.5640422	2.647166	-14.81089	11.8094
stocksl	1,497	2.44e+09	2.19e+10	15.62117	3.22e+11
stocksl_gw	1,489	.7124478	2.511485	-13.49695	23.558
msciworld_gw	4,995	.5887854	1.360897	-4.297278	3.183805
msci_usd_eur	103	1185.787	1294.197	35.79475	6473.805
bondsUS_tw	4,995	3.954146	1.474038	1.777527	6.587451
billsUS_tw	4,995	2.319415	1.917184	.0349862	5.485422
imfnew	4,995	.0084084	.0253229	0	.1666667
unsc	4,995	.0022856	.0136118	0	.0833333
unsc_start~e	4,995	.0022856	.0136118	0	.0833333
unsc24	4,995	.0530197	.2028395	0	1
unsc_full	4,995	.0601935	.227598	0	1
gdp_w	4,875	3.608382	4.314536	-12.58	17.095
cpi	4,004	.7949849	3.500414	-1.2075	133.2775
cpi_tw	4,004	.5932811	1.046176	-1.222756	8.925319
prgt	4,995	.3795796	.4853309	0	1
VIXCLS	6,567	19.66474	5.837379	11.09024	32.69553
fbsusgap	5,561	2.040252	8.681721	-175.6214	132.0947
pbsusgap0	3,545	-.696735	8.94296	-109.1942	31.43961
pbsusgap1	3,545	-.4725562	7.932255	-96.79687	32.05574
pbsusgap2	2,586	-.2764674	7.027678	-119.7522	36.24833
pbsusgap3	3,489	-5.761738	11.41485	-131.0409	31.04496
pbsusgap4	3,489	4.151848	6.540885	-62.55282	37.06666
cds5y	1,259	1705.588	22262.16	1.770833	671619.3
banking	5,544	.0189394	.1363233	0	1
currency	5,544	.0236291	.1519045	0	1
debt	5,544	.0054113	.0733686	0	1

```

.
. gen vul100 =100*vul
(1,599 missing values generated)

. gen read100=100*read
(1,518 missing values generated)

. gen Dvul100=D.vul100
(1,783 missing values generated)

.
.
. merge 1:1 imfcode period using emdat.dta, ///
> keep(1 3) nogen

```

Result	Number of obs
Not matched	264
from master	264
from using	0
Matched	6,303

```

. *** Table 1. Descriptive statistics for the variables involved
.
. // Run everything for Table 1
.
. #delimit;
delimiter now ;
. local variables "vul bonds_tw sovrate CAB GDebt GDeficit cpi_tw banking currency debt msciworld_gw bondsUS_tw V
> IXCLS";

. summ `variables' if period>1995;

```

Variable	Obs	Mean	Std. dev.	Min	Max
vul	4,784	.4416259	.0955079	.2441626	.6961713
bonds_tw	2,052	6.078127	3.655882	.4380725	23.31288
sovrate	3,300	12.36036	5.18118	1	21
CAB	4,510	-2.275641	14.00589	-147.9973	311.7461
GDebt	4,694	55.83493	44.59376	0	600.117
GDeficit	4,859	-2.020479	6.55102	-59.742	125.135
cpi_tw	3,607	.5009582	.8430271	-1.222756	8.925319
banking	4,356	.0119376	.1086175	0	1
currency	4,356	.0176768	.1317888	0	1
debt	4,356	.0061983	.0784942	0	1
msciworld_gw	4,440	.5238988	1.420379	-4.297278	3.183805
bondsUS_tw	4,440	3.685846	1.331757	1.777527	6.047619
VIXCLS	5,373	20.48322	5.903067	11.09024	32.69553

```

. #delimit cr
delimiter now cr
. outreg2 using sum if period>1995, word replace ///
> sum(detail) keep(`variables') ///
> eqkeep(N mean p25 p50 p75 sd min max) // dec(2)

```

IMF country code

Percentiles		Smallest		
1%	112	111		
5%	137	111		
10%	178	111	Obs	5,373
25%	316	111	Sum of wgt.	5,373
50%	578		Mean	557.5829
		Largest	Std. dev.	262.2953
75%	746	968		
90%	924	968	Variance	68798.82
95%	943	968	Skewness	-.0968659
99%	964	968	Kurtosis	1.803612

Period

Percentiles		Smallest		
1%	1996	1996		
5%	1997	1996		
10%	1998	1996	Obs	5,373
25%	2002	1996	Sum of wgt.	5,373
50%	2009		Mean	2009
		Largest	Std. dev.	7.789606
75%	2016	2022		
90%	2020	2022	Variance	60.67796
95%	2021	2022	Skewness	0
99%	2022	2022	Kurtosis	1.796703

Current Account Balance in % of GDP

	Percentiles	Smallest		
1%	-31.93284	-147.9973		
5%	-18.32073	-65.25674		
10%	-13.52362	-59.99555	Obs	4,510
25%	-7.166942	-56.13009	Sum of wgt.	4,510
50%	-2.772924		Mean	-2.275641
		Largest	Std. dev.	14.00589
75%	1.73874	216.8781		
90%	8.126619	225.0329	Variance	196.1651
95%	14.06517	235.7506	Skewness	6.831586
99%	34.71003	311.7461	Kurtosis	128.5685

net IIP / GDP domestic currency

	Percentiles	Smallest		
1%	-2.108638	-6.823869		
5%	-1.304612	-6.516654		
10%	-.9675279	-6.473585	Obs	3,395
25%	-.5756767	-5.818635	Sum of wgt.	3,395
50%	-.2746915		Mean	-.1916606
		Largest	Std. dev.	1.154042
75%	-.0152968	11.09214		
90%	.4744182	11.34816	Variance	1.331812
95%	.9008924	11.77008	Skewness	4.703879
99%	4.409312	12.67608	Kurtosis	44.97329

External Debt Stocks (% of GNI)

	Percentiles	Smallest		
1%	3.874053	1.154187		
5%	12.84846	1.25724		
10%	17.37263	1.321404	Obs	3,180
25%	27.83612	1.399575	Sum of wgt.	3,180
50%	45.34305		Mean	57.84397
		Largest	Std. dev.	51.203
75%	69.90713	497.9303		
90%	106.3969	506.0367	Variance	2621.748
95%	136.62	528.5996	Skewness	3.677797
99%	268.482	610.4519	Kurtosis	24.80092

Gov. Net Lending/Borrowing in % of GDP

	Percentiles	Smallest		
1%	-18.29	-59.742		
5%	-9.74	-55.406		
10%	-7.471	-41.992	Obs	4,859
25%	-4.613	-40.349	Sum of wgt.	4,859
50%	-2.335		Mean	-2.020479
		Largest	Std. dev.	6.55102
75%	.016	43.171		
90%	3.029	47.827	Variance	42.91586
95%	6.727	51.517	Skewness	2.230114
99%	21.656	125.135	Kurtosis	43.19337

General Gov. Gross Debt in % of GDP

	Percentiles	Smallest		
1%	0	0		
5%	8.607	0		
10%	16.041	0	Obs	4,694
25%	29.479	0	Sum of wgt.	4,694

50%	46.251		Mean	55.83493
		Largest	Std. dev.	44.59376
75%	69.833	468.08		
90%	101.523	488.458	Variance	1988.604
95%	131.051	543.399	Skewness	3.29178
99%	229.464	600.117	Kurtosis	24.46558

Consumer Price Inflation

Percentiles		Smallest		
1%	-2.314972	-16.85969		
5%	-.4156276	-16.11733		
10%	.368042	-10.06749	Obs	4,657
25%	1.63495	-9.797647	Sum of wgt.	4,657
50%	3.58922		Mean	8.378046
		Largest	Std. dev.	66.19835
75%	7.352812	513.9068		
90%	12.97767	557.2018	Variance	4382.221
95%	20.5836	1058.374	Skewness	53.77356
99%	63.93737	4145.106	Kurtosis	3292.583

Bank Credit to the Non-Financial Sector

Percentiles		Smallest		
1%	10.8	8.15		
5%	17.575	8.25		
10%	26.4	8.575	Obs	1,154
25%	50.4	8.75	Sum of wgt.	1,154
50%	80.575		Mean	82.55167
		Largest	Std. dev.	43.97461
75%	112.55	249.05		
90%	141.45	275.475	Variance	1933.766
95%	156.325	275.5	Skewness	.6004273
99%	195.3	278.5	Kurtosis	3.553925

Reel Effective Exchange Rate, basis 100 in 2010

Percentiles		Smallest		
1%	60.31218	42.90222		
5%	73.59288	46.24187		
10%	81.09153	47.95559	Obs	2,504
25%	92.77127	50.03793	Sum of wgt.	2,504
50%	99.70196		Mean	100.8481
		Largest	Std. dev.	25.15578
75%	105.8949	334.4452		
90%	117.5417	368.3285	Variance	632.813
95%	127.4744	511.051	Skewness	10.34149
99%	167.3001	741.6998	Kurtosis	212.9376

Fuel Export on Total Exports

Percentiles		Smallest		
1%	0	0		
5%	.0000494	0		
10%	.0067467	0	Obs	4,129
25%	.4881653	0	Sum of wgt.	4,129
50%	3.58842		Mean	15.82014
		Largest	Std. dev.	26.17739
75%	15.67077	99.78728		
90%	62.25328	99.79687	Variance	685.2555
95%	87.80417	99.88606	Skewness	1.977263
99%	97.22652	99.98648	Kurtosis	5.712573

Fuel Import on Total Imports

Percentiles		Smallest		
1%	.4896454	.0104926		
5%	1.446096	.0134593		
10%	3.4781	.0220932	Obs	4,309
25%	8.053708	.0242816	Sum of wgt.	4,309
50%	13.1856		Mean	14.04917
		Largest	Std. dev.	8.541186
75%	18.93046	55.60628		
90%	25.26662	62.62607	Variance	72.95186
95%	29.76995	66.41944	Skewness	.8386915
99%	38.50903	67.77688	Kurtosis	4.413544

Market cap. of listed domestic companies (% of GDP)

Percentiles		Smallest		
1%	.1630001	.0092022		
5%	4.251986	.0102898		
10%	8.015645	.0231675	Obs	1,747
25%	18.76657	.0455471	Sum of wgt.	1,747
50%	39.58964		Mean	65.91058
		Largest	Std. dev.	115.9082
75%	77.21398	1254.468		
90%	128.9133	1274.789	Variance	13434.71
95%	182.3595	1349.374	Skewness	7.763269
99%	508.2236	1777.226	Kurtosis	80.67973

Chinn-Ito index

Percentiles		Smallest		
1%	-1.931087	-1.931087		
5%	-1.931087	-1.931087		
10%	-1.241745	-1.931087	Obs	4,588
25%	-1.241745	-1.931087	Sum of wgt.	4,588
50%	-.0419779		Mean	.2664879
		Largest	Std. dev.	1.577925
75%	2.299361	2.299361		
90%	2.299361	2.299361	Variance	2.489849
95%	2.299361	2.299361	Skewness	.154604
99%	2.299361	2.299361	Kurtosis	1.354394

Chinn-Ito index, normalized

Percentiles		Smallest		
1%	0	0		
5%	0	0		
10%	.1629476	0	Obs	4,588
25%	.1629476	0	Sum of wgt.	4,588
50%	.4465505		Mean	.5194661
		Largest	Std. dev.	.3729925
75%	1	1		
90%	1	1	Variance	.1391234
95%	1	1	Skewness	.154604
99%	1	1	Kurtosis	1.354394

Financial development index (aggregate)

Percentiles		Smallest		
1%	.0373297	0		
5%	.0678928	0		
10%	.0862752	0	Obs	4,732
25%	.1168347	0	Sum of wgt.	4,732

50%	.222107		Mean	.2993796
		Largest	Std. dev.	.2259602
75%	.4212559	.9853314		
90%	.6742996	.9870471	Variance	.051058
95%	.7781473	.9968163	Skewness	1.076741
99%	.9160791	1	Kurtosis	3.255771

Financial Institution index

Percentiles		Smallest		
1%	.0608711	0		
5%	.1205035	0		
10%	.153026	0	Obs	4,732
25%	.1994752	0	Sum of wgt.	4,732
50%	.3409134		Mean	.3849358
		Largest	Std. dev.	.2206356
75%	.5170438	.9891428		
90%	.7334307	.9893202	Variance	.0486801
95%	.8256966	.994068	Skewness	.7572594
99%	.9442566	1	Kurtosis	2.709351

Financial Market index

Percentiles		Smallest		
1%	0	0		
5%	0	0		
10%	0	0	Obs	4,732
25%	.0047326	0	Sum of wgt.	4,732
50%	.0672409		Mean	.2020813
		Largest	Std. dev.	.2539148
75%	.3471263	.985147		
90%	.6197819	.9878856	Variance	.0644727
95%	.7538774	.9890106	Skewness	1.186791
99%	.8920792	1	Kurtosis	3.241321

ICRG index - External Conflict

Percentiles		Smallest		
1%	5.583333	2.125		
5%	7.083333	2.208333		
10%	8	2.583333	Obs	3,653
25%	9.208333	3	Sum of wgt.	3,653
50%	10		Mean	9.908691
		Largest	Std. dev.	1.431791
75%	11	12		
90%	11.5	12	Variance	2.050025
95%	12	12	Skewness	-1.04833
99%	12	12	Kurtosis	4.862078

ICRG index - Bureaucracy Quality

Percentiles		Smallest		
1%	0	0		
5%	0	0		
10%	1	0	Obs	3,653
25%	1.5	0	Sum of wgt.	3,653
50%	2		Mean	2.23037
		Largest	Std. dev.	1.092863
75%	3	4		
90%	4	4	Variance	1.19435
95%	4	4	Skewness	.0283842
99%	4	4	Kurtosis	2.366789

ICRG index - Corruption

Percentiles		Smallest		
1%	.9583333	0		
5%	1	0		
10%	1.5	0	Obs	3,653
25%	2	0	Sum of wgt.	3,653
50%	2.5	Largest	Mean	2.744776
			Std. dev.	1.211439
75%	3.5	6		
90%	4.583333	6	Variance	1.467584
95%	5	6	Skewness	.7164054
99%	6	6	Kurtosis	2.884471

ICRG index - Democratic Accountability

Percentiles		Smallest		
1%	0	0		
5%	1	0		
10%	2	0	Obs	3,653
25%	3	0	Sum of wgt.	3,653
50%	4.041667	Largest	Mean	4.027181
			Std. dev.	1.635482
75%	5.5	6		
90%	6	6	Variance	2.6748
95%	6	6	Skewness	-.5241565
99%	6	6	Kurtosis	2.29535

ICRG index - Ethnic Tensions

Percentiles		Smallest		
1%	1	0		
5%	2	0		
10%	2	0	Obs	3,653
25%	3	0	Sum of wgt.	3,653
50%	4	Largest	Mean	4.036454
			Std. dev.	1.245408
75%	5	6		
90%	6	6	Variance	1.55104
95%	6	6	Skewness	-.289059
99%	6	6	Kurtosis	2.303374

ICRG index - Government Stability

Percentiles		Smallest		
1%	5.041667	2.916667		
5%	5.791667	3.166667		
10%	6.166667	3.333333	Obs	3,653
25%	6.833333	3.333333	Sum of wgt.	3,653
50%	7.958333	Largest	Mean	8.113024
			Std. dev.	1.601062
75%	9.5	11.5		
90%	10.375	11.58333	Variance	2.563399
95%	10.91667	12	Skewness	.1453441
99%	11.08333	12	Kurtosis	2.135543

ICRG index - Internal Conflict

Percentiles		Smallest		
1%	4.083333	.4166667		
5%	6	1.333333		
10%	7	1.75	Obs	3,653
25%	8.041667	2	Sum of wgt.	3,653

50%	9.416667		Mean	9.174407
		Largest	Std. dev.	1.746908
75%	10.5	12		
90%	11.33333	12	Variance	3.051689
95%	12	12	Skewness	-.6982707
99%	12	12	Kurtosis	3.737106

ICRG index - Law and Order

Percentiles		Smallest		
1%	1	.5		
5%	2	.5		
10%	2	.5	Obs	3,653
25%	2.75	.5	Sum of wgt.	3,653
50%	4		Mean	3.76736
		Largest	Std. dev.	1.315515
75%	5	6		
90%	5.541667	6	Variance	1.730579
95%	6	6	Skewness	-.0064637
99%	6	6	Kurtosis	2.105336

ICRG index - Law and Order

Percentiles		Smallest		
1%	0	0		
5%	1	0		
10%	1.5	0	Obs	3,653
25%	2.5	0	Sum of wgt.	3,653
50%	4		Mean	3.851355
		Largest	Std. dev.	1.718088
75%	5	6		
90%	6	6	Variance	2.951826
95%	6	6	Skewness	-.4286774
99%	6	6	Kurtosis	2.168115

ICRG index - Religious Tensions

Percentiles		Smallest		
1%	1	0		
5%	2	0		
10%	2.5	0	Obs	3,653
25%	4	0	Sum of wgt.	3,653
50%	5		Mean	4.580288
		Largest	Std. dev.	1.297547
75%	5.5	6		
90%	6	6	Variance	1.683627
95%	6	6	Skewness	-.951252
99%	6	6	Kurtosis	3.308651

ICRG index - Socioeconomic Conditions

Percentiles		Smallest		
1%	.5	0		
5%	2	0		
10%	2.5	0	Obs	3,653
25%	4	0	Sum of wgt.	3,653
50%	5.5		Mean	5.716489
		Largest	Std. dev.	2.427911
75%	7.5	11		
90%	9.166667	11	Variance	5.894754
95%	9.791667	11	Skewness	.0789358
99%	10.5	11	Kurtosis	2.275913

ICRG index - Investment Profile

Percentiles		Smallest		
1%	2.5	0		
5%	4.583333	.6666667		
10%	5.833333	1	Obs	3,653
25%	7	1	Sum of wgt.	3,653
50%	8.083333		Mean	8.320901
		Largest	Std. dev.	2.178324
75%	10	12		
90%	11.5	12	Variance	4.745095
95%	12	12	Skewness	-.3010115
99%	12	12	Kurtosis	3.032726

Personal remittances, received (% of GDP)

Percentiles		Smallest		
1%	0	0		
5%	.0227784	0		
10%	.0822952	0	Obs	4,646
25%	.2956019	0	Sum of wgt.	4,646
50%	1.595145		Mean	4.211819
		Largest	Std. dev.	6.718227
75%	5.196908	57.50462		
90%	11.76216	67.98388	Variance	45.13458
95%	17.90444	87.55909	Skewness	3.564894
99%	29.68497	98.38965	Kurtosis	25.95724

Military expenditure (% of GDP)

Percentiles		Smallest		
1%	.0020281	.0003489		
5%	.0045876	.0003593		
10%	.006349	.0004822	Obs	3,859
25%	.0103253	.0004927	Sum of wgt.	3,859
50%	.0150782		Mean	.0204543
		Largest	Std. dev.	.0198928
75%	.0244296	.3249707		
90%	.0395603	.3265567	Variance	.0003957
95%	.0540164	.3354657	Skewness	6.180989
99%	.0856568	.3437777	Kurtosis	77.12751

MVF: Potential real GDP growth (percent)

Percentiles		Smallest		
1%	-2.399894	-9.163499		
5%	-.2695308	-7.215169		
10%	.6058946	-5.360445	Obs	1,962
25%	1.761614	-5.270265	Sum of wgt.	1,962
50%	3.045997		Mean	3.089487
		Largest	Std. dev.	2.309262
75%	4.278587	15.13628		
90%	5.667745	19.47046	Variance	5.33269
95%	6.761108	20.39099	Skewness	.8972885
99%	9.624201	22.28646	Kurtosis	10.01738

CF: Potential real GDP growth (percent)

Percentiles		Smallest		
1%	-2.373438	-7.105941		
5%	-.6390516	-6.262377		
10%	.3108758	-5.539729	Obs	1,962
25%	1.570193	-5.248143	Sum of wgt.	1,962

50%	2.904518		Mean	3.083705
		Largest	Std. dev.	2.509436
75%	4.490763	16.55023		
90%	5.91806	19.9486	Variance	6.29727
95%	7.118933	20.70829	Skewness	.9091014
99%	10.49936	21.55679	Kurtosis	7.95574

HP: Potential real GDP growth (percent)

Percentiles		Smallest		
1%	-1.287918	-5.802664		
5%	-.0569258	-5.316601		
10%	.6322935	-5.063198	Obs	1,962
25%	1.680679	-3.949438	Sum of wgt.	1,962
50%	2.981915		Mean	3.1137
		Largest	Std. dev.	2.24607
75%	4.318661	16.09106		
90%	5.723354	17.566	Variance	5.044832
95%	6.79561	18.39283	Skewness	1.038979
99%	10.03369	19.1487	Kurtosis	7.820674

BK: Potential real GDP growth (percent)

Percentiles		Smallest		
1%	-4.051818	-10.14352		
5%	-1.30498	-9.141929		
10%	-.0668108	-8.292189	Obs	1,962
25%	1.534777	-8.106364	Sum of wgt.	1,962
50%	3.027238		Mean	3.045545
		Largest	Std. dev.	2.775331
75%	4.570024	15.58165		
90%	6.160172	21.63985	Variance	7.702463
95%	7.302282	22.04183	Skewness	.5489641
99%	9.837472	26.96705	Kurtosis	9.00817

UCM: Potential real GDP growth (percent)

Percentiles		Smallest		
1%	-1.072861	-4.382327		
5%	.6756406	-4.278335		
10%	1.225484	-4.021938	Obs	1,977
25%	2.010305	-3.722157	Sum of wgt.	1,977
50%	2.986707		Mean	3.025026
		Largest	Std. dev.	1.664095
75%	3.867363	9.705173		
90%	4.88654	10.21828	Variance	2.769213
95%	5.940721	10.9109	Skewness	.4041248
99%	8.422082	11.48152	Kurtosis	5.616429

BW: Potential real GDP growth (percent)

Percentiles		Smallest		
1%	-1.622038	-10.62781		
5%	-.0311774	-6.105775		
10%	.6883115	-5.602174	Obs	1,962
25%	1.724086	-4.909271	Sum of wgt.	1,962
50%	2.942861		Mean	3.091324
		Largest	Std. dev.	2.194312
75%	4.241981	14.16874		
90%	5.614301	17.00577	Variance	4.815004
95%	6.706776	17.74937	Skewness	.7661922
99%	9.474479	19.2406	Kurtosis	8.122516

Exp(WEO): Potential real GDP growth (percent)

Percentiles		Smallest		
1%	.3497728	-9.042277		
5%	1.499752	-7.660636		
10%	1.9	-6.742797	Obs	4,290
25%	2.819955	-5.564719	Sum of wgt.	4,290
50%	4.048155		Mean	4.30717
		Largest	Std. dev.	2.61975
75%	5.5	44.4		
90%	6.506495	50.60827	Variance	6.863089
95%	7.251738	57.41383	Skewness	7.655733
99%	10.52881	65.03672	Kurtosis	148.2233

PFA: Potential real GDP growth (percent)

Percentiles		Smallest		
1%	-.6085138	-2.466566		
5%	.3616152	-2.443213		
10%	.8456069	-2.187064	Obs	1,942
25%	1.859857	-2.022126	Sum of wgt.	1,942
50%	3.114492		Mean	3.235463
		Largest	Std. dev.	1.977139
75%	4.444963	12.42918		
90%	5.652598	12.96244	Variance	3.909079
95%	6.474016	13.12467	Skewness	.6464445
99%	8.571218	13.82682	Kurtosis	4.517661

Contribution of capital accumulation to potential growth (percentage points)

Percentiles		Smallest		
1%	-.5525717	-2.695527		
5%	.0587102	-2.301567		
10%	.3017166	-2.296801	Obs	1,942
25%	.6465636	-1.731863	Sum of wgt.	1,942
50%	1.247598		Mean	1.470878
		Largest	Std. dev.	1.209911
75%	2.014689	8.895308		
90%	3.039899	9.050057	Variance	1.463885
95%	3.760817	9.240756	Skewness	1.565372
99%	5.233923	9.766214	Kurtosis	8.545529

Contribution of labor to potential growth (percentage points)

Percentiles		Smallest		
1%	-.6829789	-1.012055		
5%	-.2987305	-.9510471		
10%	-.0982387	-.8709149	Obs	1,942
25%	.2862923	-.8338634	Sum of wgt.	1,942
50%	.7186679		Mean	.7531268
		Largest	Std. dev.	.6740357
75%	1.1701	2.597879		
90%	1.657407	2.606983	Variance	.4543242
95%	1.908833	2.71859	Skewness	.3115375
99%	2.43173	2.858675	Kurtosis	2.916807

Potential TFP growth (percent)

Percentiles		Smallest		
1%	-1.465206	-5.27768		
5%	-.6805458	-2.884022		
10%	-.2361641	-2.436579	Obs	1,942
25%	.308033	-2.367229	Sum of wgt.	1,942

50%	.8571577		Mean	1.011458
		Largest	Std. dev.	1.179201
75%	1.560646	6.458211		
90%	2.525335	6.710578	Variance	1.390514
95%	3.205585	6.761465	Skewness	.766488
99%	4.558584	6.808414	Kurtosis	5.449165

Exchange Rate Stability Index

Percentiles		Smallest		
1%	.0628479	.001342		
5%	.1669631	.0038472		
10%	.2283633	.0044621	Obs	4,386
25%	.349238	.0049927	Sum of wgt.	4,386
50%	.6016456		Mean	.6206387
		Largest	Std. dev.	.3070486
75%	1	1		
90%	1	1	Variance	.0942789
95%	1	1	Skewness	-.005649
99%	1	1	Kurtosis	1.590039

GDP (current US\$)

Percentiles		Smallest		
1%	1.60e+08	4.76e+07		
5%	5.00e+08	6.31e+07		
10%	9.66e+08	6.51e+07	Obs	5,231
25%	4.24e+09	6.53e+07	Sum of wgt.	5,231
50%	1.85e+10		Mean	3.14e+11
		Largest	Std. dev.	1.41e+12
75%	1.28e+11	2.11e+13		
90%	4.98e+11	2.14e+13	Variance	1.97e+24
95%	1.37e+12	2.33e+13	Skewness	10.13735
99%	4.97e+12	2.54e+13	Kurtosis	126.6238

Total reserves minus gold (current US\$)

Percentiles		Smallest		
1%	2.46e+07	267707.4		
5%	6.95e+07	376466.9		
10%	1.33e+08	419970.6	Obs	4,572
25%	5.19e+08	431082.3	Sum of wgt.	4,572
50%	3.00e+09		Mean	4.30e+10
		Largest	Std. dev.	2.10e+11
75%	1.86e+10	3.33e+12		
90%	6.87e+10	3.35e+12	Variance	4.40e+22
95%	1.49e+11	3.84e+12	Skewness	12.12569
99%	6.63e+11	3.86e+12	Kurtosis	178.1305

Total reserves minus gold (% of GDP)

Percentiles		Smallest		
1%	.3608426	.0079961		
5%	1.593563	.0114261		
10%	2.806255	.0249938	Obs	4,542
25%	7.041934	.0306301	Sum of wgt.	4,542
50%	13.27799		Mean	18.1018
		Largest	Std. dev.	19.07478
75%	22.03725	162.3542		
90%	35.74814	174.6077	Variance	363.8471
95%	51.07561	189.547	Skewness	3.240108
99%	103.711	217.5606	Kurtosis	18.46138

Inflation Targeters

Percentiles		Smallest		
1%	0	0		
5%	0	0		
10%	0	0	Obs	5,373
25%	0	0	Sum of wgt.	5,373
50%	0	Largest	Mean	.13512
75%	0		Std. dev.	.3418836
90%	1	1	Variance	.1168844
95%	1	1	Skewness	2.134726
99%	1	1	Kurtosis	5.557056

ND-GAIN overall vulnerability

Percentiles		Smallest		
1%	.2657048	.2441626		
5%	.2993556	.2441857		
10%	.3208648	.2452629	Obs	4,784
25%	.3718786	.2458329	Sum of wgt.	4,784
50%	.4298115	Largest	Mean	.4416259
75%	.5166319		Std. dev.	.0955079
90%	.5742593	.6950421	Variance	.0091218
95%	.6051293	.6956675	Skewness	.282366
99%	.6523987	.6961713	Kurtosis	2.257771

ND-GAIN overall readiness

Percentiles		Smallest		
1%	.1864625	.1148045		
5%	.2388	.119845		
10%	.2641674	.1267297	Obs	4,862
25%	.3044719	.1269958	Sum of wgt.	4,862
50%	.3732921	Largest	Mean	.4043018
75%	.4739748		Std. dev.	.1331471
90%	.6101703	.8136891	Variance	.0177282
95%	.6868604	.815844	Skewness	.8496657
99%	.7651361	.8177426	Kurtosis	3.222075

Country group

Percentiles		Smallest		
1%	1	1		
5%	1	1		
10%	1	1	Obs	5,184
25%	2	1	Sum of wgt.	5,184
50%	2	Largest	Mean	1.802083
75%	2		Std. dev.	.3984674
90%	2	2	Variance	.1587763
95%	2	2	Skewness	-1.516372
99%	2	2	Kurtosis	3.299385

Geographical region

Percentiles		Smallest		
1%	1	1		
5%	1	1		
10%	1	1	Obs	5,184
25%	2	1	Sum of wgt.	5,184

50%	3		Mean	3.416667
		Largest	Std. dev.	1.852355
75%	6	7		
90%	6	7	Variance	3.431217
95%	6	7	Skewness	.3432699
99%	7	7	Kurtosis	1.709219

Income group

Percentiles		Smallest		
1%	1	1		
5%	1	1		
10%	1	1	Obs	5,184
25%	2	1	Sum of wgt.	5,184
50%	3		Mean	2.802083
		Largest	Std. dev.	1.042132
75%	4	5		
90%	4	5	Variance	1.086038
95%	4	5	Skewness	-.2335048
99%	4	5	Kurtosis	1.884275

General government gross debt, % of GDP
[Government debt sustainability]

Percentiles		Smallest		
1%	3.077	.002		
5%	10.161	.052		
10%	17.167	.055	Obs	4,662
25%	29.974	.059	Sum of wgt.	4,662
50%	46.5775		Mean	56.388
		Largest	Std. dev.	44.42299
75%	70.018	468.08		
90%	101.689	488.458	Variance	1973.402
95%	131.179	543.399	Skewness	3.335177
99%	229.464	600.117	Kurtosis	24.82328

Primary balance, % of GDP [Government debt sustainability]

Percentiles		Smallest		
1%	-16.745	-57.223		
5%	-7.639	-55.406		
10%	-5.232	-38.89	Obs	4,758
25%	-2.713	-38.689	Sum of wgt.	4,758
50%	-.6365		Mean	-.4003404
		Largest	Std. dev.	6.017783
75%	1.531	40.546		
90%	4.557	41.032	Variance	36.21371
95%	7.597	47.981	Skewness	2.038743
99%	18.639	126.464	Kurtosis	55.21375

Cyclically-adjusted balance, % of potential GDP
[Government debt sustainability]

Percentiles		Smallest		
1%	-13.27627	-27.89812		
5%	-8.583205	-27.70639		
10%	-6.915238	-25.38508	Obs	3,554
25%	-4.416863	-23.03165	Sum of wgt.	3,554
50%	-2.43723		Mean	-2.147097
		Largest	Std. dev.	4.930774
75%	-.2999961	33.82738		
90%	2.043513	34.57888	Variance	24.31253
95%	4.658547	34.84859	Skewness	1.801344
99%	16.68032	40.88641	Kurtosis	15.33119

Fiscal balance, % of GDP [Government debt sustainability]

Percentiles		Smallest		
1%	-18.29	-59.742		
5%	-9.718	-55.406		
10%	-7.471	-41.992	Obs	4,899
25%	-4.613	-40.349	Sum of wgt.	4,899
			Mean	-2.018306
50%	-2.334	Largest	Std. dev.	6.536883
75%	.024	43.171		
90%	3.062	47.827	Variance	42.73085
95%	6.711	51.517	Skewness	2.229455
99%	21.656	125.135	Kurtosis	43.2235

General government gross debt, % of average tax revenues [Government debt sustainai

Percentiles		Smallest		
1%	21.47134	.0313969		
5%	63.94805	.3947674		
10%	98.32207	.4175425	Obs	4,594
25%	188.3233	.4479092	Sum of wgt.	4,594
			Mean	455.4876
50%	294.125	Largest	Std. dev.	803.4545
75%	480.5399	9509.323		
90%	796.5107	11627.47	Variance	645539.1
95%	1201.339	18464.26	Skewness	14.53849
99%	3687.994	27964.48	Kurtosis	381.6201

Fiscal balance, % of average tax revenues [Government debt sustainability]

Percentiles		Smallest		
1%	-277.4674	-2874.928		
5%	-78.79257	-1183.078		
10%	-55.53825	-1177.813	Obs	4,804
25%	-30.84151	-1158.628	Sum of wgt.	4,804
			Mean	-8.585111
50%	-14.63123	Largest	Std. dev.	182.5883
75%	-.1773823	3024.811		
90%	16.93677	3073.121	Variance	33338.47
95%	40.22295	3399.739	Skewness	9.911097
99%	334.4995	3927.696	Kurtosis	196.6585

General government gross debt, % of 10-year moving average GDP [Government debt

Percentiles		Smallest		
1%	4.148743	.0033874		
5%	16.27242	.0664475		
10%	25.69462	.0694535	Obs	4,538
25%	45.02468	.0732803	Sum of wgt.	4,538
			Mean	83.55483
50%	67.04421	Largest	Std. dev.	81.00344
75%	99.44839	1083.625		
90%	142.7156	1118.832	Variance	6561.557
95%	191.2574	1198.527	Skewness	5.90888
99%	398.4715	1310.429	Kurtosis	60.61955

Fiscal balance, % of 10-year moving average GDP [Government debt sustainability]

Percentiles		Smallest		
1%	-27.21554	-82.48598		
5%	-14.81201	-81.85159		
10%	-11.08465	-65.5267	Obs	4,755
25%	-6.792097	-65.40052	Sum of wgt.	4,755

50%	-3.254948		Mean	-2.781126
		Largest	Std. dev.	10.70655
75%	.0416546	72.31235		
90%	4.603992	81.51528	Variance	114.6303
95%	10.76465	115.0252	Skewness	3.998177
99%	38.05156	247.6643	Kurtosis	78.54952

General government debt in foreign currency, % of total [Balance sheet compositi

Percentiles		Smallest		
1%	0	0		
5%	0	0		
10%	.0019569	0	Obs	766
25%	1.015049	0	Sum of wgt.	766
50%	20.62794		Mean	29.77175
		Largest	Std. dev.	30.13493
75%	51.88197	100		
90%	75.75573	100	Variance	908.114
95%	88.30422	100	Skewness	.6735007
99%	100	100	Kurtosis	2.183418

Debt securities held by nonresidents, % of total [Balance sheet composition]

Percentiles		Smallest		
1%	.0066404	0		
5%	.0880354	0		
10%	.147118	.0004383	Obs	965
25%	.4157351	.0006083	Sum of wgt.	965
50%	1.946375		Mean	3.393978
		Largest	Std. dev.	5.258295
75%	4.926527	23.80479		
90%	8.071179	51.24703	Variance	27.64966
95%	10.99788	67.89184	Skewness	7.245384
99%	18.28528	85.31197	Kurtosis	93.59386

General government debt held by nonresidents, % of total [Balance sheet composit

Percentiles		Smallest		
1%	3.006603	2.06399		
5%	7.069797	2.132759		
10%	12.31338	2.267759	Obs	1,303
25%	24.94576	2.277491	Sum of wgt.	1,303
50%	41.25378		Mean	41.18411
		Largest	Std. dev.	21.35915
75%	55.73731	94.63868		
90%	69.25017	95.26088	Variance	456.2133
95%	76.60153	100	Skewness	.1786694
99%	91.40111	100	Kurtosis	2.371338

Concessional external debt stocks, % of external public debt [Balance sheet comp

Percentiles		Smallest		
1%	.1132934	0		
5%	.502008	0		
10%	2.293194	0	Obs	2,958
25%	10.53843	0	Sum of wgt.	2,958
50%	33.27273		Mean	37.96831
		Largest	Std. dev.	28.99046
75%	62.89391	97.94978		
90%	80.44285	98.30277	Variance	840.4468
95%	87.31541	98.76178	Skewness	.3199552
99%	94.64228	98.83048	Kurtosis	1.805007

Sovereign debt average maturity, years [Balance sheet composition]

	Percentiles	Smallest		
1%	1.139726	.3123288		
5%	2.20274	.4273973		
10%	3.172603	.4383562	Obs	915
25%	4.810959	.4767123	Sum of wgt.	915
50%	6.416438		Mean	6.68283
		Largest	Std. dev.	2.899997
75%	8.287671	15.23562		
90%	10.39452	15.5863	Variance	8.409983
95%	12.36712	16.51781	Skewness	.6318863
99%	14.69589	20	Kurtosis	3.794645

Central government debt maturing in 12 months or less, % of GDP [Balance sheet c

	Percentiles	Smallest		
1%	.0059985	0		
5%	.1291087	.00067		
10%	.4347826	.0008026	Obs	1,533
25%	1.77156	.0008275	Sum of wgt.	1,533
50%	4.500309		Mean	7.170218
		Largest	Std. dev.	8.491593
75%	9.290467	57.33762		
90%	17.66714	58.67241	Variance	72.10714
95%	22.62424	65.35034	Skewness	2.706122
99%	45.04385	70.74007	Kurtosis	13.17954

Total external debt stocks, % of GDP [External and private sector debt]

	Percentiles	Smallest		
1%	4.52303	1.262543		
5%	13.83499	1.322676		
10%	18.77237	1.400716	Obs	4,064
25%	30.38171	1.581202	Sum of wgt.	4,064
50%	51.18033		Mean	106.1637
		Largest	Std. dev.	352.6985
75%	88.27927	5868.299		
90%	162.9898	6237.527	Variance	124396.2
95%	266.4075	6370.005	Skewness	12.8877
99%	872.529	6596.141	Kurtosis	192.0705

External debt in foreign currency, % of total [External and private sector debt]

	Percentiles	Smallest		
1%	2.476304	1.004835		
5%	13.63957	1.006385		
10%	29.58128	1.049087	Obs	631
25%	67.45039	1.880748	Sum of wgt.	631
50%	89.88615		Mean	77.58295
		Largest	Std. dev.	26.85264
75%	97.76864	100		
90%	99.63049	100	Variance	721.0644
95%	99.92233	100	Skewness	-1.419639
99%	100	100	Kurtosis	4.071995

Private external debt stocks, % of GDP [External and private sector debt]

	Percentiles	Smallest		
1%	0	0		
5%	.0672134	0		
10%	.3441063	0	Obs	3,094
25%	2.908527	0	Sum of wgt.	3,094

50%	13.02752		Mean	84.60899
		Largest	Std. dev.	395.0326
75%	49.01646	5670.924		
90%	132.3988	6053.606	Variance	156050.8
95%	274.4993	6196.125	Skewness	11.29491
99%	973.5942	6385.398	Kurtosis	146.1574

Domestic credit to private sector, % of GDP
[External and private sector debt]

Percentiles		Smallest		
1%	1.49942	.0014018		
5%	5.042955	.0014262		
10%	7.972025	.0015927	Obs	4,762
25%	16.16158	.0016961	Sum of wgt.	4,762
50%	37.25171		Mean	59.18188
		Largest	Std. dev.	62.74191
75%	73.67991	388.8		
90%	156.9	397.1	Variance	3936.547
95%	192.7	403.5	Skewness	1.861216
99%	271.5	407.4	Kurtosis	6.719105

Short-term external debt stocks, % of total
[External and private sector debt]

Percentiles		Smallest		
1%	0	0		
5%	.0920025	0		
10%	1.102934	0	Obs	4,050
25%	4.633544	0	Sum of wgt.	4,050
50%	12.02196		Mean	17.21466
		Largest	Std. dev.	16.48348
75%	25.6557	86.72114		
90%	39.40639	87.87831	Variance	271.7052
95%	49.87817	88.19446	Skewness	1.413675
99%	72.95538	88.43844	Kurtosis	4.926633

Short-term external debt stocks, % of reserves
[External and private sector debt]

Percentiles		Smallest		
1%	0	0		
5%	.2720667	0		
10%	2.406077	0	Obs	3,677
25%	14.68347	0	Sum of wgt.	3,677
50%	41.18527		Mean	1934.197
		Largest	Std. dev.	20294.29
75%	129.3685	343758.7		
90%	897.0849	349702.2	Variance	4.12e+08
95%	2141.907	527646.8	Skewness	17.87947
99%	28166.64	567255.3	Kurtosis	386.0725

Total external debt stocks, % of reserves
[External and private sector debt]

Percentiles		Smallest		
1%	23.2479	2.604098		
5%	70.472	2.749778		
10%	105.0557	2.962797	Obs	3,709
25%	206.5718	3.101955	Sum of wgt.	3,709
50%	365.48		Mean	5193.823
		Largest	Std. dev.	43541.8
75%	840.7016	632130.4		
90%	3094.789	643848.3	Variance	1.90e+09
95%	6877.468	987436	Skewness	14.37281
99%	114188.1	1059468	Kurtosis	250.2552

Total external debt stocks, % of reserves
excluding gold [External and private s

Percentiles		Smallest		
1%	23.2479	2.694032		
5%	74.89661	2.883313		
10%	109.7502	3.073855	Obs	3,709
25%	213.3253	3.228835	Sum of wgt.	3,709
50%	379.8818		Mean	7101.149
		Largest	Std. dev.	56704.72
75%	935.7491	876723.5		
90%	5198.115	1059468	Variance	3.22e+09
95%	11695.81	1246200	Skewness	14.37642
99%	146828	1411828	Kurtosis	256.3189

Foreign currency long-term sovereign debt ratings,
index from 1-21 [best] [Marke

Percentiles		Smallest		
1%	2.82878	1		
5%	5.319178	1		
10%	6.063927	1	Obs	3,300
25%	8	1	Sum of wgt.	3,300
50%	11.86548		Mean	12.36036
		Largest	Std. dev.	5.18118
75%	16.66667	21		
90%	20.66667	21	Variance	26.84463
95%	21	21	Skewness	.2375252
99%	21	21	Kurtosis	1.898261

ND-GAIN overall score

Percentiles		Smallest		
1%	29.31041	25.41625		
5%	33.92504	25.65922		
10%	35.51145	25.77942	Obs	4,784
25%	40.02007	25.88258	Sum of wgt.	4,784
50%	46.73949		Mean	48.04338
		Largest	Std. dev.	10.49908
75%	54.51261	76.49272		
90%	64.02494	76.55679	Variance	110.2307
95%	68.85712	76.97101	Skewness	.5367207
99%	73.18661	76.99231	Kurtosis	2.660591

IMF IFS, Exchange Rate, National Currency Per USD

Percentiles		Smallest		
1%	.2910881	.001		
5%	.709	.001		
10%	1	.004	Obs	4,000
25%	2.7	.0100833	Sum of wgt.	4,000
50%	13.62889		Mean	1681428
		Largest	Std. dev.	1.06e+08
75%	202.5463	33226.3		
90%	1129.68	40864.33	Variance	1.13e+16
95%	2899.484	45885.51	Skewness	63.22183
99%	14045.42	6.72e+09	Kurtosis	3998

IMF IFS, Exchange Rate, National Currency Per USD,
growth rate (winsorized)

Percentiles		Smallest		
1%	-1.446371	-2.606477		
5%	-.925133	-2.606477		
10%	-.6522531	-2.606477	Obs	3,996
25%	-.0183226	-2.606477	Sum of wgt.	3,996

50%	.0811281		Mean	.3417188
		Largest	Std. dev.	1.031324
75%	.6975903	7.973751		
90%	1.308949	7.991317	Variance	1.06363
95%	2.037568	11.68413	Skewness	2.640653
99%	4.476201	12.41035	Kurtosis	19.16684

IMF IFS, Interest Rates, Government Bonds, yield,
percentage

Percentiles		Smallest		
1%	.31	-.49		
5%	1.002833	-.3625		
10%	1.842083	-.1816667	Obs	1,320
25%	3.430417	-.0725	Sum of wgt.	1,320
50%	5.144167		Mean	6.294379
		Largest	Std. dev.	4.783413
75%	7.94125	38.50833		
90%	12.02039	39.25	Variance	22.88104
95%	14.74667	39.25	Skewness	2.554819
99%	24.89833	42.66667	Kurtosis	14.23773

JPM EMBI sovereign bond index, yield, percentage

Percentiles		Smallest		
1%	2.291917	.801		
5%	3.290583	.801		
10%	3.932	.801	Obs	1,011
25%	4.821083	.801	Sum of wgt.	1,011
50%	6.284833		Mean	7.671235
		Largest	Std. dev.	5.887525
75%	8.385	43.84708		
90%	11.95408	44.52842	Variance	34.66295
95%	16.14792	53.115	Skewness	5.837058
99%	31.12292	97.011	Kurtosis	65.1106

Sovereign bonds,IMF IFS or JPM EMBI, yield,
percentage

Percentiles		Smallest		
1%	.4666667	-.49		
5%	1.555	-.3625		
10%	2.546	-.1816667	Obs	2,052
25%	4.134167	-.0725	Sum of wgt.	2,052
50%	5.549773		Mean	6.760906
		Largest	Std. dev.	5.330575
75%	7.935875	43.84708		
90%	11.67392	44.52842	Variance	28.41503
95%	14.7	53.115	Skewness	4.894848
99%	29.02	97.011	Kurtosis	54.52696

Government bonds, JPM EMBI & IMF, yields %,
100*x/(x+100)(winsorized)

Percentiles		Smallest		
1%	.5197462	.4380725		
5%	1.531111	.4380725		
10%	2.482784	.4380725	Obs	2,052
25%	3.969811	.4380725	Sum of wgt.	2,052
50%	5.255799		Mean	6.078127
		Largest	Std. dev.	3.655882
75%	7.350521	23.31288		
90%	10.43521	23.31288	Variance	13.36547
95%	12.7261	23.31288	Skewness	1.828243
99%	21.84875	23.31288	Kurtosis	8.151658

IMF IFS, Treasury Bills, interest rates,
percentage

Percentiles		Smallest		
1%	-.2076667	-1.028333		
5%	.3116667	-.9916667		
10%	.8516667	-.91	Obs	2,030
25%	2.951818	-.668	Sum of wgt.	2,030
50%	5.943333		Mean	8.689899
		Largest	Std. dev.	15.40656
75%	10.47167	185.1075		
90%	16.54875	248.7667	Variance	237.3621
95%	23.89667	322.3558	Skewness	12.67164
99%	50.48417	340	Kurtosis	232.8128

IMF IFS, Treasury Bills, yields %, $100*x/(x+100)$ (winsorized)

Percentiles		Smallest		
1%	-.0984338	-.1101211		
5%	.3106492	-.1101211		
10%	.8442593	-.1101211	Obs	2,030
25%	2.855242	-.1101211	Sum of wgt.	2,030
50%	5.597156		Mean	6.983875
		Largest	Std. dev.	6.209331
75%	9.468165	36.30573		
90%	14.18937	36.30573	Variance	38.5558
95%	19.20794	36.30573	Skewness	1.873222
99%	32.28616	36.30573	Kurtosis	7.743436

IMF AREAER coded float or free float

Percentiles		Smallest		
1%	0	0		
5%	0	0		
10%	0	0	Obs	4,070
25%	0	0	Sum of wgt.	4,070
50%	1		Mean	.607371
		Largest	Std. dev.	.4883954
75%	1	1		
90%	1	1	Variance	.2385301
95%	1	1	Skewness	-.4397429
99%	1	1	Kurtosis	1.193374

MSCI country index, price, USD, growth rate
(winsorized)

Percentiles		Smallest		
1%	-7.447968	-14.81089		
5%	-4.356797	-10.12296		
10%	-2.456522	-9.597633	Obs	1,371
25%	-.896078	-9.196276	Sum of wgt.	1,371
50%	.573491		Mean	.5081706
		Largest	Std. dev.	2.666995
75%	2.075698	8.048136		
90%	3.578624	8.167474	Variance	7.11286
95%	4.754222	8.80011	Skewness	-.5150846
99%	6.604248	11.35375	Kurtosis	4.991226

MSCI country index, price, local currency units

Percentiles		Smallest		
1%	57.77392	15.62117		
5%	129.6016	17.02433		
10%	184.5759	20.03658	Obs	1,379
25%	433.6577	23.87892	Sum of wgt.	1,379

50%	838.794		Mean	2.63e+09
		Largest	Std. dev.	2.28e+10
75%	1461.515	2.30e+11		
90%	4700.586	2.34e+11	Variance	5.19e+20
95%	80939.35	2.73e+11	Skewness	9.65287
99%	1.58e+11	3.22e+11	Kurtosis	100.4755

MSCI country index, price, local currency units,
growth rate (winsorized)

Percentiles		Smallest		
1%	-6.368054	-13.49695		
5%	-3.571022	-8.668351		
10%	-2.06984	-8.627056	Obs	1,371
25%	-.6085364	-8.333346	Sum of wgt.	1,371
50%	.7173584		Mean	.629952
		Largest	Std. dev.	2.401406
75%	1.88685	7.920711		
90%	3.365425	8.708986	Variance	5.76675
95%	4.448584	9.482744	Skewness	-.4096105
99%	6.801702	10.6606	Kurtosis	5.29307

MSCI world index, price, USD, growth rate
(winsorized)

Percentiles		Smallest		
1%	-4.297278	-4.297278		
5%	-1.53118	-4.297278		
10%	-1.296199	-4.297278	Obs	4,440
25%	.0059069	-4.297278	Sum of wgt.	4,440
50%	.8902051		Mean	.5238988
		Largest	Std. dev.	1.420379
75%	1.285237	3.183805		
90%	1.690515	3.183805	Variance	2.017477
95%	1.801037	3.183805	Skewness	-1.489686
99%	3.183805	3.183805	Kurtosis	6.447081

MSCI stock prices at country level, in USD or EUR

Percentiles		Smallest		
1%	35.79475	35.79475		
5%	55.26767	41.10342		
10%	72.1858	42.52333	Obs	94
25%	120.0942	48.2663	Sum of wgt.	94
50%	669.3626		Mean	1240.041
		Largest	Std. dev.	1335.124
75%	1977.926	3974.04		
90%	3046.783	4257.545	Variance	1782557
95%	3476.142	5500.946	Skewness	1.378278
99%	6473.805	6473.805	Kurtosis	4.949186

US government bonds, yields %, $100 * x / (x + 100)$ (winsorized)

Percentiles		Smallest		
1%	1.777527	1.777527		
5%	1.82222	1.777527		
10%	2.090967	1.777527	Obs	4,440
25%	2.38644	1.777527	Sum of wgt.	4,440
50%	3.697286		Mean	3.685846
		Largest	Std. dev.	1.331757
75%	4.674675	6.047619		
90%	5.685096	6.047619	Variance	1.773577
95%	5.972183	6.047619	Skewness	.2141418
99%	6.047619	6.047619	Kurtosis	1.816234

US government bills, yields %, $100 \times x / (x + 100)$ (winsorized)

Percentiles		Smallest		
1%	.0349862	.0349862		
5%	.0524357	.0349862		
10%	.0582784	.0349862	Obs	4,440
25%	.1468368	.0349862	Sum of wgt.	4,440
50%	1.510106		Mean	2.099115
		Largest	Std. dev.	1.894353
75%	4.334991	5.485422		
90%	4.783746	5.485422	Variance	3.588574
95%	4.825291	5.485422	Skewness	.4071199
99%	5.485422	5.485422	Kurtosis	1.602258

Dummy, 1 if new program approved

Percentiles		Smallest		
1%	0	0		
5%	0	0		
10%	0	0	Obs	4,440
25%	0	0	Sum of wgt.	4,440
50%	0		Mean	.0078641
		Largest	Std. dev.	.0245564
75%	0	.0833333		
90%	0	.1666667	Variance	.000603
95%	.0833333	.1666667	Skewness	2.854033
99%	.0833333	.1666667	Kurtosis	9.493725

Dummy, 1 for election date of non-permanent member of UNSC

Percentiles		Smallest		
1%	0	0		
5%	0	0		
10%	0	0	Obs	4,440
25%	0	0	Sum of wgt.	4,440
50%	0		Mean	.0022898
		Largest	Std. dev.	.013624
75%	0	.0833333		
90%	0	.0833333	Variance	.0001856
95%	0	.0833333	Skewness	5.78115
99%	.0833333	.0833333	Kurtosis	34.4217

Dummy, 1 for 24 months of serving as non-permanent member of UNSC

Percentiles		Smallest		
1%	0	0		
5%	0	0		
10%	0	0	Obs	4,440
25%	0	0	Sum of wgt.	4,440
50%	0		Mean	.0022898
		Largest	Std. dev.	.013624
75%	0	.0833333		
90%	0	.0833333	Variance	.0001856
95%	0	.0833333	Skewness	5.78115
99%	.0833333	.0833333	Kurtosis	34.4217

Dummy, UNSC election month until 24 months after election date

Percentiles		Smallest		
1%	0	0		
5%	0	0		
10%	0	0	Obs	4,440
25%	0	0	Sum of wgt.	4,440

50%	0		Mean	.0558934
		Largest	Std. dev.	.2078403
75%	0	1		
90%	0	1	Variance	.0431976
95%	.75	1	Skewness	3.783211
99%	1	1	Kurtosis	15.95571

Dummy, UNSC election month until end of serving time

Percentiles		Smallest		
1%	0	0		
5%	0	0		
10%	0	0	Obs	4,440
25%	0	0	Sum of wgt.	4,440
50%	0		Mean	.061524
		Largest	Std. dev.	.2298754
75%	0	1		
90%	0	1	Variance	.0528427
95%	1	1	Skewness	3.719403
99%	1	1	Kurtosis	15.14397

IMF WEO, GDP, Constant Prices, Change y/y, annual (Jan value), winsorized

Percentiles		Smallest		
1%	-9.53	-12.58		
5%	-2.965	-12.58		
10%	-.657	-12.58	Obs	4,375
25%	1.729	-12.58	Sum of wgt.	4,375
50%	3.774		Mean	3.719619
		Largest	Std. dev.	4.128382
75%	5.965	17.095		
90%	8.192	17.095	Variance	17.04354
95%	10.011	17.095	Skewness	-.322549
99%	16.815	17.095	Kurtosis	5.680649

CPI % yoy, national sources

Percentiles		Smallest		
1%	-.2233333	-1.2075		
5%	-.0433333	-1.13		
10%	.0241667	-1.07	Obs	3,607
25%	.1325	-.8616667	Sum of wgt.	3,607
50%	.2933333		Mean	.5962535
		Largest	Std. dev.	2.210161
75%	.5958333	36.25		
90%	1.036667	38.19167	Variance	4.884812
95%	1.5225	47.68333	Skewness	20.97597
99%	5.0225	80.63333	Kurtosis	602.9627

CPI % yoy, national sources, 100*x/(x+100) (winsorized)

Percentiles		Smallest		
1%	-.2106859	-1.222756		
5%	-.0419056	-1.147153		
10%	.0239828	-.8694805	Obs	3,607
25%	.1318631	-.8555685	Sum of wgt.	3,607
50%	.2914182		Mean	.5009582
		Largest	Std. dev.	.8430271
75%	.5915999	8.925319		
90%	1.039431	8.925319	Variance	.7106948
95%	1.489243	8.925319	Skewness	5.451201
99%	4.430053	8.925319	Kurtosis	43.52914

PRGT member, dummy

Percentiles		Smallest		
1%	0	0		
5%	0	0		
10%	0	0	Obs	4,440
25%	0	0	Sum of wgt.	4,440
50%	0	Largest	Mean	.3702703
			Std. dev.	.4829314
75%	1	1		
90%	1	1	Variance	.2332227
95%	1	1	Skewness	.53732
99%	1	1	Kurtosis	1.288713

CBOE Volatility Index: VIX

Percentiles		Smallest		
1%	11.09024	11.09024		
5%	12.80653	11.09024		
10%	12.80706	11.09024	Obs	5,373
25%	15.47964	11.09024	Sum of wgt.	5,373
50%	19.65615	Largest	Mean	20.48322
			Std. dev.	5.903067
75%	25.60298	32.69553		
90%	29.2513	32.69553	Variance	34.8462
95%	31.47901	32.69553	Skewness	.3419639
99%	32.69553	32.69553	Kurtosis	2.07609

Sustainability gap, fiscal balance, % of GDP
[Government debt sustainability]

Percentiles		Smallest		
1%	-21.73286	-86.8037		
5%	-9.355527	-56.11696		
10%	-5.436617	-53.82725	Obs	4,874
25%	-1.546192	-51.75674	Sum of wgt.	4,874
50%	1.652069	Largest	Mean	1.96431
			Std. dev.	8.447719
75%	5.10411	54.02088		
90%	9.807907	55.85075	Variance	71.36396
95%	14.22867	61.4126	Skewness	.8542242
99%	30.09697	132.0947	Kurtosis	23.72544

Sustainability gap, primary balance,
country-specific conditions, % of GDP [Gove

Percentiles		Smallest		
1%	-17.36774	-109.1942		
5%	-6.844877	-92.00324		
10%	-4.935696	-88.85724	Obs	3,079
25%	-2.388597	-86.57724	Sum of wgt.	3,079
50%	-.0557358	Largest	Mean	-.7166823
			Std. dev.	8.778306
75%	2.027613	28.97297		
90%	4.411727	29.00714	Variance	77.05866
95%	6.801768	30.70907	Skewness	-6.953518
99%	13.41642	31.43961	Kurtosis	68.09932

Sustainability gap, primary balance, historical
conditions, % of GDP [Government

Percentiles		Smallest		
1%	-15.02081	-96.79687		
5%	-6.949777	-79.60587		
10%	-4.85141	-76.45987	Obs	3,079
25%	-2.336819	-74.17987	Sum of wgt.	3,079

50%	.0631814		Mean	-.4869112
		Largest	Std. dev.	7.824074
75%	2.184554	29.00634		
90%	4.614746	29.13913	Variance	61.21614
95%	6.890136	31.43453	Skewness	-6.019284
99%	13.62256	32.05574	Kurtosis	58.12937

Sustainability gap, primary balance, current conditions, % of GDP [Government de

Percentiles		Smallest		
1%	-19.56306	-119.7522		
5%	-10.15645	-87.30103		
10%	-6.569861	-79.14776	Obs	2,416
25%	-2.777908	-63.35097	Sum of wgt.	2,416
50%	.2439993		Mean	-.1907465
		Largest	Std. dev.	7.150662
75%	3.082523	26.70363		
90%	6.011062	31.13219	Variance	51.13197
95%	8.528568	36.14865	Skewness	-3.820548
99%	16.9357	36.24833	Kurtosis	55.11127

Sustainability gap, primary balance, stressed conditions, % of GDP [Government d

Percentiles		Smallest		
1%	-49.16211	-131.0409		
5%	-14.0961	-113.8499		
10%	-10.65817	-110.7039	Obs	3,029
25%	-7.25944	-108.4239	Sum of wgt.	3,029
50%	-4.352862		Mean	-5.773356
		Largest	Std. dev.	11.1188
75%	-1.773129	19.42416		
90%	.5184238	25.61727	Variance	123.6277
95%	2.35886	27.62186	Skewness	-6.513167
99%	7.870265	31.04496	Kurtosis	57.21818

Sustainability gap, primary balance, benign conditions, % of GDP [Government deb

Percentiles		Smallest		
1%	-11.03823	-62.55282		
5%	-3.056353	-45.36182		
10%	-1.280799	-42.21582	Obs	3,029
25%	1.267271	-39.93582	Sum of wgt.	3,029
50%	3.993323		Mean	4.131097
		Largest	Std. dev.	6.593363
75%	6.645238	33.20406		
90%	10.10223	33.53385	Variance	43.47243
95%	13.25885	33.90206	Skewness	-1.216796
99%	26.12506	37.06666	Kurtosis	18.89964

5-year sovereign CDS spreads, basis points [Market perception]

Percentiles		Smallest		
1%	2.697917	1.770833		
5%	7.866667	1.979167		
10%	19.17233	2	Obs	1,259
25%	48.33333	2	Sum of wgt.	1,259
50%	118.5		Mean	1705.588
		Largest	Std. dev.	22262.16
75%	285	63724.43		
90%	583.1429	269915.8	Variance	4.96e+08
95%	934.0767	269937.7	Skewness	24.55715
99%	21978.63	671619.3	Kurtosis	686.1286

1 = Laeven and Valencia 2020: systemic banking crisis

Percentiles		Smallest		
1%	0	0		
5%	0	0		
10%	0	0	Obs	4,356
25%	0	0	Sum of wgt.	4,356
50%	0		Mean	.0119376
		Largest	Std. dev.	.1086175
75%	0	1		
90%	0	1	Variance	.0117978
95%	0	1	Skewness	8.987842
99%	1	1	Kurtosis	81.78131

1 = Laeven and Valencia 2020: currency crisis

Percentiles		Smallest		
1%	0	0		
5%	0	0		
10%	0	0	Obs	4,356
25%	0	0	Sum of wgt.	4,356
50%	0		Mean	.0176768
		Largest	Std. dev.	.1317888
75%	0	1		
90%	0	1	Variance	.0173683
95%	0	1	Skewness	7.32048
99%	1	1	Kurtosis	54.58942

1 = Laeven and Valencia 2020: debt crisis

Percentiles		Smallest		
1%	0	0		
5%	0	0		
10%	0	0	Obs	4,356
25%	0	0	Sum of wgt.	4,356
50%	0		Mean	.0061983
		Largest	Std. dev.	.0784942
75%	0	1		
90%	0	1	Variance	.0061613
95%	0	1	Skewness	12.58331
99%	0	1	Kurtosis	159.3396

vul100

Percentiles		Smallest		
1%	26.57048	24.41626		
5%	29.93555	24.41857		
10%	32.08648	24.52629	Obs	4,784
25%	37.18786	24.5833	Sum of wgt.	4,784
50%	42.98115		Mean	44.16259
		Largest	Std. dev.	9.550787
75%	51.66319	69.40924		
90%	57.42593	69.5042	Variance	91.21753
95%	60.51293	69.56675	Skewness	.282366
99%	65.23988	69.61713	Kurtosis	2.257771

read100

Percentiles		Smallest		
1%	18.64626	11.48045		
5%	23.88	11.9845		
10%	26.41674	12.67297	Obs	4,862
25%	30.44719	12.69958	Sum of wgt.	4,862

50%	37.32921		Mean	40.43018
		Largest	Std. dev.	13.31471
75%	47.39748	81.36891		
90%	61.01702	81.5844	Variance	177.2816
95%	68.68604	81.77425	Skewness	.8496657
99%	76.5136	82.31771	Kurtosis	3.222075

Dvul100

Percentiles		Smallest		
1%	-1.281883	-5.751526		
5%	-.6577339	-4.976135		
10%	-.4553719	-3.833935	Obs	4,784
25%	-.2283936	-3.392979	Sum of wgt.	4,784
50%	-.070013		Mean	-.0827945
		Largest	Std. dev.	.4340853
75%	.0641041	3.932522		
90%	.2586479	4.092171	Variance	.18843
95%	.4628105	4.546429	Skewness	.0529663
99%	1.205624	5.317837	Kurtosis	32.3251

Number of Disaster

Percentiles		Smallest		
1%	0	0		
5%	0	0		
10%	0	0	Obs	5,157
25%	0	0	Sum of wgt.	5,157
50%	0		Mean	.1072329
		Largest	Std. dev.	.3094391
75%	0	1		
90%	1	1	Variance	.0957526
95%	1	1	Skewness	2.538821
99%	1	1	Kurtosis	7.44561

Following variable is string, not included:

country ccode countryname names
sum.rtf
dir : seeout

```
.
. // Prepare the data for the heatplot
.
. encode country, generate(CN)

. kcountry country, from(iso3c)
```

The command has finished.
The new variable is named NAMES_STD.

```
. rename NAMES_STD cn
. encode cn, generate(cnum)

.
. sum vul, detail
```

ND-GAIN overall vulnerability

Percentiles		Smallest		
1%	.2658107	.2441626		
5%	.2996005	.2441857		
10%	.3210096	.2452629	Obs	4,968
25%	.3722011	.2458329	Sum of wgt.	4,968

50%	.4300644		Mean	.4420377
		Largest	Std. dev.	.0955
75%	.5174853	.6940924		
90%	.5748193	.6950421	Variance	.0091203
95%	.6052048	.6956675	Skewness	.2783735
99%	.6523987	.6961713	Kurtosis	2.256964

```
.
. format vul %4.2f
```

```
.
. by imfcode: egen mean_vul = mean(vul)
(495 missing values generated)
```

```
.
. *** Figure 1. Heat plot for the low vulnerability score
```

```
.
. // Run everything between preserve and restore
```

```
.
. preserve
```

```
. drop if mean_vul > .3722011
(5,148 observations deleted)
```

```
. drop if period<1995
(215 observations deleted)
```

```
. drop if period>2021
(43 observations deleted)
```

```
.
. summ period
```

Variable	Obs	Mean	Std. dev.	Min	Max
period	1,161	2008	7.792238	1995	2021

```
. local x1 = `r(min)'
```

```
. local x2 = `r(max)'
```

```
. heatmap vul i.cnum period if mean_vul< .3722011 , ///
> yscale(noline) ///
> ylabel(, nogrid labsize(*0.5)) ///
> xlabel(`x1'(5)`x2', labsize(*0.75) angle(vertical) nogrid) ///
> color(viridis) ///
> levels(8) ///
> ramp(right space(14) format(%4.2f)) ///
> p(lcolor(black%10) lwidth(*0.1)) ///
> ytitle("") ///
> xtitle("", size(vsmall)) ///
> xdiscrete name(vulQ1, replace) ///
> title("Vulnerability Index (below Q1)") ///
> note("Data source: Notre Dame Global Adaptation Initiative.", ///
> size(vsmall))
```

```
. graph export vulQ1.png, as(png) width(4000) replace
file vulQ1.png saved as PNG format
```

```
. restore
```

```
.
. *** Figure 2. Heat plot for the high vulnerability score
.
. // Run everything between preserve and restore
.
. preserve

. drop if mean_vul < .5174853
(4,554 observations deleted)

. drop if period<1995
(305 observations deleted)

. drop if period>2021
(61 observations deleted)

.
. summ period
```

Variable	Obs	Mean	Std. dev.	Min	Max
period	1,647	2008	7.791247	1995	2021

```
. local x1 = `r(min)'
. local x2 = `r(max)'

. heatmap vul i.cnum period if mean_vul> .5174853 , ///
> yscale(noline) ///
> ylabel(, nogrid labsize(*0.5)) ///
> xlabel(`x1'(5)`x2', labsize(*0.75) angle(vertical) nogrid) ///
> color(inferno, reverse) ///
> levels(8) ///
> ramp(right space(14) format(%4.2f)) ///
> p(lcolor(black%10) lwidth(*0.1)) ///
> ytitle("") ///
> xtitle("", size(vsmall)) ///
> xdiscrete name(vulQ3, replace) ///
> title("Vulnerability Index (above Q3)") ///
> note("Data source: Notre Dame Global Adaptation Initiative.", ///
> size(vsmall))

. graph export vulQ3.png, as(png) width(4000) replace
file vulQ3.png saved as PNG format

. restore

.
.
. *** Table 2. Comparing fundamentals and institutional features for different levels of vulnerability
.
. gen vul100Q4 = 0

.
. summ vul100 if period>1995, detail
```

vul100				
Percentiles		Smallest		
1%	26.57048	24.41626		
5%	29.93555	24.41857		
10%	32.08648	24.52629	Obs	4,784
25%	37.18786	24.5833	Sum of wgt.	4,784
50%	42.98115		Mean	44.16259
		Largest	Std. dev.	9.550787
75%	51.66319	69.40924		
90%	57.42593	69.5042	Variance	91.21753
95%	60.51293	69.56675	Skewness	.282366
99%	65.23988	69.61713	Kurtosis	2.257771

```

. local x1 = `r(p75)'

. replace vul100Q4 = 1 if vul100 < `x1'
(3,721 real changes made)

. replace vul100Q4 = . if vul100 ==.
(1,599 real changes made, 1,599 to missing)

.
. label define vul100Q4lab 0 "VUL High" 1 "VUL Low"

. label values vul100Q4 vul100Q4lab

.
. lab var milpol "ICRG index - Military in Politics"

.
. putdocx clear

. putdocx begin, font("Times New Roman", 7)

.
. #delimit;
delimiter now ;
. local variables "vul read bonds_tw bills_tw sovrates ka_open ers FI FM extconf corruption bureau demoacc ethnictens
> ns govstab intconf laworder milpol reletensions";

. summ `variables' if period>1995;

```

Variable	Obs	Mean	Std. dev.	Min	Max
vul	4,784	.4416259	.0955079	.2441626	.6961713
read	4,862	.4043018	.1331471	.1148045	.8231771
bonds_tw	2,052	6.078127	3.655882	.4380725	23.31288
bills_tw	2,030	6.983875	6.209331	-.1101211	36.30573
sovrates	3,300	12.36036	5.18118	1	21
ka_open	4,588	.5194661	.3729925	0	1
ers	4,386	.6206387	.3070486	.001342	1
FI	4,732	.3849358	.2206356	0	1
FM	4,732	.2020813	.2539148	0	1
extconf	3,653	9.908691	1.431791	2.125	12
corruption	3,653	2.744776	1.211439	0	6
bureau	3,653	2.23037	1.092863	0	4
demoacc	3,653	4.027181	1.635482	0	6
ethnictens	3,653	4.036454	1.245408	0	6
govstab	3,653	8.113024	1.601062	2.916667	12
intconf	3,653	9.174407	1.746908	.4166667	12
laworder	3,653	3.76736	1.315515	.5	6
milpol	3,653	3.851355	1.718088	0	6
reletensions	3,653	4.580288	1.297547	0	6

```

. dtable `variables' if period>1995,
> by(vul100Q4, totals tests)
> column(by(hide))
> sample(, place(seplabels))
> title(Descriptive statistics Full sample after 1995)
> titlestyles(font(, size(12) color(blue) bold))
> nformat(%6.0fc frequency)
> nformat(%6.2f mean sd cv)
> continuous(, statistic(mean sd cv))
> halign(right)
> note(Mean (Standard deviation) Coefficient of Variation:
> p-value from a pooled t-test.)
> note(Frequency (Percent%): p-value from Pearson test.);
note: using test regress across levels of vul100Q4 for vul, read, bonds_tw, bills_tw, sovrates, ka_open, ers, FI,
FM, extconf, corruption, bureau, demoacc, ethnictens, govstab, intconf, laworder, milpol, and reletensions.

```

Descriptive statistics Full sample after 1995

	VUL High 1,196 (25.0%)		VUL Low 3,588 (75.0%)		
ND-GAIN overall vulnerability	0.57	(0.04)	0.07	0.40	(0.06) 0.16
ND-GAIN overall readiness	0.31	(0.07)	0.22	0.43	(0.13) 0.31
Government bonds, JPM EMBI & IMF, yields %, $100*x/(x+100)$ (winsorized)	7.67	(3.55)	0.46	5.93	(3.63) 0.61
IMF IFS, Treasury Bills, yields %, $100*x/(x+100)$ (winsorized)	9.83	(6.34)	0.65	6.36	(6.03) 0.95
Foreign currency long-term sovereign debt ratings, index from 1-21 [best] [Marke	7.30	(1.38)	0.19	12.64	(5.14) 0.41
Chinn-Ito index, normalized	0.34	(0.32)	0.94	0.57	(0.37) 0.65
Exchange Rate Stability Index	0.57	(0.28)	0.48	0.62	(0.31) 0.50
Financial Institution index	0.18	(0.07)	0.41	0.45	(0.21) 0.48
Financial Market index	0.03	(0.08)	2.48	0.26	(0.26) 1.03
ICRG index - External Conflict	9.11	(1.64)	0.18	10.10	(1.32) 0.13
ICRG index - Corruption	1.95	(0.78)	0.40	2.91	(1.22) 0.42
ICRG index - Bureaucracy Quality	1.22	(0.81)	0.66	2.44	(1.03) 0.42
ICRG index - Democratic Accountability	3.19	(1.32)	0.41	4.22	(1.65) 0.39
ICRG index - Ethnic Tensions	3.08	(1.06)	0.34	4.24	(1.19) 0.28
ICRG index - Government Stability	8.09	(1.66)	0.21	8.17	(1.58) 0.19
ICRG index - Internal Conflict	7.76	(1.62)	0.21	9.48	(1.64) 0.17
ICRG index - Law and Order	2.80	(0.98)	0.35	3.98	(1.29) 0.32
ICRG index - Military in Politics	2.08	(1.32)	0.64	4.24	(1.56) 0.37
ICRG index - Religious Tensions	3.67	(1.34)	0.36	4.78	(1.21) 0.25

Mean (Standard deviation) Coefficient of Variation: p-value from a pooled t-test.

Frequency (Percent%): p-value from Pearson test.

```
. collect style putdocx, halign(center)
> layout(autofitcontents);

. putdocx collect;
(collection DTable posted to putdocx)

. putdocx save ds_table2.docx, replace;
successfully replaced "C:/Users/jamel/Dropbox/Latex/PROJECTS/24-03-emft-adb-private/manuscript/SI_EE/Replication_E
> E/ds_table2.docx"

. #delimit cr
delimiter now cr

.
. *** Figure 3. Scatter plot for the vulnerability score and bond yields
.
. twoway (lfit vul bonds_tw) (scatter vul bonds_tw), ///
> name(bonds, replace)

.
. graph export bonds.png, as(png) width(4000) replace
file bonds.png saved as PNG format

.
. *** Figure 4. Scatter plot for the vulnerability score and the sovereign ratings
.
. lab var sovrage "Foreign currency long-term sovereign debt ratings, index from 1-21 [best]"

.
. twoway (lfit vul sovrage) (scatter vul sovrage), ///
> name(sovrage, replace) xlabel(0(5)27)

.
. graph export sovrage.png, as(png) replace width(4000)
file sovrage.png saved as PNG format
```

```

.
. *** Figure 5a. Ridgeline for the changes in climate vulnerability scores
.
. set scheme white_tableau

. graph set window fontface "Arial Narrow"

.
. ridgeline Dvul100 if mean_vul< .3722011 & period<2018, ///
> by(period) name(JoyplotQ1, replace) ///
> overlap(8) bwid(0.1) palette(CET C1) alpha(100) ///
> lc(white) lw(0.2) xlabel(-1.5(0.5)1.5) laboff(-0.25) norm(local) ///
> xtitle("Change in vulnerability") ytitle("Period") ///
> title("{fontface Arial Bold:Ridgeline for the low vulnerability score}") ///
> subtitle("Below Q1") ///
> note("Data source: Notre Dame Global Adaptation Initiative.", size(vsmall))

. graph export joyQ1.png, as(png) width(4000) replace
file joyQ1.png saved as PNG format

.
. *** Figure 5b. Ridgeline for the changes in climate vulnerability scores
.
. ridgeline Dvul100 if mean_vul> .5174853 & period<2018, ///
> by(period) name(JoyplotQ3, replace) ///
> overlap(8) bwid(0.25) palette(CET C1) alpha(100) ///
> lc(white) lw(0.2) xlabel(-6(1)6) off(-0.25) norm(local) ///
> xtitle("Change in vulnerability") ytitle("Period") ///
> title("{fontface Arial Bold:Ridgeline for the high vulnerability score}") ///
> subtitle("Above Q3") ///
> note("Data source: Notre Dame Global Adaptation Initiative.", size(vsmall))

. graph export joyQ3.png, as(png) width(4000) replace
file joyQ3.png saved as PNG format

.
. *** Table 3. Reverse causality
.
. // Run everything
.
. areg D.vul100 L(0/4).bonds_tw i.period, absorb(imfcode) robust

Linear regression, absorbing indicators      Number of obs      =   1,670
Absorbed variable: imfcode                  No. of categories   =    111
                                             F(27, 1532)         =    3.12
                                             Prob > F             =  0.0000
                                             R-squared            =  0.1009
                                             Adj R-squared        =  0.0205
                                             Root MSE             =  0.3197

```

		Robust				
D.vul100	Coefficient	std. err.	t	P> t	[95% conf. interval]	
bonds_tw						
--	.0064704	.0067884	0.95	0.341	-.0068452	.0197859
L1.	-.0090161	.0108829	-0.83	0.408	-.030363	.0123308
L2.	.01091	.0110457	0.99	0.323	-.0107562	.0325762
L3.	-.0113992	.0093492	-1.22	0.223	-.0297378	.0069395
L4.	.002586	.0074665	0.35	0.729	-.0120596	.0172316
period						
1998	.0119626	.0375203	0.32	0.750	-.061634	.0855593
1999	.0268691	.0442682	0.61	0.544	-.0599636	.1137018
2000	.0150892	.0454678	0.33	0.740	-.0740965	.1042748
2001	.0125183	.0509633	0.25	0.806	-.0874469	.1124835
2002	.1208534	.100425	1.20	0.229	-.0761316	.3178385
2003	.0069355	.0681123	0.10	0.919	-.1266678	.1405388
2004	.0466256	.0482961	0.97	0.334	-.048108	.1413591
2005	-.0455202	.0484062	-0.94	0.347	-.1404696	.0494291
2006	-.0179604	.0459237	-0.39	0.696	-.1080404	.0721195
2007	-.0343289	.055173	-0.62	0.534	-.1425515	.0738937
2008	.0210138	.046559	0.45	0.652	-.0703123	.1123399

2009	.0297911	.0619478	0.48	0.631	-.0917203	.1513024
2010	-.1346533	.0484215	-2.78	0.005	-.2296328	-.0396738
2011	-.099792	.0569771	-1.75	0.080	-.2115533	.0119693
2012	-.0318001	.0503514	-0.63	0.528	-.1305651	.0669649
2013	.0462755	.049311	0.94	0.348	-.0504486	.1429996
2014	-.0029592	.0527675	-0.06	0.955	-.1064635	.100545
2015	.0283336	.0578935	0.49	0.625	-.0852253	.1418924
2016	.0446499	.0509138	0.88	0.381	-.0552182	.144518
2017	.0530213	.0508573	1.04	0.297	-.0467359	.1527785
2018	.0062959	.0543263	0.12	0.908	-.1002658	.1128576
2019	.1624279	.0532387	3.05	0.002	.0579996	.2668563
_cons	-.0917515	.0609989	-1.50	0.133	-.2114017	.0278987

```
.
. estimate store m1
```

```
.
. local switches "dec(2) word pvalue e(rmse)"
```

```
. outreg2 [m1] ///
> using "bonds.rtf", replace `switches'
bonds.rtf
dir : seeout
```

```
. areg D.vul100 L(0/4).sovrates i.period, absorb(imfcode) robust
```

Linear regression, absorbing indicators	Number of obs	=	2,632
Absorbed variable: imfcode	No. of categories	=	138
	F(30, 2464)	=	4.61
	Prob > F	=	0.0000
	R-squared	=	0.1135
	Adj R-squared	=	0.0534
	Root MSE	=	0.2802

D.vul100	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
sovrates						
--.	-.0116197	.0100592	-1.16	0.248	-.031345	.0081055
L1.	.011075	.0168303	0.66	0.511	-.0219279	.044078
L2.	-.006535	.0184534	-0.35	0.723	-.0427208	.0296508
L3.	.023873	.0171244	1.39	0.163	-.0097067	.0574527
L4.	-.0105837	.0095065	-1.11	0.266	-.0292252	.0080578
period						
1997	-.0771944	.0357107	-2.16	0.031	-.1472203	-.0071684
1998	-.0725689	.0338133	-2.15	0.032	-.1388743	-.0062636
1999	-.0599636	.0346905	-1.73	0.084	-.1279892	.008062
2000	-.0762175	.0353703	-2.15	0.031	-.145576	-.0068589
2001	-.0625579	.0364362	-1.72	0.086	-.1340065	.0088908
2002	-.0411787	.0403168	-1.02	0.307	-.120237	.0378796
2003	-.04193	.037136	-1.13	0.259	-.1147509	.030891
2004	-.0815995	.0411994	-1.98	0.048	-.1623885	-.0008105
2005	-.1403184	.0377993	-3.71	0.000	-.21444	-.0661967
2006	-.1169994	.0329374	-3.55	0.000	-.1815874	-.0524115
2007	-.0569619	.0340522	-1.67	0.094	-.1237357	.009812
2008	-.047897	.034431	-1.39	0.164	-.1154137	.0196196
2009	-.0641022	.0433348	-1.48	0.139	-.1490787	.0208743
2010	-.1698717	.0361658	-4.70	0.000	-.2407902	-.0989531
2011	-.1470052	.0413635	-3.55	0.000	-.228116	-.0658944
2012	-.1176198	.0379799	-3.10	0.002	-.1920957	-.0431439
2013	-.0437349	.0381606	-1.15	0.252	-.118565	.0310952
2014	-.0682998	.0357302	-1.91	0.056	-.1383641	.0017646
2015	-.0643886	.0436995	-1.47	0.141	-.1500801	.0213029
2016	-.0209129	.0340313	-0.61	0.539	-.0876457	.0458199
2017	.0024652	.0369902	0.07	0.947	-.07007	.0750004
2018	-.0340619	.0331435	-1.03	0.304	-.0990539	.0309301
2019	.0638792	.0329751	1.94	0.053	-.0007826	.1285409
2020	.0375303	.0327902	1.14	0.253	-.026769	.1018295

2021		.0026953	.0281917	0.10	0.924	-.0525866	.0579772
_cons		-.108235	.0517153	-2.09	0.036	-.2096449	-.0068252

```
. estimate store m2
```

```
. local switches "dec(2) word pvalue e(rmse)"
```

```
. outreg2 [m2] ///
> using "sov.rtf", replace `switches'
sov.rtf
dir : seeout
```

```
. *** Figure 6. Panel LP - impact of change in vulnerability on bond yields (Baseline)
```

```
. locproj bonds_tw, shock(Dvul100) ///
> z h(5) yl(1) sl(3) ///
> c(f(1/5).Dvul100 i.period) ///
> fe cluster(imfcode) conf(90 95) ///
> ttitle("Horizon") ///
> title("Impact of Change in Vulnerability on Bonds Yields") ///
> save irfname(bonds_TZ) noisily stats gname(TZ1)
bonds_tw_h(0)
```

		Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Dvul100							
--		.1969226	.0951677	2.07	0.039	.0103974	.3834478
L1.		.4518202	.1223067	3.69	0.000	.2121035	.6915369
L2.		.3360053	.1345511	2.50	0.013	.07229	.5997206
L3.		.0197633	.1105525	0.18	0.858	-.1969156	.2364422
bonds_tw							
L1.		.6970993	.0452437	15.41	0.000	.6084232	.7857754
Dvul100							
F1.		.1174104	.130548	0.90	0.368	-.1384589	.3732798
F2.		.1914235	.1781349	1.07	0.283	-.1577145	.5405615
F3.		-.0164867	.1594324	-0.10	0.918	-.3289685	.2959952
F4.		-.1228312	.1605075	-0.77	0.444	-.4374202	.1917578
F5.		-.0669783	.1234098	-0.54	0.587	-.3088571	.1749005
period							
1999		0 (empty)					
2000		-.1822767	.3016931	-0.60	0.546	-.7735843	.4090308
2001		-.7257514	.3231662	-2.25	0.025	-1.359145	-.0923574
2002		-.8567085	.3462413	-2.47	0.013	-1.535329	-.1780881
2003		-1.559519	.323329	-4.82	0.000	-2.193232	-.9258054
2004		-1.070534	.3307339	-3.24	0.001	-1.718761	-.4223075
2005		-1.504292	.3442044	-4.37	0.000	-2.17892	-.8296634
2006		-.9480199	.3353529	-2.83	0.005	-1.6053	-.2907403
2007		-.8594597	.3241498	-2.65	0.008	-1.494782	-.2241377
2008		-.5707816	.3022444	-1.89	0.059	-1.16317	.0216066
2009		-.3877145	.3019624	-1.28	0.199	-.97955	.204121
2010		-2.016004	.3763471	-5.36	0.000	-2.753631	-1.278377
2011		-.9975136	.3527005	-2.83	0.005	-1.688794	-.3062334
2012		-1.459862	.3463397	-4.22	0.000	-2.138675	-.7810486
2013		-1.687155	.3111576	-5.42	0.000	-2.297012	-1.077297
2014		-1.794599	.3269802	-5.49	0.000	-2.435469	-1.15373
2015		-1.385771	.334295	-4.15	0.000	-2.040977	-.730565
2016		-1.567108	.3524763	-4.45	0.000	-2.257949	-.8762675
_cons		2.918295	.4643071	6.29	0.000	2.008269	3.82832

```
bonds_tw_h(1)
```

		Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Dvul100							

--.	.5134614	.1437519	3.57	0.000	.2317129	.79521
L1.	.5815895	.1780178	3.27	0.001	.2326811	.930498
L2.	.1918925	.1400255	1.37	0.171	-.0825524	.4663374
L3.	.1032169	.1493474	0.69	0.489	-.1894987	.3959324
bonds_tw						
L1.	.3617234	.0774456	4.67	0.000	.2099328	.513514
Dvul100						
F1.	.2576502	.1786049	1.44	0.149	-.0924089	.6077093
F2.	.2508659	.2206769	1.14	0.256	-.1816529	.6833848
F3.	.1533128	.2421908	0.63	0.527	-.3213724	.627998
F4.	-.100107	.2229031	-0.45	0.653	-.5369891	.336775
F5.	-.1686571	.1847008	-0.91	0.361	-.5306641	.1933499
period						
1999	0	(empty)				
2000	-.7732672	.2407591	-3.21	0.001	-1.245146	-.3013881
2001	-1.117557	.3675565	-3.04	0.002	-1.837954	-.397159
2002	-2.047617	.3724825	-5.50	0.000	-2.777669	-1.317565
2003	-2.137068	.3639276	-5.87	0.000	-2.850353	-1.423783
2004	-2.362185	.3628806	-6.51	0.000	-3.073418	-1.650952
2005	-2.112556	.3797191	-5.56	0.000	-2.856792	-1.36832
2006	-1.719416	.3882096	-4.43	0.000	-2.480293	-.9585388
2007	-1.366527	.3584746	-3.81	0.000	-2.069124	-.6639295
2008	-.9699604	.3778206	-2.57	0.010	-1.710475	-.2294456
2009	-2.451212	.3845372	-6.37	0.000	-3.204891	-1.697533
2010	-2.494054	.4166051	-5.99	0.000	-3.310585	-1.677523
2011	-2.374176	.4333921	-5.48	0.000	-3.223609	-1.524743
2012	-2.926764	.3606599	-8.12	0.000	-3.633644	-2.219883
2013	-3.202862	.3715325	-8.62	0.000	-3.931053	-2.474672
2014	-2.949206	.4213318	-7.00	0.000	-3.775001	-2.123411
2015	-2.904199	.4252716	-6.83	0.000	-3.737716	-2.070682
2016	-3.395433	.4387063	-7.74	0.000	-4.255282	-2.535585
_cons	5.843653	.6481621	9.02	0.000	4.573278	7.114027

bonds_tw_h(2)						
	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Dvul100						
--.	.5647578	.1813333	3.11	0.002	.2093511	.9201646
L1.	.3046979	.1758962	1.73	0.083	-.0400523	.6494482
L2.	.099303	.1898561	0.52	0.601	-.2728081	.4714141
L3.	.1231428	.1842056	0.67	0.504	-.2378936	.4841791
bonds_tw						
L1.	.1840183	.0892877	2.06	0.039	.0090177	.3590189
Dvul100						
F1.	.5142857	.1886861	2.73	0.006	.1444678	.8841037
F2.	.2955134	.2217689	1.33	0.183	-.1391457	.7301726
F3.	.2368115	.2380038	0.99	0.320	-.2296674	.7032905
F4.	.1482692	.2348096	0.63	0.528	-.3119492	.6084877
F5.	-.0919816	.2132331	-0.43	0.666	-.5099108	.3259476
period						
1999	0	(empty)				
2000	-.5962831	.2540868	-2.35	0.019	-1.094284	-.0982821
2001	-1.698187	.3101444	-5.48	0.000	-2.306059	-1.090315
2002	-1.90876	.3385724	-5.64	0.000	-2.57235	-1.24517
2003	-2.396887	.3017878	-7.94	0.000	-2.98838	-1.805394
2004	-2.115246	.3826749	-5.53	0.000	-2.865275	-1.365216
2005	-1.864394	.404848	-4.61	0.000	-2.657882	-1.070907
2006	-1.339502	.3879723	-3.45	0.001	-2.099913	-.57909
2007	-1.051307	.3966263	-2.65	0.008	-1.82868	-.2739334
2008	-2.248481	.4255988	-5.28	0.000	-3.082639	-1.414323
2009	-2.302136	.4338964	-5.31	0.000	-3.152557	-1.451714
2010	-2.719362	.4523921	-6.01	0.000	-3.606034	-1.83269
2011	-2.958712	.414121	-7.14	0.000	-3.770375	-2.14705
2012	-3.409416	.4117284	-8.28	0.000	-4.216389	-2.602444
2013	-3.224142	.4871161	-6.62	0.000	-4.178872	-2.269412

2014	-3.29783	.511386	-6.45	0.000	-4.300128	-2.295532
2015	-3.67802	.5336939	-6.89	0.000	-4.724041	-2.631999
2016	-3.383	.5292519	-6.39	0.000	-4.420315	-2.345686
_cons	6.962877	.7506154	9.28	0.000	5.491698	8.434056

bonds_tw_h(3)

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Dvul100						
--	.3009406	.1866883	1.61	0.107	-.0649617	.6668429
L1.	.1636609	.2194158	0.75	0.456	-.2663861	.593708
L2.	.1662895	.2224384	0.75	0.455	-.2696817	.6022607
L3.	.2964742	.197766	1.50	0.134	-.09114	.6840884
bonds_tw						
L1.	.0612393	.0923097	0.66	0.507	-.1196844	.2421629
Dvul100						
F1.	.5631062	.2123735	2.65	0.008	.1468619	.9793506
F2.	.495886	.2114929	2.34	0.019	.0813675	.9104046
F3.	.3134813	.2376089	1.32	0.187	-.1522236	.7791862
F4.	.1941774	.227893	0.85	0.394	-.2524846	.6408395
F5.	.2011131	.2208043	0.91	0.362	-.2316553	.6338815
period						
1999	0	(empty)				
2000	-1.224593	.1699669	-7.20	0.000	-1.557723	-.8914643
2001	-1.560393	.239719	-6.51	0.000	-2.030234	-1.090552
2002	-2.089875	.2872027	-7.28	0.000	-2.652782	-1.526968
2003	-1.988167	.3968746	-5.01	0.000	-2.766027	-1.210307
2004	-1.708828	.4648869	-3.68	0.000	-2.61999	-.7976664
2005	-1.281748	.4541847	-2.82	0.005	-2.171933	-.3915619
2006	-.8974089	.5073712	-1.77	0.077	-1.891838	.0970204
2007	-2.14074	.5196469	-4.12	0.000	-3.15923	-1.122251
2008	-2.049619	.5509794	-3.72	0.000	-3.129518	-.969719
2009	-2.518106	.5409517	-4.65	0.000	-3.578352	-1.45786
2010	-2.974811	.4562493	-6.52	0.000	-3.869043	-2.080579
2011	-3.280288	.5309172	-6.18	0.000	-4.320867	-2.239709
2012	-3.212573	.5848464	-5.49	0.000	-4.358851	-2.066295
2013	-3.329786	.6153563	-5.41	0.000	-4.535862	-2.123709
2014	-3.76861	.6433079	-5.86	0.000	-5.02947	-2.507749
2015	-3.405026	.6828802	-4.99	0.000	-4.743446	-2.066605
2016	-3.69843	.6200903	-5.96	0.000	-4.913785	-2.483075
_cons	7.633982	.9051066	8.43	0.000	5.860005	9.407958

bonds_tw_h(4)

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Dvul100						
--	.186815	.218593	0.85	0.393	-.2416194	.6152494
L1.	.1434561	.2398731	0.60	0.550	-.3266865	.6135986
L2.	.3673087	.2521478	1.46	0.145	-.126892	.8615093
L3.	.1974527	.1986243	0.99	0.320	-.1918438	.5867492
bonds_tw						
L1.	-.0356566	.0655004	-0.54	0.586	-.164035	.0927218
Dvul100						
F1.	.3333295	.1976623	1.69	0.092	-.0540815	.7207405
F2.	.4884703	.2217948	2.20	0.028	.0537605	.9231801
F3.	.5277535	.2242045	2.35	0.019	.0883208	.9671861
F4.	.2724235	.2384772	1.14	0.253	-.1949833	.7398302
F5.	.2338852	.2146586	1.09	0.276	-.1868379	.6546083
period						
1999	0	(empty)				
2000	-.3983593	.165722	-2.40	0.016	-.7231684	-.0735501
2001	-.9714546	.2215341	-4.39	0.000	-1.405654	-.5372557
2002	-.9858446	.3242278	-3.04	0.002	-1.621319	-.3503697

2003	-.8106111	.3538498	-2.29	0.022	-1.504144	-.1170783
2004	-.4021419	.3697967	-1.09	0.277	-1.12693	.3226463
2005	-.1076066	.3933605	-0.27	0.784	-.878579	.6633658
2006	-1.286673	.4544462	-2.83	0.005	-2.177371	-.3959747
2007	-1.202542	.5107047	-2.35	0.019	-2.203505	-.2015795
2008	-1.623624	.5325947	-3.05	0.002	-2.66749	-.5797574
2009	-2.138841	.4527383	-4.72	0.000	-3.026192	-1.25149
2010	-2.432151	.4521877	-5.38	0.000	-3.318423	-1.54588
2011	-2.366276	.5694346	-4.16	0.000	-3.482347	-1.250204
2012	-2.592687	.5654439	-4.59	0.000	-3.700936	-1.484437
2013	-2.993175	.5834026	-5.13	0.000	-4.136623	-1.849727
2014	-2.673384	.6309005	-4.24	0.000	-3.909926	-1.436842
2015	-2.977258	.6145771	-4.84	0.000	-4.181807	-1.772708
_cons	7.344787	.6940008	10.58	0.000	5.984571	8.705004

bonds_tw_h(5)

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Dvul100						
--	.1275825	.2203937	0.58	0.563	-.3043812	.5595463
L1.	.3290782	.2649804	1.24	0.214	-.1902738	.8484301
L2.	.2383116	.2622968	0.91	0.364	-.2757806	.7524038
L3.	.1221966	.2081284	0.59	0.557	-.2857275	.5301208
bonds_tw						
L1.	-.0633481	.0499357	-1.27	0.205	-.1612202	.034524
Dvul100						
F1.	.2189784	.1999817	1.09	0.274	-.1729784	.6109353
F2.	.1908444	.2031019	0.94	0.347	-.207228	.5889168
F3.	.4889544	.2318717	2.11	0.035	.0344942	.9434147
F4.	.4941343	.2304443	2.14	0.032	.0424717	.9457968
F5.	.3287786	.2234148	1.47	0.141	-.1091064	.7666635
period						
1999	0	(empty)				
2000	-.5936845	.1832844	-3.24	0.001	-.9529153	-.2344537
2001	-.6541751	.3078192	-2.13	0.034	-1.25749	-.0508606
2002	-.5442346	.3206675	-1.70	0.090	-1.172731	.0842621
2003	-.2230289	.3142102	-0.71	0.478	-.8388695	.3928117
2004	.0487771	.3679026	0.13	0.895	-.6722988	.769853
2005	-1.118959	.3877013	-2.89	0.004	-1.87884	-.3590785
2006	-.9759225	.477669	-2.04	0.041	-1.912137	-.0397085
2007	-1.50712	.513146	-2.94	0.003	-2.512868	-.5013726
2008	-1.911978	.4598615	-4.16	0.000	-2.81329	-1.010666
2009	-2.283479	.467323	-4.89	0.000	-3.199415	-1.367543
2010	-2.099389	.5447071	-3.85	0.000	-3.166995	-1.031782
2011	-2.360392	.5832122	-4.05	0.000	-3.503467	-1.217317
2012	-2.757189	.5826197	-4.73	0.000	-3.899103	-1.615276
2013	-2.401685	.6438392	-3.73	0.000	-3.663586	-1.139783
2014	-2.752076	.6419219	-4.29	0.000	-4.01022	-1.493932
_cons	7.243032	.6302653	11.49	0.000	6.007735	8.478329

Statistics by step

	N	R2	psR2	F	Chi2	Prob
0	1513	0.657	.	64.49	.	0.000
1	1505	0.418	.	20.32	.	0.000
2	1486	0.313	.	20.63	.	0.000
3	1464	0.253	.	17.98	.	0.000
4	1353	0.215	.	13.56	.	0.000
5	1244	0.207	.	9.21	.	0.000

Impulse Response Function

	IRF	Std.Err.	IRF LO~90	IRF UP 90	IRF LO~95	IRF UP 95
0	0.19692	0.09517	0.03908	0.35477	0.00836	0.38549
1	0.51346	0.14375	0.27502	0.75190	0.22861	0.79832
2	0.56476	0.18133	0.26398	0.86553	0.20543	0.92408
3	0.30094	0.18669	-0.00877	0.61065	-0.06907	0.67095
4	0.18682	0.21859	-0.17585	0.54948	-0.24647	0.62010
5	0.12758	0.22039	-0.23813	0.49329	-0.30937	0.56453

```
.
. graph export bond_vul_jan25_TZ.png, as(png) width(4000) replace
file bond_vul_jan25_TZ.png saved as PNG format
```

```
.
. *** Figure 7. Impact of change in vulnerability on bond yields (Climate vulnerability)
```

```
. egen vul100_p25 = pctlile(vul100), p(25)
```

```
. sum vul100, detail
```

vul100				
Percentiles	Smallest			
1%	26.58107	24.41626		
5%	29.96005	24.41857		
10%	32.10096	24.52629	Obs	4,968
25%	37.22011	24.5833	Sum of wgt.	4,968
50%	43.00644		Mean	44.20377
		Largest	Std. dev.	9.550001
75%	51.74853	69.40924		
90%	57.48193	69.5042	Variance	91.20253
95%	60.52048	69.56675	Skewness	.2783735
99%	65.23988	69.61713	Kurtosis	2.256964

```
.
. cap drop D1
```

```
. gen D1 = 0
```

```
. replace D1 = 1 if l.vul100>l.vul100_p25
(5,126 real changes made)
```

```
.
. locproj bonds_tw D1#c.Dvul100, lcs(1.D1#c.Dvul100) ///
> h(5) y1(1) sl(1) ///
> c(f(1/5).Dvul100 i.period) ///
> fe cluster(imfcode) conf(95) ///
> title("High Vulnerability") ///
> save irfname(belowbv) zero
```

Impulse Response Function

	IRF	Std.Err.	IRF LOW	IRF UP
0	0.22858	0.09892	0.03259	0.42456
1	0.59017	0.17624	0.24097	0.93937
2	0.69625	0.22064	0.25903	1.13346
3	0.39075	0.19615	0.00199	0.77950
4	0.21481	0.21584	-0.21303	0.64265
5	0.08954	0.21453	-0.33573	0.51481

```
. graph rename Graph belowbv, replace
```

```
.
. locproj bonds_tw D1#c.Dvul100, lcs(0.D1#c.Dvul100) ///
> h(5) yl(1) sl(1) ///
> c(f(1/5).Dvul100 i.period) ///
> fe cluster(imfcode) conf(95) ///
> title(`"Low Vulnerability"') ///
> save irfname(highbv) zero
```

Impulse Response Function

	IRF	Std.Err.	IRF LOW	IRF UP
0	-0.11172	0.22154	-0.55065	0.32720
1	0.20794	0.16187	-0.11278	0.52867
2	0.24437	0.25884	-0.26853	0.75728
3	0.06640	0.39399	-0.71449	0.84728
4	0.10128	0.36899	-0.63013	0.83268
5	0.24256	0.41988	-0.58980	1.07492

```
. graph rename Graph highbv, replace
```

```
.
. graph close belowbv highbv
.
. graph combine belowbv highbv, row(1) ///
> name(vul_bonds, replace) ycommon ///
> title("State-Dependent LP for Change in Vulnerability on Bond Yields") ///
> note("Note: High/Low Vulnerability is defined as above/below Q1 for VUL." "State dependence is measured with a
> dummy for High/Low Vulnerability score." "The shock is on D.vul100. Time FE included.", size(vsmall))
```

```
. graph export bondvulSLPQ1_apr25.png, as(png) width(4000) replace
file bondvulSLPQ1_apr25.png saved as PNG format
```

```
. *** Figure 8. Panel LP - impact of change in vulnerability on sovereign ratings (Baseline)
```

```
. locproj sovrates, shock(Dvul100) ///
> z h(5) yl(1) sl(3) ///
> c(f(1/5).Dvul100 i.period) ///
> fe cluster(imfcode) conf(90 95) ///
> ttitle("Horizon") ///
> title(`"Impact of Change in Vulnerability on Ratings"') ///
> save irfname(sovrates_TZ) noisily stats grname(TZ2)
sovrates_h(0)
```

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Dvul100						
--.	-.0796016	.0551841	-1.44	0.149	-.1877604	.0285573
L1.	-.1281329	.0455794	-2.81	0.005	-.2174669	-.0387989
L2.	-.108154	.0580508	-1.86	0.062	-.2219315	.0056235
L3.	-.0310634	.0593149	-0.52	0.600	-.1473183	.0851916
sovrates						
L1.	.9125379	.0170397	53.55	0.000	.8791408	.9459351
Dvul100						
F1.	-.0780762	.0514088	-1.52	0.129	-.1788355	.0226832
F2.	-.0525287	.0598011	-0.88	0.380	-.1697367	.0646793
F3.	.0491424	.0492585	1.00	0.318	-.0474025	.1456874
F4.	.0173326	.0519421	0.33	0.739	-.084472	.1191373
F5.	.1010519	.0514432	1.96	0.049	.0002251	.2018787
period						
1999	0 (empty)					
2000	.2660349	.0852036	3.12	0.002	.0990389	.4330309
2001	.2301702	.103228	2.23	0.026	.027847	.4324935
2002	.2317125	.117555	1.97	0.049	.0013089	.4621161
2003	.3531973	.1218875	2.90	0.004	.1143021	.5920924

2004	.3434693	.107675	3.19	0.001	.1324302	.5545083
2005	.3362188	.1178328	2.85	0.004	.1052708	.5671668
2006	.4156011	.1127367	3.69	0.000	.1946413	.6365609
2007	.4305009	.1031591	4.17	0.000	.2283128	.632689
2008	.3059257	.0988419	3.10	0.002	.1121992	.4996523
2009	.0392565	.1066775	0.37	0.713	-.1698276	.2483406
2010	.2643274	.120302	2.20	0.028	.0285399	.5001149
2011	.1280557	.1499375	0.85	0.393	-.1658163	.4219278
2012	-.0024363	.1414263	-0.02	0.986	-.2796267	.2747542
2013	.0536713	.1186244	0.45	0.651	-.1788284	.2861709
2014	.1343808	.1012846	1.33	0.185	-.0641333	.332895
2015	.138075	.0978372	1.41	0.158	-.0536823	.3298324
2016	.0840941	.1000641	0.84	0.401	-.1120279	.280216
_cons	.8389055	.2005798	4.18	0.000	.4457763	1.232035

sovrates_h(1)

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Dvul100						
--	-.1963298	.0796315	-2.47	0.014	-.3524047	-.0402549
L1.	-.2145928	.0858052	-2.50	0.012	-.3827678	-.0464177
L2.	-.1261296	.0890137	-1.42	0.156	-.3005933	.0483341
L3.	-.0452033	.0969069	-0.47	0.641	-.2351373	.1447306
sovrates						
L1.	.7547213	.0322668	23.39	0.000	.6914795	.8179631
Dvul100						
F1.	-.1399855	.0942727	-1.48	0.138	-.3247566	.0447857
F2.	-.1292501	.0906495	-1.43	0.154	-.3069199	.0484197
F3.	-.0061934	.080558	-0.08	0.939	-.1640841	.1516974
F4.	.057778	.084125	0.69	0.492	-.107104	.2226599
F5.	.1113621	.082352	1.35	0.176	-.050045	.2727691
period						
1999	0 (empty)					
2000	.2004063	.0949834	2.11	0.035	.0142423	.3865702
2001	.181493	.1494199	1.21	0.224	-.1113646	.4743506
2002	.2974364	.1617088	1.84	0.066	-.019507	.6143798
2003	.4289676	.1567047	2.74	0.006	.1218319	.7361032
2004	.4048467	.1520477	2.66	0.008	.1068387	.7028546
2005	.4984867	.1439717	3.46	0.001	.2163073	.7806662
2006	.6028907	.1425736	4.23	0.000	.3234515	.8823299
2007	.502834	.1392161	3.61	0.000	.2299756	.7756925
2008	.133363	.1523878	0.88	0.381	-.1653116	.4320376
2009	.1183806	.1618859	0.73	0.465	-.1989099	.435671
2010	.174096	.2178147	0.80	0.424	-.2528131	.601005
2011	-.0695176	.2411073	-0.29	0.773	-.5420792	.4030439
2012	-.1421833	.2126607	-0.67	0.504	-.5589907	.274624
2013	-.026074	.1723841	-0.15	0.880	-.3639405	.3117925
2014	.0580202	.1583431	0.37	0.714	-.2523266	.3683671
2015	-.0051133	.1504421	-0.03	0.973	-.2999744	.2897478
2016	-.0727695	.1634356	-0.45	0.656	-.3930974	.2475585
_cons	2.78924	.3759766	7.42	0.000	2.05234	3.526141

sovrates_h(2)

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Dvul100						
--	-.2698537	.1099828	-2.45	0.014	-.4854159	-.0542914
L1.	-.2171074	.1047259	-2.07	0.038	-.4223664	-.0118485
L2.	-.1440576	.1189787	-1.21	0.226	-.3772516	.0891364
L3.	-.1528685	.1034806	-1.48	0.140	-.3556868	.0499498
sovrates						
L1.	.590264	.0447664	13.19	0.000	.5025234	.6780046
Dvul100						
F1.	-.243335	.1154541	-2.11	0.035	-.469621	-.0170491

Dvul100						
--.	-.2777262	.1481449	-1.87	0.061	-.5680848	.0126325
L1.	-.2819684	.1427746	-1.97	0.048	-.5618016	-.0021353
L2.	-.2815986	.1462513	-1.93	0.054	-.568246	.0050488
L3.	-.2183613	.1312511	-1.66	0.096	-.4756088	.0388862
sovrata						
L1.	.3130486	.0462235	6.77	0.000	.2224523	.4036449
Dvul100						
F1.	-.3059396	.1553588	-1.97	0.049	-.6104372	-.001442
F2.	-.3276756	.1576843	-2.08	0.038	-.6367311	-.0186201
F3.	-.2877191	.1480352	-1.94	0.052	-.5778627	.0024245
F4.	-.1892086	.1465238	-1.29	0.197	-.4763899	.0979727
F5.	.0012383	.1207831	0.01	0.992	-.2354923	.2379689
period						
1999	0	(empty)				
2000	.2365768	.0774195	3.06	0.002	.0848374	.3883163
2001	.3786361	.1101539	3.44	0.001	.1627385	.5945338
2002	.5122827	.1332966	3.84	0.000	.2510262	.7735391
2003	.6647576	.1532358	4.34	0.000	.3644209	.9650943
2004	.7014743	.165962	4.23	0.000	.3761949	1.026754
2005	.4709481	.1906099	2.47	0.013	.0973596	.8445366
2006	.3900991	.2203927	1.77	0.077	-.0418626	.8220608
2007	.1725309	.2848901	0.61	0.545	-.3858434	.7309052
2008	-.1187926	.3314759	-0.36	0.720	-.7684735	.5308882
2009	-.2655195	.3500726	-0.76	0.448	-.9516492	.4206101
2010	-.2484428	.3418667	-0.73	0.467	-.9184893	.4216036
2011	-.2654856	.334648	-0.79	0.428	-.9213837	.3904125
2012	-.3167659	.3078668	-1.03	0.304	-.9201736	.2866419
2013	-.3358597	.2977338	-1.13	0.259	-.9194072	.2476878
2014	-.2478293	.2871641	-0.86	0.388	-.8106605	.3150019
2015	-.1939139	.2766898	-0.70	0.483	-.736216	.3483882
2016	-.2885788	.26553	-1.09	0.277	-.8090081	.2318504
_cons	8.292386	.5378155	15.42	0.000	7.238287	9.346485

sovrata_h(5)

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Dvul100						
--.	-.3306346	.1561641	-2.12	0.034	-.6367106	-.0245587
L1.	-.2871887	.1566348	-1.83	0.067	-.5941874	.01981
L2.	-.30061	.1507896	-1.99	0.046	-.5961522	-.0050678
L3.	-.2350706	.1200273	-1.96	0.050	-.4703198	.0001787
sovrata						
L1.	.1983633	.0437948	4.53	0.000	.1125272	.2841995
Dvul100						
F1.	-.2882886	.1651498	-1.75	0.081	-.6119763	.0353991
F2.	-.3163158	.1655371	-1.91	0.056	-.6407625	.008131
F3.	-.3160174	.1623393	-1.95	0.052	-.6341967	.0021618
F4.	-.2882517	.1487608	-1.94	0.053	-.5798175	.0033141
F5.	-.0998058	.132151	-0.76	0.450	-.3588171	.1592055
period						
1999	0	(empty)				
2000	.2381015	.0616439	3.86	0.000	.1172817	.3589212
2001	.3581672	.0970287	3.69	0.000	.1679945	.54834
2002	.5181032	.1117939	4.63	0.000	.2989913	.7372152
2003	.5914291	.1258052	4.70	0.000	.3448555	.8380028
2004	.3700526	.1668676	2.22	0.027	.0429982	.6971071
2005	.3170898	.1975477	1.61	0.108	-.0700965	.7042761
2006	.1418304	.2694428	0.53	0.599	-.3862678	.6699285
2007	-.1496645	.3188362	-0.47	0.639	-.7745719	.4752428
2008	-.3373555	.3418142	-0.99	0.324	-1.007299	.3325881
2009	-.383659	.3402097	-1.13	0.259	-1.050458	.2831397
2010	-.3623793	.340332	-1.06	0.287	-1.029418	.3046592
2011	-.4401145	.3276133	-1.34	0.179	-1.082225	.2019957
2012	-.4989602	.3227965	-1.55	0.122	-1.13163	.1337093

2013		-.420057	.3077763	-1.36	0.172	-1.023236	.1832247
2014		-.3549639	.2941854	-1.21	0.228	-.9315566	.2216289
2015		-.4453495	.2827611	-1.58	0.115	-.999551	.108852
2016		-.5736021	.2875051	-2.00	0.046	-1.137102	-.0101024
_cons		9.838166	.5448807	18.06	0.000	8.770219	10.90611

Statistics by step

	N	R2	psR2	F	Chi2	Prob
0	2066	0.814	.	788.14	.	0.000
1	2057	0.563	.	134.51	.	0.000
2	2048	0.364	.	28.78	.	0.000
3	2040	0.232	.	19.06	.	0.000
4	2033	0.145	.	13.80	.	0.000
5	2026	0.102	.	8.83	.	0.000

Impulse Response Function

	IRF	Std.Err.	IRF LO~90	IRF UP 90	IRF LO~95	IRF UP 95
0	-0.07960	0.05518	-0.17099	0.01179	-0.18873	0.02953
1	-0.19633	0.07963	-0.32821	-0.06445	-0.35381	-0.03885
2	-0.26985	0.10998	-0.45201	-0.08770	-0.48737	-0.05234
3	-0.28168	0.12562	-0.48972	-0.07363	-0.53011	-0.03325
4	-0.27773	0.14814	-0.52309	-0.03237	-0.57071	0.01526
5	-0.33063	0.15616	-0.58932	-0.07195	-0.63954	-0.02173

```
.
. graph export sov_vul_jan25_TZ.png, as(png) width(4000) replace
file sov_vul_jan25_TZ.png saved as PNG format
```

```
.
. *** Figure 9. Impact of change in vulnerability on sovereign ratings (Climate vulnerability)
.
. locproj sovrates D1#c.Dvul100, lcs(1.D1#c.Dvul100) ///
> h(5) yl(1) sl(1) ///
> c(f(1/5).Dvul100 i.period) ///
> fe cluster(imfcode) conf(95) ///
> title(`"High Vulnerability"') ///
> save irfname(belowbv) zero
```

Impulse Response Function

	IRF	Std.Err.	IRF LOW	IRF UP
0	-0.02108	0.05244	-0.12479	0.08262
1	-0.13650	0.07983	-0.29437	0.02136
2	-0.23546	0.10866	-0.45035	-0.02056
3	-0.22351	0.11708	-0.45506	0.00803
4	-0.19480	0.13900	-0.46969	0.08010
5	-0.25663	0.14483	-0.54312	0.02986

```
. graph rename Graph belowsv, replace
.
. locproj sovrates D1#c.Dvul100, lcs(0.D1#c.Dvul100) ///
> h(5) yl(1) sl(1) ///
> c(f(1/5).Dvul100 i.period) ///
> fe cluster(imfcode) conf(95) ///
> title(`"Low Vulnerability"') ///
> save irfname(highbv) zero
```

Impulse Response Function

	IRF	Std.Err.	IRF LOW	IRF UP
0	-0.20957	0.13457	-0.47569	0.05655
1	-0.16148	0.20080	-0.55858	0.23561
2	-0.12166	0.25962	-0.63512	0.39180
3	-0.13819	0.28489	-0.70161	0.42522
4	-0.21076	0.27940	-0.76334	0.34181
5	-0.25711	0.31029	-0.87090	0.35667

. graph rename Graph highsv, replace

. graph close belowbv highsv

. graph combine belowsv highsv, row(1) ///

> name(vul_bonds, replace) ycommon ///

> title("State-Dependent LP for Change in Vulnerability on Sovereign Rating") ///

> note("Note: High/Low Vulnerability is defined as above/below Q1 for VUL." "State dependence is measured with a dummy for High/Low Vulnerability score." "The shock is on D.vul100. Time FE included.", size(vsmall))

. graph export sovratevulSLPQ1_apr25.png, as(png) width(4000) replace
file sovratevulSLPQ1_apr25.png saved as PNG format

. *** Figure 10. Panel LP - impact of change in vulnerability on bond yields (Financial Institutions)

. egen FI_p75 = pctlile(FI), p(75)

. sum FI, detail

Financial Institution index				
Percentiles		Smallest		
1%	0	0		
5%	.095033	0		
10%	.1406377	0	Obs	5,824
25%	.1930571	0	Sum of wgt.	5,824
50%	.3141608		Mean	.3683459
		Largest	Std. dev.	.2219489
75%	.4979673	.9891428		
90%	.7187247	.9893202	Variance	.0492613
95%	.81482	.994068	Skewness	.7537904
99%	.9402609	1	Kurtosis	2.8112

. cap drop D2

. gen D2 = 0

. replace D2 = 1 if l.FI>l.FI_p75
(2,000 real changes made)

. locproj bonds_tw D2#c.Dvul100, lcs(1.D2#c.Dvul100) ///

> h(5) y1(1) sl(1) ///

> c(f(1/5).Dvul100 i.period) ///

> fe cluster(imfcode) conf(95) ///

> title("High FI") ///

> save irfname(belowbf) zero

Impulse Response Function

	IRF	Std.Err.	IRF LOW	IRF UP
0	-0.11092	0.22237	-0.55147	0.32963
1	-0.10411	0.27218	-0.64340	0.43518
2	0.01156	0.25087	-0.48554	0.50867
3	0.04706	0.23720	-0.42307	0.51718
4	0.33347	0.29866	-0.25854	0.92547
5	0.49868	0.31617	-0.12808	1.12545

```
. graph rename Graph belowbf, replace
```

```
.
. locproj bonds_tw D2#c.Dvul100, lcs(0.D2#c.Dvul100) ///
> h(5) yl(1) sl(1) ///
> c(f(1/5).Dvul100 i.period) ///
> fe cluster(imfcode) conf(95) ///
> title(`"Low FI"') ///
> save irfname(highbf) zero
```

Impulse Response Function

	IRF	Std.Err.	IRF LOW	IRF UP
0	0.26146	0.10664	0.05019	0.47273
1	0.70972	0.18172	0.34966	1.06978
2	0.78842	0.24523	0.30247	1.27436
3	0.40768	0.22354	-0.03537	0.85073
4	0.15179	0.22858	-0.30129	0.60488
5	0.00271	0.23104	-0.45530	0.46073

```
. graph rename Graph highbf, replace
```

```
. graph close belowbf highbf
```

```
.
. graph combine belowbf highbf, row(1) ///
> name(vul_fi, replace) ycommon ///
> title(`"State-Dependent LP for Change in Vulnerability on Bond Yields"') ///
> note("Note: High/Low Financial Institutions is defined as above/below Q3 for FI." "State dependence is measured with a dummy for High/Low Financial Institutions." "The shock is on D.vul100. Time FE included.", size(vsmall))
```

```
. graph export bondFISLPQ3_apr25.png, as(png) width(4000) replace
file bondFISLPQ3_apr25.png saved as PNG format
```

```
. *** Figure 11. Panel LP - impact of change in vulnerability on sovereign ratings (Financial Institutions)
```

```
. locproj sovrates D2#c.Dvul100, lcs(1.D2#c.Dvul100) ///
> h(5) yl(1) sl(1) ///
> c(f(1/5).Dvul100 i.period) ///
> fe cluster(imfcode) conf(95) ///
> title(`"High FI"') ///
> save irfname(belowbf) zero
```

Impulse Response Function

	IRF	Std.Err.	IRF LOW	IRF UP
0	0.04707	0.13351	-0.21696	0.31110
1	0.00308	0.17561	-0.34420	0.35037
2	0.19560	0.19418	-0.18843	0.57964
3	0.27313	0.19942	-0.12127	0.66753
4	0.22115	0.19673	-0.16791	0.61021
5	0.04300	0.21333	-0.37899	0.46499

```
. graph rename Graph belowbf, replace

.
. locproj sovrates D2#c.Dvul100, lcs(0.D2#c.Dvul100) ///
> h(5) y1(1) sl(1) ///
> c(f(1/5).Dvul100 i.period) ///
> fe cluster(imfcode) conf(95) ///
> title(`"Low FI"') ///
> save irfname(highbf) zero
```

Impulse Response Function

	IRF	Std.Err.	IRF LOW	IRF UP
0	-0.08259	0.05187	-0.18518	0.01999
1	-0.18717	0.08218	-0.34969	-0.02466
2	-0.34577	0.11138	-0.56604	-0.12551
3	-0.36254	0.11976	-0.59939	-0.12570
4	-0.32982	0.14645	-0.61945	-0.04018
5	-0.35369	0.15691	-0.66406	-0.04331

```
. graph rename Graph highbf, replace
```

```
.
. graph close belowbf highbf

.
. graph combine belowbf highbf, row(1) ///
> name(vul_fi, replace) ycommon ///
> title(`"State-Dependent LP for Change in Vulnerability on Sovereign Ratings"') ///
> note("Note: High/Low Financial Institutions is defined as above/below Q3 for FI." "State dependence is measured
> with a dummy for High/Low Financial Institutions." "The shock is on D.vul100. Time FE included.", size(vsmall))

.
```

```
. graph export sovFISLPQ3_apr25.png, as(png) width(4000) replace
file sovFISLPQ3_apr25.png saved as PNG format
```

```
.
. *** Figure 12. Panel LP - impact of change in vulnerability on bond yields (Religious Tensions)
.
. egen reltensions_p50 = pctlile(reltensions), p(50)
```

```
. sum reltensions, detail
```

ICRG index - Religious Tensions				
Percentiles		Smallest		
1%	1	0		
5%	2	0		
10%	2.5	0	Obs	4,401
25%	4	0	Sum of wgt.	4,401
50%	5	Largest	Mean	4.588569
			Std. dev.	1.29756
75%	5.625	6		
90%	6	6	Variance	1.683663
95%	6	6	Skewness	-.9444002
99%	6	6	Kurtosis	3.2888

```
.
```

```
. cap drop D3
. gen D3 = 0
. replace D3 = 1 if l.reltensions<l.reltensions_p50
(1,672 real changes made)
```

```
.
. locproj bonds_tw D3#c.Dvul100, lcs(1.D3#c.Dvul100) ///
> h(5) yl(1) sl(1) ///
> c(f(1/5).Dvul100 i.period) ///
> fe cluster(imfcode) conf(95) ///
> title(`"High Religious Tensions"') ///
> save irfname(belowbr) zero
```

Impulse Response Function

	IRF	Std.Err.	IRF LOW	IRF UP
0	0.53523	0.19283	0.15320	0.91726
1	0.75070	0.26133	0.23291	1.26849
2	0.95373	0.28627	0.38646	1.52100
3	0.85248	0.28471	0.28820	1.41676
4	0.66984	0.28393	0.10705	1.23264
5	0.44547	0.29025	-0.12990	1.02085

```
. graph rename Graph belowbr, replace
```

```
.
. locproj bonds_tw D3#c.Dvul100, lcs(0.D3#c.Dvul100) ///
> h(5) yl(1) sl(1) ///
> c(f(1/5).Dvul100 i.period) ///
> fe cluster(imfcode) conf(95) ///
> title(`"Low Religious Tensions"') ///
> save irfname(highbr) zero
```

Impulse Response Function

	IRF	Std.Err.	IRF LOW	IRF UP
0	0.08235	0.09356	-0.10301	0.26770
1	0.45803	0.16324	0.13459	0.78146
2	0.52193	0.20564	0.11443	0.92942
3	0.20226	0.18184	-0.15815	0.56266
4	0.08248	0.21728	-0.34821	0.51317
5	0.03152	0.23454	-0.43344	0.49648

```
. graph rename Graph highbr, replace
```

```
.
. graph close belowbr highbr
```

```
.
. graph combine belowbr highbr, row(1) ///
> name(vul_rel, replace) ycommon ///
> title(`"State-Dependent LP for Change in Vulnerability on Bond Yields"') ///
> note("Note: High/Low Religious Tensions is defined as below/above Q2 for reltensions." "State dependence is mea
> sured with a dummy for High/Low Religious Tensions." "The shock is on D.vul100. Time FE included.", size(vsmall)
> )
```

. graph export bondreлтensSLPQ2_apr25.png, as(png) width(4000) replace
file bondreлтensSLPQ2_apr25.png saved as PNG format

```
.
. // Wald test for religious tensions regimes
.
. forvalues v = 0(1)5{
.   2.
.   qui locproj bonds_tw D3#c.Dvul100, lcs(1.D3#c.Dvul100) ///
.   > h(`v') yl(1) sl(1) ///
.   > c(f(1/5).Dvul100 i.period) ///
.   > fe cluster(imfcode) conf(95) ///
.   > title(`"High Religious Tensions"') ///
.   > save irfname(belowbv) zero ///
.   > stats nograph
.   3.
.   test (_b[0.D3#c.Dvul100] = ///
.   > _b[1.D3#c.Dvul100])
.   4.
. }
```

(1) 0b.D3#c.Dvul100 - 1.D3#c.Dvul100 = 0

chi2(1) = 4.48
Prob > chi2 = 0.0342

(1) 0b.D3#c.Dvul100 - 1.D3#c.Dvul100 = 0

chi2(1) = 0.88
Prob > chi2 = 0.3478

(1) 0b.D3#c.Dvul100 - 1.D3#c.Dvul100 = 0

chi2(1) = 1.55
Prob > chi2 = 0.2129

(1) 0b.D3#c.Dvul100 - 1.D3#c.Dvul100 = 0

chi2(1) = 4.19
Prob > chi2 = 0.0406

(1) 0b.D3#c.Dvul100 - 1.D3#c.Dvul100 = 0

chi2(1) = 3.06
Prob > chi2 = 0.0802

(1) 0b.D3#c.Dvul100 - 1.D3#c.Dvul100 = 0

chi2(1) = 1.14
Prob > chi2 = 0.2847

```
.
. *** Figure 13. Panel LP - impact of change in vulnerability on sovereign ratings (Religious Tensions)
.
. locproj sovrate D3#c.Dvul100, lcs(1.D3#c.Dvul100) ///
. > h(5) yl(1) sl(1) ///
. > c(f(1/5).Dvul100 i.period) ///
. > fe cluster(imfcode) conf(95) ///
. > title(`"High Religious Tensions"') ///
. > save irfname(belowbr) zero
```

Impulse Response Function

	IRF	Std.Err.	IRF LOW	IRF UP
0	-0.14759	0.09735	-0.34011	0.04494
1	-0.18072	0.14797	-0.47335	0.11191
2	-0.26442	0.17488	-0.61028	0.08144
3	-0.37189	0.19709	-0.76167	0.01789
4	-0.45861	0.21985	-0.89341	-0.02382
5	-0.44806	0.24763	-0.93790	0.04178

```
. graph rename Graph belowbr, replace

.
. locproj sovrates D3#c.Dvul100, lcs(0.D3#c.Dvul100) ///
> h(5) y1(1) sl(1) ///
> c(f(1/5).Dvul100 i.period) ///
> fe cluster(imfcode) conf(95) ///
> title(`"Low Religious Tensions")'') ///
> save irfname(highbr) zero
```

Impulse Response Function

	IRF	Std.Err.	IRF LOW	IRF UP
0	-0.01970	0.06030	-0.13896	0.09955
1	-0.12509	0.08997	-0.30300	0.05283
2	-0.19698	0.11840	-0.43114	0.03717
3	-0.15045	0.12105	-0.38985	0.08894
4	-0.10791	0.13807	-0.38097	0.16515
5	-0.19117	0.14188	-0.47182	0.08948

```
. graph rename Graph highbr, replace
```

```
.
. graph close belowbr highbr

.
. graph combine belowbr highbr, row(1) ///
> name(vul_rel, replace) ycommon ///
> title(`"State-Dependent LP for Change in Vulnerability on Sovereign Ratings")'') ///
> note("Note: High/Low Religious Tensions is defined as below/above Q2 for reltensions." "State dependence is mea
> sured with a dummy for High/Low Religious Tensions." "The shock is on D.vul100. Time FE included.", size(vsmall)
> )
```

```
.
. graph export sovratereltensSLPQ3_apr25.png, as(png) width(4000) replace
file sovratereltensSLPQ3_apr25.png saved as PNG format
```

```
. *** Figure 14. Panel LP - impact of change in VUL_N on bond yields
```

```
. // Data preparation
```

```
. merge 1:1 imfcode period using VUL_N.dta, ///
```

```
> keep(1 3) nogen force
```

```
(note: variable cn was long in the using data, but will be str32 now)
```

Result	Number of obs
Not matched	1,359
from master	1,359
from using	0
Matched	5,208

```
. // PCA (first)
```

```
. pca FOOD_03 WATE_03 ECOS_ 04 ECOS_ 05 ECOS_ 06 INFR_ 03 ///
```

```
> INFR_ 04
```

Principal components/correlation	Number of obs	=	3,948
	Number of comp.	=	7
	Trace	=	7
Rotation: (unrotated = principal)	Rho	=	1.0000

Component	Eigenvalue	Difference	Proportion	Cumulative
Comp1	2.25196	.761245	0.3217	0.3217
Comp2	1.49072	.478781	0.2130	0.5347
Comp3	1.01194	.110332	0.1446	0.6792
Comp4	.901607	.334797	0.1288	0.8080
Comp5	.56681	.105731	0.0810	0.8890
Comp6	.461079	.145197	0.0659	0.9549
Comp7	.315882	.	0.0451	1.0000

Principal components (eigenvectors)

Variable	Comp1	Comp2	Comp3	Comp4	Comp5	Comp6	Comp7	Unexplained
FOOD_03	0.4133	0.1661	-0.0067	0.6614	-0.5012	-0.1071	-0.3185	0
WATE_03	0.0192	0.6989	-0.0719	-0.2516	-0.3424	0.5212	0.2320	0
ECOS_04	-0.2686	0.6146	-0.1952	-0.0475	0.1778	-0.6682	-0.1777	0
ECOS_05	0.5820	0.0583	-0.0691	0.0356	0.1524	-0.3156	0.7274	0
ECOS_06	0.5084	0.1535	-0.2679	-0.0666	0.6012	0.2964	-0.4386	0
INFR_03	-0.3989	0.0845	-0.2813	0.6798	0.3468	0.2837	0.3029	0
INFR_04	0.0178	0.2685	0.8950	0.1708	0.3083	0.0489	-0.0024	0

```
. pca FOOD_03 WATE_03 ECOS_04 ECOS_05 ECOS_06 INFR_03 ///
> INFR_04, mineigen(1)
```

Principal components/correlation

Number of obs = 3,948
 Number of comp. = 3
 Trace = 7
 Rho = 0.6792

Component	Eigenvalue	Difference	Proportion	Cumulative
Comp1	2.25196	.761245	0.3217	0.3217
Comp2	1.49072	.478781	0.2130	0.5347
Comp3	1.01194	.110332	0.1446	0.6792
Comp4	.901607	.334797	0.1288	0.8080
Comp5	.56681	.105731	0.0810	0.8890
Comp6	.461079	.145197	0.0659	0.9549
Comp7	.315882	.	0.0451	1.0000

Principal components (eigenvectors)

Variable	Comp1	Comp2	Comp3	Unexplained
FOOD_03	0.4133	0.1661	-0.0067	.5741
WATE_03	0.0192	0.6989	-0.0719	.2658
ECOS_04	-0.2686	0.6146	-0.1952	.2358
ECOS_05	0.5820	0.0583	-0.0691	.2273
ECOS_06	0.5084	0.1535	-0.2679	.3102
INFR_03	-0.3989	0.0845	-0.2813	.5509
INFR_04	0.0178	0.2685	0.8950	.08127


```
.
. cap drop pc*
. estat loadings
```

Principal component loadings (unrotated)
component normalization: sum of squares(column) = 1

	Comp1	Comp2	Comp3
FOOD_03	.4133	.1661	-.006672
WATE_03	.01919	.6989	-.07191
ECOS_04	-.2686	.6146	-.1952
ECOS_05	.582	.05829	-.06912
ECOS_06	.5084	.1535	-.2679
INFR_03	-.3989	.08454	-.2813
INFR_04	.01777	.2685	.895

```
. predict pc1 pc2 pc3, score
```

Scoring coefficients
sum of squares(column-loading) = 1

Variable	Comp1	Comp2	Comp3
FOOD_03	0.4133	0.1661	-0.0067
WATE_03	0.0192	0.6989	-0.0719
ECOS_04	-0.2686	0.6146	-0.1952
ECOS_05	0.5820	0.0583	-0.0691
ECOS_06	0.5084	0.1535	-0.2679
INFR_03	-0.3989	0.0845	-0.2813
INFR_04	0.0178	0.2685	0.8950

```
.
. // Local Projection (Second)
.
. locproj bonds_tw, shock(D.pc1) ///
> z h(4) yl(1) sl(5) ///
> c(F(1/5).D.pc1 l(1).CAB l(1).cpi_tw l(1).GDebt l(1).GDeficit ///
> l(1).banking l(1).currency l(1).debt i.period) ///
> title("Impact of change in the first component of less correlated vulnerability") ///
> ttitle("Horizon") ///
> fe conf(90 95) noisily stats gname(TZ3)
bonds_tw_h(0)
```

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
pc1						
D1.	.9235362	.3565286	2.59	0.010	.224753	1.622319
LD.	1.058773	.3482559	3.04	0.002	.3762036	1.741342
L2D.	.4323651	.3470682	1.25	0.213	-.2478761	1.112606
L3D.	.477657	.3339169	1.43	0.153	-.1768081	1.132122
L4D.	.4172245	.3244146	1.29	0.198	-.2186164	1.053065
L5D.	.6303384	.3152845	2.00	0.046	.0123921	1.248285
bonds_tw						
L1.	.6289632	.0220008	28.59	0.000	.5858424	.672084
pc1						
FD.	.1445687	.3587733	0.40	0.687	-.5586141	.8477514
F2D.	-.2874022	.3548026	-0.81	0.418	-.9828025	.4079982
F3D.	-.6021223	.3466526	-1.74	0.082	-1.281549	.0773042
F4D.	-.5316373	.3327383	-1.60	0.110	-1.183792	.1205179
F5D.	-.4653206	.3086494	-1.51	0.132	-1.070262	.1396211
CAB						
L1.	-.0340609	.0093334	-3.65	0.000	-.052354	-.0157679
cpi_tw						

L1.	.3600583	.1083245	3.32	0.001	.1477462	.5723704
GDebt						
L1.	.0070163	.0030107	2.33	0.020	.0011154	.0129172
GDeficit						
L1.	-.0077871	.0138649	-0.56	0.574	-.0349618	.0193877
banking						
L1.	.6354686	.2467747	2.58	0.010	.151799	1.119138
currency						
L1.	.678489	.3092958	2.19	0.028	.0722804	1.284698
debt						
L1.	.3327677	.4614018	0.72	0.471	-.5715632	1.237099
period						
2001	0	(empty)				
2002	-.2015352	.2370437	-0.85	0.395	-.6661324	.263062
2003	-.8302595	.235166	-3.53	0.000	-1.291176	-.3693427
2004	-.1534181	.2408651	-0.64	0.524	-.625505	.3186688
2005	-.5723056	.2376294	-2.41	0.016	-1.038051	-.1065605
2006	-.0120008	.2365058	-0.05	0.960	-.4755436	.4515421
2007	-.0757946	.2326374	-0.33	0.745	-.5317555	.3801662
2008	.0775848	.2333321	0.33	0.740	-.3797377	.5349074
2009	-.1658627	.2366017	-0.70	0.483	-.6295935	.2978682
2010	-1.318074	.2383148	-5.53	0.000	-1.785163	-.850986
2011	-.4072174	.2434958	-1.67	0.094	-.8844605	.0700256
2012	-1.097779	.2400376	-4.57	0.000	-1.568244	-.6273141
2013	-1.164853	.2417296	-4.82	0.000	-1.638634	-.6910719
2014	-1.289348	.2448515	-5.27	0.000	-1.769248	-.8094475
2015	-.9040679	.2536445	-3.56	0.000	-1.401202	-.4069339
2016	-.9985518	.2535367	-3.94	0.000	-1.495475	-.5016289
2017	-1.406268	.2541614	-5.53	0.000	-1.904415	-.9081203
_cons	1.885443	.2634713	7.16	0.000	1.369048	2.401837

bonds_tw_h(1)						

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	

pc1						
D1.	1.684601	.4483745	3.76	0.000	.8058026	2.563398
LD.	1.171226	.4359424	2.69	0.007	.3167946	2.025657
L2D.	1.079056	.4336725	2.49	0.013	.2290739	1.929039
L3D.	.9010572	.4182656	2.15	0.031	.0812717	1.720843
L4D.	1.124465	.4108333	2.74	0.006	.3192462	1.929683
L5D.	1.209364	.3967442	3.05	0.002	.4317594	1.986968
bonds_tw						
L1.	.2985774	.0276335	10.80	0.000	.2444166	.3527381
pc1						
FD.	.7158142	.4487859	1.60	0.111	-.16379	1.595418
F2D.	.1137499	.443439	0.26	0.798	-.7553747	.9828744
F3D.	-.2584451	.4332452	-0.60	0.551	-1.10759	.5906998
F4D.	-.8189854	.4174188	-1.96	0.050	-1.637111	-.0008595
F5D.	-.5477124	.3860409	-1.42	0.156	-1.304339	.2089139
CAB						
L1.	-.0400428	.0117655	-3.40	0.001	-.0631028	-.0169829
cpi_tw						
L1.	.2938812	.1464839	2.01	0.045	.006778	.5809845
GDebt						
L1.	.0034894	.0038018	0.92	0.359	-.0039619	.0109408
GDeficit						
L1.	-.0174538	.0175184	-1.00	0.319	-.0517892	.0168817
banking						

L1.	.9992834	.30708	3.25	0.001	.3974178	1.601149
currency						
L1.	.4136069	.3868478	1.07	0.285	-.3446008	1.171815
debt						
L1.	-1.072693	.5740635	-1.87	0.062	-2.197836	.0524512
period						
2001	0 (empty)					
2002	-.7774899	.2949153	-2.64	0.008	-1.355513	-.1994664
2003	-.605229	.2926224	-2.07	0.039	-1.178758	-.0316996
2004	-.7447057	.299785	-2.48	0.013	-1.332274	-.1571378
2005	-.4603169	.295876	-1.56	0.120	-1.040223	.1195893
2006	-.2227368	.2944651	-0.76	0.449	-.7998778	.3544041
2007	.0338046	.2901914	0.12	0.907	-.53496	.6025692
2008	.1219042	.2917861	0.42	0.676	-.449986	.6937944
2009	-1.645648	.2945518	-5.59	0.000	-2.222959	-1.068337
2010	-1.221524	.2968206	-4.12	0.000	-1.803281	-.6397661
2011	-1.432489	.3031222	-4.73	0.000	-2.026597	-.8383803
2012	-1.925362	.2989616	-6.44	0.000	-2.511316	-1.339408
2013	-2.158784	.3012726	-7.17	0.000	-2.749267	-1.5683
2014	-2.016973	.3050806	-6.61	0.000	-2.61492	-1.419026
2015	-1.680824	.3171419	-5.30	0.000	-2.302411	-1.059237
2016	-2.214571	.3158308	-7.01	0.000	-2.833588	-1.595554
2017	-1.896251	.3218038	-5.89	0.000	-2.526974	-1.265527
_cons	4.240454	.3291846	12.88	0.000	3.595264	4.885644

bonds_tw_h(2)						

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	

pc1						
D1.	1.705183	.4597973	3.71	0.000	.8039973	2.60637
LD.	1.1677	.4470667	2.61	0.009	.2914653	2.043935
L2D.	1.004738	.4459432	2.25	0.024	.1307053	1.878771
L3D.	1.481575	.4352649	3.40	0.001	.6284718	2.334679
L4D.	1.419196	.4222491	3.36	0.001	.5916033	2.246789
L5D.	.8969438	.409352	2.19	0.028	.0946287	1.699259
bonds_tw						
L1.	.1669631	.028458	5.87	0.000	.1111865	.2227398
pc1						
FD.	1.171673	.4598455	2.55	0.011	.2703923	2.072954
F2D.	.8115609	.45524	1.78	0.075	-.0806932	1.703815
F3D.	.3183761	.4436519	0.72	0.473	-.5511657	1.187918
F4D.	-.2643564	.4282965	-0.62	0.537	-1.103802	.5750893
F5D.	-.556498	.3962932	-1.40	0.160	-1.333218	.2202225
CAB						
L1.	-.0416749	.0121228	-3.44	0.001	-.0654351	-.0179146
cpi_tw						
L1.	.0838827	.1682164	0.50	0.618	-.2458154	.4135808
GDebt						
L1.	-.0062086	.003951	-1.57	0.116	-.0139525	.0015353
GDeficit						
L1.	-.0124282	.0180514	-0.69	0.491	-.0478084	.022952
banking						
L1.	1.078055	.3137546	3.44	0.001	.4631074	1.693003
currency						
L1.	-.0442487	.3986085	-0.11	0.912	-.8255069	.7370096
debt						
L1.	-.349242	.5863548	-0.60	0.551	-1.498476	.7999922
period						

2001	0	(empty)				
2002	.1619357	.3006819	0.54	0.590	-.4273899	.7512613
2003	-.1000552	.2984172	-0.34	0.737	-.6849422	.4848318
2004	.2251758	.3058977	0.74	0.462	-.3743726	.8247243
2005	.2733907	.3020168	0.91	0.365	-.3185514	.8653327
2006	.7613382	.3014944	2.53	0.012	.1704201	1.352256
2007	.8016015	.2969956	2.70	0.007	.2195007	1.383702
2008	-.3933231	.2976657	-1.32	0.186	-.9767371	.1900909
2009	-.7700453	.3009996	-2.56	0.011	-1.359994	-.1800969
2010	-.9696201	.3030302	-3.20	0.001	-1.563548	-.3756919
2011	-1.200146	.3093527	-3.88	0.000	-1.806466	-.5938262
2012	-1.649929	.3052139	-5.41	0.000	-2.248137	-1.051721
2013	-1.546357	.3083863	-5.01	0.000	-2.150783	-.9419304
2014	-1.397865	.3124051	-4.47	0.000	-2.010168	-.7855623
2015	-1.602914	.3243131	-4.94	0.000	-2.238556	-.9672719
2016	-1.337633	.3288868	-4.07	0.000	-1.982239	-.6930269
2017	-1.593293	.3294655	-4.84	0.000	-2.239033	-.9475523
_cons	4.903412	.3373903	14.53	0.000	4.242139	5.564685

bonds_tw_h(3)

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
pc1						
D1.	1.402571	.4778343	2.94	0.003	.4660331	2.339109
LD.	1.20312	.4659848	2.58	0.010	.2898071	2.116434
L2D.	1.668331	.4681665	3.56	0.000	.7507413	2.58592
L3D.	1.948135	.4538484	4.29	0.000	1.058608	2.837661
L4D.	1.149284	.4414292	2.60	0.009	.284099	2.01447
L5D.	.9527067	.4255506	2.24	0.025	.1186427	1.786771
bonds_tw						
L1.	.0917743	.0295736	3.10	0.002	.0338111	.1497375
pc1						
FD.	1.321772	.4886605	2.70	0.007	.3640146	2.279529
F2D.	1.36965	.4676626	2.93	0.003	.4530482	2.286252
F3D.	.8312839	.457766	1.82	0.069	-.065921	1.728489
F4D.	.3870906	.4387017	0.88	0.378	-.4727489	1.24693
F5D.	.1312157	.4090718	0.32	0.748	-.6705504	.9329818
CAB						
L1.	-.0495315	.0126884	-3.90	0.000	-.0744002	-.0246627
cpi_tw						
L1.	.1200899	.1815207	0.66	0.508	-.2356841	.4758639
GDebt						
L1.	-.007979	.0042188	-1.89	0.059	-.0162477	.0002896
GDeficit						
L1.	.0146225	.0188227	0.78	0.437	-.0222694	.0515143
banking						
L1.	.93546	.308342	3.03	0.002	.3311208	1.539799
currency						
L1.	-.2549575	.4280606	-0.60	0.551	-1.093941	.5840259
debt						
L1.	-.5633799	.5784894	-0.97	0.330	-1.697198	.5704386
period						
2001	0	(empty)				
2002	-.2188655	.2950099	-0.74	0.458	-.7970743	.3593432
2003	.0300623	.2939198	0.10	0.919	-.5460099	.6061346
2004	.182597	.302642	0.60	0.546	-.4105705	.7757645
2005	.4729251	.2989296	1.58	0.114	-.1129661	1.058816
2006	.7178073	.2989066	2.40	0.016	.1319612	1.303653
2007	-.5301971	.292993	-1.81	0.070	-1.104453	.0440586
2008	-.4550873	.2948783	-1.54	0.123	-1.033038	.1228635
2009	-1.44408	.2972608	-4.86	0.000	-2.0267	-.8614594

2010	-1.430154	.3018564	-4.74	0.000	-2.021782	-.8385263
2011	-1.723773	.3087627	-5.58	0.000	-2.328937	-1.11861
2012	-1.835109	.3040088	-6.04	0.000	-2.430955	-1.239262
2013	-1.679478	.3070228	-5.47	0.000	-2.281232	-1.077724
2014	-2.039661	.3127367	-6.52	0.000	-2.652614	-1.426708
2015	-1.536007	.3319056	-4.63	0.000	-2.18653	-.8854837
2016	-1.799023	.3293317	-5.46	0.000	-2.444501	-1.153545
_cons	5.581581	.3446317	16.20	0.000	4.906115	6.257047

bonds_tw_h(4)

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
pc1						
D1.	1.03215	.4949658	2.09	0.037	.0620348	2.002265
LD.	1.929538	.4901399	3.94	0.000	.9688815	2.890195
L2D.	1.689156	.4825319	3.50	0.000	.7434112	2.634902
L3D.	1.34944	.4654155	2.90	0.004	.4372427	2.261638
L4D.	.9414721	.449545	2.09	0.036	.0603801	1.822564
L5D.	-.1612065	.4482333	-0.36	0.719	-1.039728	.7173147
bonds_tw						
L1.	-.0000665	.0306786	-0.00	0.998	-.0601955	.0600624
pc1						
FD.	.7959764	.5052016	1.58	0.115	-.1942004	1.786153
F2D.	1.443218	.4881901	2.96	0.003	.4863825	2.400053
F3D.	1.151066	.4627952	2.49	0.013	.2440038	2.058127
F4D.	.5816114	.4439579	1.31	0.190	-.28853	1.451753
F5D.	.6251938	.4153447	1.51	0.132	-.1888668	1.439254
CAB						
L1.	-.0515315	.0131162	-3.93	0.000	-.0772387	-.0258242
cpi_tw						
L1.	.3125349	.1855839	1.68	0.092	-.0512029	.6762727
GDebt						
L1.	-.003695	.0045201	-0.82	0.414	-.0125542	.0051642
GDeficit						
L1.	.0496723	.0192915	2.57	0.010	.0118617	.087483
banking						
L1.	.4483334	.2992198	1.50	0.134	-.1381266	1.034793
currency						
L1.	.158508	.5726615	0.28	0.782	-.9638879	1.280904
debt						
L1.	-.8312481	.5670238	-1.47	0.143	-1.942594	.2800981
period						
2001	0 (empty)					
2002	.0219498	.2854039	0.08	0.939	-.5374316	.5813311
2003	.1976103	.2848496	0.69	0.488	-.3606847	.7559052
2004	.5785428	.2953085	1.96	0.050	-.0002513	1.157337
2005	.5846132	.2920973	2.00	0.045	.012113	1.157113
2006	-.4268787	.2907673	-1.47	0.142	-.9967721	.1430148
2007	-.2940437	.2858976	-1.03	0.304	-.8543927	.2663054
2008	-1.06548	.2869558	-3.71	0.000	-1.627903	-.5030573
2009	-1.721073	.2896293	-5.94	0.000	-2.288736	-1.15341
2010	-1.663763	.2972663	-5.60	0.000	-2.246394	-1.081132
2011	-1.683308	.3041205	-5.54	0.000	-2.279373	-1.087243
2012	-1.784509	.2989736	-5.97	0.000	-2.370486	-1.198532
2013	-2.160051	.3039551	-7.11	0.000	-2.755792	-1.56431
2014	-1.612654	.3163748	-5.10	0.000	-2.232737	-.9925707
2015	-1.940914	.3306176	-5.87	0.000	-2.588912	-1.292915
_cons	5.799634	.3439351	16.86	0.000	5.125533	6.473734

Statistics by step

	N	R2	psR2	F	Chi2	Prob
0	1233	0.698	.	72.83	.	0.000
1	1213	0.450	.	25.30	.	0.000
2	1193	0.365	.	17.49	.	0.000
3	1098	0.351	.	15.43	.	0.000
4	1004	0.348	.	14.23	.	0.000

Impulse Response Function

	IRF	Std.Err.	IRF LO~90	IRF UP 90	IRF LO~95	IRF UP 95
0	0.92354	0.35653	0.33661	1.51047	0.22399	1.62309
1	1.68460	0.44837	0.94646	2.42274	0.80482	2.56438
2	1.70518	0.45980	0.94823	2.46214	0.80297	2.60740
3	1.40257	0.47783	0.61585	2.18929	0.46487	2.34028
4	1.03215	0.49497	0.21715	1.84715	0.06070	2.00360

```
.
. graph export bond_pca_jan25_TZ.png, as(png) width(4000) replace
file bond_pca_jan25_TZ.png saved as PNG format
```

```
.
. *** Figure 15. Panel LP - impact of change in VUL_N on sovereign ratings
.
. locproj sovrates, shock(D.pc1) ///
> z h(4) yl(1) sl(5) ///
> c(F(1/5).D.pc1 l(1).CAB l(1).cpi_tw l(1).GDebt l(1).GDeficit ///
> l(1).banking l(1).currency l(1).debt i.period) ///
> title("Impact of change in the first component of less correlated vulnerability") ///
> ttitle("Horizon") ///
> fe conf(90 95) noisily grname(TZ4)
sovrates_h(0)
```

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
pc1						
D1.	-.3571554	.1537533	-2.32	0.020	-.6585063	-.0558046
LD.	-.4476618	.1496975	-2.99	0.003	-.7410634	-.1542601
L2D.	-.4032402	.1489418	-2.71	0.007	-.6951607	-.1113196
L3D.	-.1099112	.1441856	-0.76	0.446	-.3925098	.1726874
L4D.	-.0847823	.139034	-0.61	0.542	-.357284	.1877195
L5D.	-.209898	.1326028	-1.58	0.113	-.4697947	.0499986
sovrates						
L1.	.8669923	.0120432	71.99	0.000	.843388	.8905966
pc1						
FD.	.1027352	.1509996	0.68	0.496	-.1932186	.3986891
F2D.	.4018413	.1505432	2.67	0.008	.106782	.6969005
F3D.	.5701629	.1482243	3.85	0.000	.2796486	.8606772
F4D.	.5738387	.1421735	4.04	0.000	.2951837	.8524936
F5D.	.50359	.1388213	3.63	0.000	.2315053	.7756748
CAB						
L1.	.00544	.003419	1.59	0.112	-.001261	.012141
cpi_tw						
L1.	-.0705392	.0404693	-1.74	0.081	-.1498576	.0087792
GDebt						
L1.	-.0099428	.0013318	-7.47	0.000	-.0125531	-.0073326
GDeficit						
L1.	.0488872	.0049799	9.82	0.000	.0391269	.0586476
banking						
L1.	-.3770625	.1116191	-3.38	0.001	-.5958319	-.158293
currency						
L1.	-.8891579	.139202	-6.39	0.000	-1.161989	-.6163269

debt						
L1.	-.4401769	.192723	-2.28	0.022	-.817907	-.0624468
period						
2001	0 (empty)					
2002	.0456289	.0918141	0.50	0.619	-.1343234	.2255812
2003	.1827562	.0929013	1.97	0.049	.0006731	.3648393
2004	.0802961	.09434	0.85	0.395	-.104607	.2651991
2005	.0538607	.0933249	0.58	0.564	-.1290528	.2367741
2006	.0344501	.0901893	0.38	0.702	-.1423176	.2112178
2007	-.0646215	.0893315	-0.72	0.469	-.239708	.1104651
2008	-.0502492	.0895326	-0.56	0.575	-.2257298	.1252315
2009	-.1008794	.0915772	-1.10	0.271	-.2803675	.0786086
2010	.134762	.0926227	1.45	0.146	-.0467752	.3162991
2011	-.0450866	.0913755	-0.49	0.622	-.2241792	.1340061
2012	-.1971261	.0901562	-2.19	0.029	-.3738291	-.0204231
2013	-.0517343	.0891737	-0.58	0.562	-.2265115	.1230429
2014	.087442	.0886538	0.99	0.324	-.0863163	.2612003
2015	.0203493	.0919247	0.22	0.825	-.1598199	.2005184
2016	.0070189	.0917393	0.08	0.939	-.1727868	.1868245
2017	-.0108673	.091921	-0.12	0.906	-.1910292	.1692946
_cons	2.349983	.2052362	11.45	0.000	1.947727	2.752238

sovrata_h(1)						

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	

pc1						
D1.	-.6787575	.2500907	-2.71	0.007	-1.168926	-.1885888
LD.	-.7768602	.2437135	-3.19	0.001	-1.25453	-.2991905
L2D.	-.6825989	.2426085	-2.81	0.005	-1.158103	-.207095
L3D.	-.2906063	.2346637	-1.24	0.216	-.7505387	.1693261
L4D.	-.314059	.2265272	-1.39	0.166	-.7580442	.1299262
L5D.	-.4139541	.2163932	-1.91	0.056	-.8380769	.0101688
sovrata						
L1.	.6740122	.0195767	34.43	0.000	.6356425	.7123819
pc1						
FD.	-.1342845	.2458652	-0.55	0.585	-.6161714	.3476024
F2D.	.5451525	.2453868	2.22	0.026	.0642032	1.026102
F3D.	.9974594	.2418405	4.12	0.000	.5234608	1.471458
F4D.	.9718538	.2310533	4.21	0.000	.5189977	1.42471
F5D.	.9977427	.2265448	4.40	0.000	.5537231	1.441762
CAB						
L1.	.018823	.0055559	3.39	0.001	.0079337	.0297123
cpi_tw						
L1.	-.0860645	.0657943	-1.31	0.191	-.215019	.0428901
GDebt						
L1.	-.0173744	.0021655	-8.02	0.000	-.0216188	-.0131301
GDeficit						
L1.	.0777738	.0081038	9.60	0.000	.0618907	.0936569
banking						
L1.	-.6304494	.1815102	-3.47	0.001	-.9862029	-.2746959
currency						
L1.	-.8561454	.2263784	-3.78	0.000	-1.299839	-.4124518
debt						
L1.	.1353179	.3131069	0.43	0.666	-.4783604	.7489962
period						
2001	0 (empty)					
2002	.0805149	.1491674	0.54	0.589	-.2118479	.3728777
2003	.2419244	.1509787	1.60	0.109	-.0539885	.5378372
2004	.106726	.1532871	0.70	0.486	-.1937112	.4071631

2005	.1379411	.1516768	0.91	0.363	-.15934	.4352222
2006	.1207867	.1465455	0.82	0.410	-.1664371	.4080105
2007	-.0642679	.1451658	-0.44	0.658	-.3487876	.2202519
2008	-.2570213	.1454668	-1.77	0.077	-.542131	.0280884
2009	-.0072152	.1495493	-0.05	0.962	-.3003265	.2858961
2010	.1057857	.1504853	0.70	0.482	-.1891601	.4007314
2011	-.1802106	.1485184	-1.21	0.225	-.4713013	.1108802
2012	-.2340214	.1464988	-1.60	0.110	-.5211538	.053111
2013	.0234	.1452634	0.16	0.872	-.261311	.3081109
2014	.1390101	.1440381	0.97	0.334	-.1432994	.4213196
2015	-.1194068	.1494933	-0.80	0.424	-.4124083	.1735948
2016	-.0518934	.1490781	-0.35	0.728	-.344081	.2402943
2017	.0737913	.1493574	0.49	0.621	-.2189438	.3665264
_cons	5.275984	.3338773	15.80	0.000	4.621597	5.930372

sovrates_h(2)						

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	

pc1						
D1.	-.7946784	.3093991	-2.57	0.010	-1.40109	-.1882674
LD.	-.8014737	.3012377	-2.66	0.008	-1.391889	-.2110587
L2D.	-.8241021	.2996802	-2.75	0.006	-1.411464	-.2367398
L3D.	-.5930236	.29039	-2.04	0.041	-1.162178	-.0238696
L4D.	-.5372089	.2807867	-1.91	0.056	-1.087541	.0131228
L5D.	-.5974433	.2669734	-2.24	0.025	-1.120702	-.074185
sovrates						
L1.	.4955915	.0241764	20.50	0.000	.4482067	.5429763
pc1						
FD.	-.5174033	.3038607	-1.70	0.089	-1.112959	.0781528
F2D.	.1235025	.3041253	0.41	0.685	-.4725721	.719577
F3D.	.9245653	.2990576	3.09	0.002	.3384232	1.510707
F4D.	1.055322	.2868321	3.68	0.000	.4931416	1.617503
F5D.	1.211785	.2798925	4.33	0.000	.6632063	1.760365
CAB						
L1.	.0319039	.0068595	4.65	0.000	.0184596	.0453483
cpi_tw						
L1.	-.0065601	.0811935	-0.08	0.936	-.1656964	.1525763
GDebt						
L1.	-.0190987	.0026778	-7.13	0.000	-.0243471	-.0138503
GDeficit						
L1.	.0851283	.0100454	8.47	0.000	.0654397	.1048169
banking						
L1.	-.9834851	.2239265	-4.39	0.000	-1.422373	-.5445972
currency						
L1.	-.6216197	.2798658	-2.22	0.026	-1.170147	-.0730929
debt						
L1.	-.1239042	.3862895	-0.32	0.748	-.8810178	.6332093
period						
2001	0	(empty)				
2002	.0173542	.1840288	0.09	0.925	-.3433357	.3780441
2003	.2354208	.18627	1.26	0.206	-.1296618	.6005033
2004	.1245832	.1891133	0.66	0.510	-.2460721	.4952385
2005	.182862	.1872149	0.98	0.329	-.1840724	.5497964
2006	.19451	.1808908	1.08	0.282	-.1600294	.5490494
2007	-.2728248	.1792083	-1.52	0.128	-.6240666	.0784169
2008	-.2855702	.1802612	-1.58	0.113	-.6388758	.0677353
2009	-.0796613	.184529	-0.43	0.666	-.4413315	.282009
2010	-.0937824	.1856592	-0.51	0.613	-.4576679	.270103
2011	-.2528127	.1832817	-1.38	0.168	-.6120382	.1064128
2012	-.1872306	.1811293	-1.03	0.301	-.5422376	.1677764
2013	.0170588	.179222	0.10	0.924	-.3342099	.3683274

2014	-.0325304	.1777194	-0.18	0.855	-.3808541	.3157933
2015	-.319215	.1845063	-1.73	0.084	-.6808407	.0424106
2016	-.0810266	.1839706	-0.44	0.660	-.4416023	.2795491
2017	.0588158	.1842695	0.32	0.750	-.3023459	.4199774
_cons	7.70597	.4128276	18.67	0.000	6.896843	8.515097

sovrata_h(3)						

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	

pc1						
D1.	-.9376192	.34445	-2.72	0.006	-1.612729	-.2625097
LD.	-.874699	.3351933	-2.61	0.009	-1.531666	-.2177322
L2D.	-1.041876	.3347109	-3.11	0.002	-1.697898	-.385855
L3D.	-.9967643	.3236561	-3.08	0.002	-1.631119	-.36241
L4D.	-.7363597	.3121464	-2.36	0.018	-1.348155	-.1245641
L5D.	-.5974962	.297273	-2.01	0.044	-1.180141	-.0148518
sovrata						
L1.	.3488076	.0269027	12.97	0.000	.2960793	.401536
pc1						
FD.	-.6128805	.3381243	-1.81	0.070	-1.275592	.0498309
F2D.	-.3863001	.3386753	-1.14	0.254	-1.050091	.2774912
F3D.	.2281147	.3338632	0.68	0.494	-.4262451	.8824745
F4D.	.6893637	.3193869	2.16	0.031	.0633768	1.315351
F5D.	1.084872	.3134049	3.46	0.001	.4706095	1.699134
CAB						
L1.	.0428661	.0076416	5.61	0.000	.0278889	.0578434
cpi_tw						
L1.	.0411163	.0902355	0.46	0.649	-.1357421	.2179747
GDebt						
L1.	-.0185311	.0029857	-6.21	0.000	-.024383	-.0126791
GDeficit						
L1.	.0834146	.0114254	7.30	0.000	.0610211	.105808
banking						
L1.	-1.367356	.2489117	-5.49	0.000	-1.855214	-.8794978
currency						
L1.	-.7068076	.3126047	-2.26	0.024	-1.319502	-.0941136
debt						
L1.	-.1750969	.429376	-0.41	0.683	-1.016658	.6664646
period						
2001	0 (empty)					
2002	.0479512	.204558	0.23	0.815	-.3529751	.4488774
2003	.2742022	.206996	1.32	0.185	-.1315025	.6799069
2004	.2346071	.2102246	1.12	0.264	-.1774254	.6466397
2005	.2894446	.2083234	1.39	0.165	-.1188618	.6977511
2006	.0536423	.2015187	0.27	0.790	-.341327	.4486117
2007	-.2365207	.1995926	-1.19	0.236	-.627715	.1546735
2008	-.3963868	.2004759	-1.98	0.048	-.7893124	-.0034612
2009	-.1376816	.2051264	-0.67	0.502	-.539722	.2643587
2010	-.1168659	.2064348	-0.57	0.571	-.5214707	.2877389
2011	-.1368003	.204161	-0.67	0.503	-.5369485	.263348
2012	-.0996914	.2013248	-0.50	0.620	-.4942808	.2948981
2013	-.0830487	.1991946	-0.42	0.677	-.4734629	.3073655
2014	-.1920857	.197532	-0.97	0.331	-.5792413	.19507
2015	-.3158557	.2051532	-1.54	0.124	-.7179486	.0862373
2016	-.0120523	.2044976	-0.06	0.953	-.4128603	.3887556
2017	-.0317522	.2047883	-0.16	0.877	-.4331299	.3696255
_cons	9.553396	.4601124	20.76	0.000	8.651592	10.4552

sovrata_h(4)

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
pc1						
D1.	-.9660613	.3663051	-2.64	0.008	-1.684006	-.2481166
LD.	-1.005582	.3582921	-2.81	0.005	-1.707821	-.3033421
L2D.	-1.410289	.3563898	-3.96	0.000	-2.108801	-.7117782
L3D.	-1.326174	.3431866	-3.86	0.000	-1.998808	-.6535411
L4D.	-.8232095	.3318928	-2.48	0.013	-1.473708	-.1727115
L5D.	-.7177914	.3164164	-2.27	0.023	-1.337956	-.0976266
sovrates						
L1.	.2182207	.0285808	7.64	0.000	.1622034	.274238
pc1						
FD.	-.7879998	.3589626	-2.20	0.028	-1.491553	-.0844461
F2D.	-.4559667	.3594443	-1.27	0.205	-1.160465	.2485312
F3D.	-.4537626	.3554747	-1.28	0.202	-1.15048	.242955
F4D.	-.202481	.3376879	-0.60	0.549	-.8643371	.459375
F5D.	.6081983	.3309656	1.84	0.066	-.0404823	1.256879
CAB						
L1.	.0559941	.0081215	6.89	0.000	.0400762	.071912
cpi_tw						
L1.	.0091937	.0958858	0.10	0.924	-.178739	.1971265
GDebt						
L1.	-.0178957	.0031717	-5.64	0.000	-.024112	-.0116794
GDeficit						
L1.	.0682206	.0121394	5.62	0.000	.0444277	.0920135
banking						
L1.	-1.122113	.264447	-4.24	0.000	-1.64042	-.6038065
currency						
L1.	-.7831302	.3368937	-2.32	0.020	-1.44343	-.1228308
debt						
L1.	-.1845681	.4564588	-0.40	0.686	-1.079211	.7100748
period						
2001	0 (empty)					
2002	.0540781	.2174609	0.25	0.804	-.3721375	.4802936
2003	.3098597	.2200074	1.41	0.159	-.1213469	.7410664
2004	.3423275	.2233031	1.53	0.125	-.0953385	.7799935
2005	.1202725	.2222726	0.54	0.588	-.3153739	.5559189
2006	.0739068	.214388	0.34	0.730	-.3462859	.4940996
2007	-.3729446	.2120548	-1.76	0.079	-.7885644	.0426752
2008	-.5418074	.2130881	-2.54	0.011	-.9594524	-.1241624
2009	-.1745276	.2179152	-0.80	0.423	-.6016335	.2525784
2010	-.068398	.2204093	-0.31	0.756	-.5003923	.3635964
2011	-.07826	.2169272	-0.36	0.718	-.5034295	.3469094
2012	-.1567738	.2138595	-0.73	0.464	-.5759308	.2623832
2013	-.2206187	.2116578	-1.04	0.297	-.6354603	.1942229
2014	-.2293151	.2094457	-1.09	0.274	-.6398212	.1811909
2015	-.2877562	.2180603	-1.32	0.187	-.7151465	.1396341
2016	-.0856069	.2171616	-0.39	0.693	-.5112359	.340022
2017	-.2089597	.217617	-0.96	0.337	-.6354811	.2175617
_cons	11.20022	.4889348	22.91	0.000	10.24193	12.15852

Impulse Response Function

	IRF	Std.Err.	IRF LO~90	IRF UP 90	IRF LO~95	IRF UP 95
0	-0.35716	0.15375	-0.61022	-0.10409	-0.65876	-0.05555
1	-0.67876	0.25009	-1.09039	-0.26713	-1.16935	-0.18817
2	-0.79468	0.30940	-1.30393	-0.28543	-1.40161	-0.18775
3	-0.93762	0.34445	-1.50456	-0.37068	-1.61331	-0.26193
4	-0.96606	0.36631	-1.56898	-0.36315	-1.68462	-0.24750

```
.
. graph export sovrata_pca_jan25_TZ.png, as(png) width(4000) replace
file sovrata_pca_jan25_TZ.png saved as PNG format
```

```
.
. *** Figure C1. Panel LP - impact of change in vulnerability on bond yields
```

```
. set scheme stcolor
```

```
.
. xtvar bonds_tw Dvul100 if vul>.3722011, lags(3) maxldep(4) ///
> fd collapse
note: 1 panel dropped due to no within-panel variation in bonds_tw.
note: the estimation sample has gaps; see xtdescribe.
```

```
Panel-data vector autoregression          Number of obs   = 853
Group variable: imfcode                   Number of groups = 72
Time variable: period                     Obs per group:
                                          min = 1
Number of moment conditions = 16 (collapsed)      avg = 11.8
                                          max = 20
Fixed-effects transform: FD
Two-step results
```

(Std. err. adjusted for 72 clusters in imfcode)

		Coefficient	WC robust std. err.	z	P> z	[95% conf. interval]	

bonds_tw							
bonds_tw							
	L1.	1.0208	.1015578	10.05	0.000	.8217506	1.21985
	L2.	-.1863864	.0650739	-2.86	0.004	-.3139289	-.0588439
	L3.	.1227504	.0599996	2.05	0.041	.0051534	.2403474
Dvul100							
	L1.	.6957627	.3012898	2.31	0.021	.1052457	1.28628
	L2.	.5064831	.2632867	1.92	0.054	-.0095494	1.022516
	L3.	.039305	.1483725	0.26	0.791	-.2514997	.3301097

Dvul100							
bonds_tw							
	L1.	.0041215	.0142783	0.29	0.773	-.0238634	.0321063
	L2.	-.0009502	.0146222	-0.06	0.948	-.029609	.0277087
	L3.	-.0096578	.0118449	-0.82	0.415	-.0328734	.0135578
Dvul100							
	L1.	-.1471378	.0723559	-2.03	0.042	-.2889528	-.0053228
	L2.	.0331688	.0581449	0.57	0.568	-.0807931	.1471308
	L3.	-.003307	.0352081	-0.09	0.925	-.0723135	.0656996

```
Hansen's test of overid. restrictions: chi2(4) = 5.63      Prob > chi2 = 0.229
GMM-type instruments: L(2/5).(bonds_tw Dvul100)
```

```
.
. vargranger
```

Granger causality Wald tests

Equation	Excluded	chi2	df	Prob > chi2

bonds_tw	Dvul100	7.9933	3	0.046
bonds_tw	ALL	7.9933	3	0.046

Dvul100	bonds_tw	2.1208	3	0.548
Dvul100	ALL	2.1208	3	0.548

```

.
. irf create lags, set(example1) step(6) replace
(file example1.irf now active)
(file example1.irf updated)

.

. irf graph irf, irf(lags) name(irf, replace)

.
. *irf cgraph (lags Dvul100 bonds_tw irf) ///
> (lags bonds_tw Dvul100 irf), ///
> ycommon name(oirf, replace)
.
. graph export bonds_var_irf.png, as(png) width(4000) replace
file bonds_var_irf.png saved as PNG format

.
. *** Figure D1. Panel LP - impact of natural disasters on bond yields
.
. locproj bonds_tw, shock(disaster) ///
> z h(5) yl(5) sl(3) ///
> c(f(1/7).disaster i.period) ///
> fe cluster(imfcode) conf(90 95) ///
> ttitle("Horizon") ///
> title("Impact of Disasters on Bonds Yields") ///
> save irfname(bonds_DTZ) noisily stats grname(DTZ1)
bonds_tw_h(0)

```

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
disaster						
--	.215997	.1035164	2.09	0.037	.0131085	.4188855
L1.	.2511794	.0976852	2.57	0.010	.05972	.4426387
L2.	.189483	.1172757	1.62	0.106	-.0403731	.4193391
L3.	.2246682	.1472435	1.53	0.127	-.0639239	.5132602
bonds_tw						
L1.	.9387898	.0691707	13.57	0.000	.8032177	1.074362
L2.	-.3064827	.0671738	-4.56	0.000	-.438141	-.1748244
L3.	.0870604	.0715182	1.22	0.223	-.0531127	.2272336
L4.	-.0395247	.0661395	-0.60	0.550	-.1691556	.0901063
L5.	-.0415189	.0412655	-1.01	0.314	-.1223978	.0393601
disaster						
F1.	-.0383553	.1483936	-0.26	0.796	-.3292014	.2524908
F2.	-.0124041	.095179	-0.13	0.896	-.1989515	.1741434
F3.	.0201583	.1306323	0.15	0.877	-.2358762	.2761929
F4.	-.0274313	.0938835	-0.29	0.770	-.2114396	.1565771
F5.	.0208757	.1147581	0.18	0.856	-.2040462	.2457975
F6.	.2083073	.1612928	1.29	0.197	-.1078208	.5244354
F7.	.0918586	.1331182	0.69	0.490	-.1690483	.3527656
period						
1998	0 (empty)					
1999	.2370128	.2806794	0.84	0.398	-.3131086	.7871342
2000	.516841	.2835071	1.82	0.068	-.0388227	1.072505
2001	-.2457174	.229717	-1.07	0.285	-.6959545	.2045198
2002	-.0077342	.2298526	-0.03	0.973	-.4582371	.4427687
2003	-1.030385	.2406288	-4.28	0.000	-1.502009	-.5587617
2004	-.3691474	.2517481	-1.47	0.143	-.8625647	.1242699
2005	-1.001029	.3064661	-3.27	0.001	-1.601691	-.4003662
2006	-.5352521	.3224612	-1.66	0.097	-1.167264	.0967602
2007	-.6078274	.2877133	-2.11	0.035	-1.171735	-.0439196
2008	-.3229239	.2572416	-1.26	0.209	-.8271081	.1812603
2009	-.3709655	.2766801	-1.34	0.180	-.9132485	.1713175
2010	-1.713737	.3774982	-4.54	0.000	-2.453619	-.9738537
2011	-.5105043	.3178592	-1.61	0.108	-1.133497	.1124883
2012	-1.398012	.3067175	-4.56	0.000	-1.999167	-.7968567
2013	-1.279391	.3218396	-3.98	0.000	-1.910185	-.6485971
2014	-1.330329	.3476004	-3.83	0.000	-2.011613	-.649045
2015	-1.061541	.37928	-2.80	0.005	-1.804917	-.3181663
_cons	2.624816	.5221873	5.03	0.000	1.601348	3.648284

bonds_tw_h(1)						
	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
disaster						
--.	.3573013	.143982	2.48	0.013	.0751018	.6395007
L1.	.3941065	.1853688	2.13	0.033	.0307903	.7574227
L2.	.4254535	.2138787	1.99	0.047	.006259	.844648
L3.	.2073402	.2056619	1.01	0.313	-.1957497	.6104302
bonds_tw						
L1.	.55185	.1034696	5.33	0.000	.3490532	.7546467
L2.	-.2098396	.0827384	-2.54	0.011	-.3720038	-.0476754
L3.	.045185	.0487776	0.93	0.354	-.0504173	.1407873
L4.	-.0723808	.0587648	-1.23	0.218	-.1875577	.0427961
L5.	-.0510376	.06373	-0.80	0.423	-.1759461	.0738708
disaster						
F1.	.2115488	.1969262	1.07	0.283	-.1744194	.597517
F2.	-.0073058	.191824	-0.04	0.970	-.383274	.3686624
F3.	.0118805	.1612437	0.07	0.941	-.3041512	.3279123
F4.	-.0084455	.1597253	-0.05	0.958	-.3215012	.3046103
F5.	-.0047444	.166706	-0.03	0.977	-.3314821	.3219933
F6.	.1854828	.2157501	0.86	0.390	-.2373797	.6083453
F7.	.2230784	.2289824	0.97	0.330	-.2257189	.6718756
period						
1998	0 (empty)					
1999	.8945939	.3374777	2.65	0.008	.2331498	1.556038
2000	.3768289	.2777734	1.36	0.175	-.167597	.9212547
2001	.0057323	.2747103	0.02	0.983	-.53269	.5441547
2002	-.8835831	.2201648	-4.01	0.000	-1.315098	-.4520679
2003	-1.154719	.2698007	-4.28	0.000	-1.683518	-.6259188
2004	-1.164656	.3641414	-3.20	0.001	-1.87836	-.4509522
2005	-1.3151	.4904725	-2.68	0.007	-2.276409	-.3537918
2006	-.9767373	.4718636	-2.07	0.038	-1.901573	-.0519017
2007	-.8455385	.4185266	-2.02	0.043	-1.665836	-.0252414
2008	-.6146395	.4216322	-1.46	0.145	-1.441023	.2117445
2009	-1.988091	.4238799	-4.69	0.000	-2.81888	-1.157302
2010	-1.980634	.488827	-4.05	0.000	-2.938717	-1.022551
2011	-1.853289	.476346	-3.89	0.000	-2.78691	-.9196685
2012	-2.492569	.431417	-5.78	0.000	-3.338131	-1.647007
2013	-2.454164	.4767619	-5.15	0.000	-3.3886	-1.519728
2014	-2.199376	.5856696	-3.76	0.000	-3.347267	-1.051484
2015	-2.351534	.5988363	-3.93	0.000	-3.525232	-1.177837
_cons	5.190584	.8755145	5.93	0.000	3.474607	6.90656
bonds_tw_h(2)						
	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
disaster						
--.	.4075309	.1923311	2.12	0.034	.0305689	.784493
L1.	.5618901	.2799179	2.01	0.045	.0132611	1.110519
L2.	.3076418	.2539203	1.21	0.226	-.1900327	.8053164
L3.	-.0260007	.1912319	-0.14	0.892	-.4008084	.348807
bonds_tw						
L1.	.2955733	.1134287	2.61	0.009	.0732571	.5178894
L2.	-.1013158	.0634165	-1.60	0.110	-.2256098	.0229782
L3.	-.0640288	.0399252	-1.60	0.109	-.1422808	.0142231
L4.	.0203433	.0649495	0.31	0.754	-.1069554	.1476419
L5.	-.1096069	.0857604	-1.28	0.201	-.2776943	.0584805
disaster						
F1.	.3749769	.2167835	1.73	0.084	-.049911	.7998649
F2.	.1479067	.2087938	0.71	0.479	-.2613217	.5571351
F3.	.0403694	.1857358	0.22	0.828	-.323666	.4044049
F4.	-.0387046	.1779209	-0.22	0.828	-.3874232	.3100141
F5.	-.0609003	.222529	-0.27	0.784	-.4970491	.3752484
F6.	-.0040501	.2042118	-0.02	0.984	-.404298	.3961977

F7.	.1734354	.2236794	0.78	0.438	-.264968	.6118389
period						
1998	0	(empty)				
1999	.1801059	.3045657	0.59	0.554	-.416832	.7770438
2000	-.0039328	.3078084	-0.01	0.990	-.6072261	.5993605
2001	-1.238801	.2703101	-4.58	0.000	-1.768599	-.7090032
2002	-1.65367	.2290592	-7.22	0.000	-2.102618	-1.204722
2003	-2.136736	.3600868	-5.93	0.000	-2.842493	-1.430979
2004	-1.88645	.5078989	-3.71	0.000	-2.881913	-.890986
2005	-1.819981	.5402329	-3.37	0.001	-2.878818	-.7611444
2006	-1.54427	.5281605	-2.92	0.003	-2.579446	-.5090949
2007	-1.356693	.5189519	-2.61	0.009	-2.37382	-.3395655
2008	-2.581954	.5099011	-5.06	0.000	-3.581342	-1.582566
2009	-2.605918	.5266899	-4.95	0.000	-3.638212	-1.573625
2010	-3.311086	.5451853	-6.07	0.000	-4.379629	-2.242542
2011	-3.325798	.5087102	-6.54	0.000	-4.322852	-2.328744
2012	-3.666248	.5626978	-6.52	0.000	-4.769115	-2.56338
2013	-3.384851	.6501287	-5.21	0.000	-4.65908	-2.110622
2014	-3.404106	.6994205	-4.87	0.000	-4.774945	-2.033267
2015	-3.925119	.7314669	-5.37	0.000	-5.358768	-2.49147
_cons	7.32099	1.024345	7.15	0.000	5.31331	9.32867

bonds_tw_h(3)

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
disaster						
--.	.5413983	.276297	1.96	0.050	-.0001338	1.08293
L1.	.381724	.3113499	1.23	0.220	-.2285106	.9919587
L2.	-.0187455	.2285251	-0.08	0.935	-.4666465	.4291555
L3.	-.0225291	.1933792	-0.12	0.907	-.4015454	.3564873
bonds_tw						
L1.	.1762161	.1127446	1.56	0.118	-.0447593	.3971914
L2.	-.1530481	.0498313	-3.07	0.002	-.2507158	-.0553805
L3.	.0029206	.0816423	0.04	0.971	-.1570955	.1629366
L4.	.0587265	.0671919	0.87	0.382	-.0729672	.1904202
L5.	-.1768553	.0746376	-2.37	0.018	-.3231423	-.0305684
disaster						
F1.	.4258167	.252253	1.69	0.091	-.0685901	.9202236
F2.	.238914	.199824	1.20	0.232	-.1527338	.6305618
F3.	.0766381	.1825647	0.42	0.675	-.2811821	.4344583
F4.	-.0897369	.1730837	-0.52	0.604	-.4289746	.2495008
F5.	-.0841008	.2094357	-0.40	0.688	-.4945872	.3263856
F6.	-.0467252	.2338358	-0.20	0.842	-.5050349	.4115846
F7.	-.0807747	.1801141	-0.45	0.654	-.4337919	.2722425
period						
1998	0	(empty)				
1999	.1573711	.3661907	0.43	0.667	-.5603494	.8750917
2000	-.6507772	.3458696	-1.88	0.060	-1.328669	.0271149
2001	-1.238941	.3190496	-3.88	0.000	-1.864267	-.6136154
2002	-1.984333	.3561391	-5.57	0.000	-2.682353	-1.286313
2003	-1.999881	.4890103	-4.09	0.000	-2.958324	-1.041439
2004	-1.646485	.5406515	-3.05	0.002	-2.706143	-.5868276
2005	-1.545627	.5234598	-2.95	0.003	-2.571589	-.5196645
2006	-1.192721	.5744144	-2.08	0.038	-2.318553	-.0668897
2007	-2.34804	.5484878	-4.28	0.000	-3.423056	-1.273024
2008	-2.459192	.5630746	-4.37	0.000	-3.562798	-1.355586
2009	-3.158069	.5376319	-5.87	0.000	-4.211808	-2.10433
2010	-3.728058	.5286457	-7.05	0.000	-4.764185	-2.691932
2011	-3.634377	.605449	-6.00	0.000	-4.821035	-2.447719
2012	-3.663143	.6662596	-5.50	0.000	-4.968988	-2.357298
2013	-3.709185	.6987816	-5.31	0.000	-5.078772	-2.339598
2014	-3.900717	.7251355	-5.38	0.000	-5.321956	-2.479478
2015	-3.757684	.7998387	-4.70	0.000	-5.325339	-2.190029
_cons	8.20624	1.084411	7.57	0.000	6.080833	10.33165

bonds_tw_h(4)

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
disaster						
--.	.2680684	.3369894	0.80	0.426	-.3924187	.9285555
L1.	.0846237	.2534195	0.33	0.738	-.4120693	.5813167
L2.	.037061	.2243321	0.17	0.869	-.4026218	.4767437
L3.	.0740249	.2148858	0.34	0.730	-.3471436	.4951934
bonds_tw						
L1.	.0036853	.0942523	0.04	0.969	-.1810458	.1884164
L2.	.015147	.0816733	0.19	0.853	-.1449297	.1752237
L3.	-.0048923	.0769098	-0.06	0.949	-.1556328	.1458483
L4.	.0149033	.0364631	0.41	0.683	-.0565631	.0863698
L5.	-.1769182	.0845953	-2.09	0.036	-.3427219	-.0111146
disaster						
F1.	.4942978	.2609986	1.89	0.058	-.0172499	1.005846
F2.	.2573597	.2005011	1.28	0.199	-.1356152	.6503347
F3.	-.0025431	.157454	-0.02	0.987	-.3111472	.3060611
F4.	-.0895515	.1619734	-0.55	0.580	-.4070136	.2279105
F5.	-.1004779	.1853454	-0.54	0.588	-.4637483	.2627925
F6.	-.1042202	.2079982	-0.50	0.616	-.5118891	.3034487
F7.	-.1876739	.2051074	-0.92	0.360	-.5896771	.2143293
period						
1998	0 (empty)					
1999	-.419606	.2786054	-1.51	0.132	-.9656626	.1264506
2000	-.5824434	.2908941	-2.00	0.045	-1.152585	-.0123014
2001	-1.477139	.2870349	-5.15	0.000	-2.039717	-.9145612
2002	-1.775406	.3679297	-4.83	0.000	-2.496535	-1.054277
2003	-1.461321	.3527041	-4.14	0.000	-2.152609	-.7700341
2004	-1.272147	.3665605	-3.47	0.001	-1.990593	-.5537017
2005	-1.049365	.4227932	-2.48	0.013	-1.878024	-.2207055
2006	-1.904375	.41779	-4.56	0.000	-2.723228	-1.085521
2007	-1.976644	.4092695	-4.83	0.000	-2.778797	-1.17449
2008	-2.744423	.3980127	-6.90	0.000	-3.524514	-1.964333
2009	-3.249911	.3700024	-8.78	0.000	-3.975102	-2.524719
2010	-3.617291	.4417584	-8.19	0.000	-4.483122	-2.751461
2011	-3.364875	.5329453	-6.31	0.000	-4.409429	-2.320322
2012	-3.566789	.4919923	-7.25	0.000	-4.531076	-2.602502
2013	-3.84355	.5273482	-7.29	0.000	-4.877133	-2.809967
2014	-3.294732	.6291228	-5.24	0.000	-4.52779	-2.061674
2015	-3.7555	.6460874	-5.81	0.000	-5.021808	-2.489192
_cons	8.355876	.9028312	9.26	0.000	6.586359	10.12539

bonds_tw_h(5)

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
disaster						
--.	.0836465	.2760875	0.30	0.762	-.457475	.6247679
L1.	.1003653	.2518745	0.40	0.690	-.3932997	.5940304
L2.	.1452096	.2616228	0.56	0.579	-.3675617	.6579809
L3.	.2403918	.2463526	0.98	0.329	-.2424504	.7232339
bonds_tw						
L1.	.0156013	.0770951	0.20	0.840	-.1355023	.1667048
L2.	.0093241	.0593171	0.16	0.875	-.1069353	.1255834
L3.	-.0334951	.051043	-0.66	0.512	-.1335375	.0665473
L4.	.0008672	.0449446	0.02	0.985	-.0872226	.0889569
L5.	-.1706474	.0864653	-1.97	0.048	-.3401163	-.0011784
disaster						
F1.	.2509594	.2854222	0.88	0.379	-.3084577	.8103766
F2.	.36998	.2119378	1.75	0.081	-.0454104	.7853704
F3.	.1101377	.2209022	0.50	0.618	-.3228226	.5430979
F4.	-.0638871	.1702741	-0.38	0.708	-.3976182	.269844
F5.	-.080673	.1782292	-0.45	0.651	-.4299959	.2686499
F6.	-.0844478	.1844712	-0.46	0.647	-.4460048	.2771092
F7.	-.1071401	.1879927	-0.57	0.569	-.4755991	.2613189

period						
1998	0	(empty)				
1999	.1955494	.2664535	0.73	0.463	-.3266897	.7177886
2000	-.4120681	.2665986	-1.55	0.122	-.9345917	.1104555
2001	-.7631717	.3357612	-2.27	0.023	-1.421251	-.1050919
2002	-.8183057	.3409716	-2.40	0.016	-1.486598	-.1500136
2003	-.4535989	.2993435	-1.52	0.130	-1.040301	.1331037
2004	-.4269248	.3342945	-1.28	0.202	-1.08213	.2282805
2005	-1.04713	.2840159	-3.69	0.000	-1.603791	-.4904695
2006	-1.076318	.3538221	-3.04	0.002	-1.769797	-.3828394
2007	-1.763132	.3780559	-4.66	0.000	-2.504108	-1.022156
2008	-2.464442	.3915428	-6.29	0.000	-3.231852	-1.697032
2009	-2.761448	.4008101	-6.89	0.000	-3.547022	-1.975875
2010	-2.659653	.5118985	-5.20	0.000	-3.662956	-1.656351
2011	-2.855388	.4572313	-6.24	0.000	-3.751545	-1.959231
2012	-3.129391	.4817383	-6.50	0.000	-4.073581	-2.185201
2013	-2.673624	.5756477	-4.64	0.000	-3.801873	-1.545376
2014	-2.785881	.5964579	-4.67	0.000	-3.954917	-1.616845
_cons	7.709898	.949705	8.12	0.000	5.848511	9.571286

Statistics by step

	N	R2	psR2	F	Chi2	Prob
0	1174	0.686	.	53.03	.	0.000
1	1166	0.425	.	23.63	.	0.000
2	1164	0.340	.	20.58	.	0.000
3	1148	0.286	.	17.41	.	0.000
4	1130	0.237	.	14.90	.	0.000
5	1040	0.216	.	14.27	.	0.000

Impulse Response Function

	IRF	Std.Err.	IRF LO~90	IRF UP 90	IRF LO~95	IRF UP 95
0	0.21600	0.10352	0.04405	0.38794	0.01049	0.42150
1	0.35730	0.14398	0.11809	0.59651	0.07138	0.64322
2	0.40753	0.19233	0.08799	0.72707	0.02560	0.78946
3	0.54140	0.27630	0.08236	1.00044	-0.00727	1.09007
4	0.26807	0.33699	-0.29181	0.82794	-0.40113	0.93726
5	0.08365	0.27609	-0.37510	0.54239	-0.46469	0.63198

```
. locproj bonds_tw if incl=4, shock(disaster) ///
> z h(5) yl(7) sl(6) ///
> c(f(1/7).disaster i.period) ///
> fe cluster(imfcode) conf(90 95) ///
> ttitle("Horizon") ///
> title("`Impact of Disasters on Bonds Yields (Other than high income countries)`) ///
> save irfname(bonds_DTZ) noisily stats grname(DT22)
bonds_tw_h(0)
```

	Coefficient	Std. err.	z	P> z	[95% conf. interval]
disaster					
--.	.4919001	.1857959	2.65	0.008	.1277468 .8560534
L1.	.3097735	.2150462	1.44	0.150	-.1117093 .7312562
L2.	.3671011	.2320928	1.58	0.114	-.0877925 .8219947
L3.	.6191223	.3499767	1.77	0.077	-.0668194 1.305064
L4.	.0167062	.2648199	0.06	0.950	-.5023314 .5357437
L5.	-.076666	.2569632	-0.30	0.765	-.5803047 .4269727
L6.	.0877387	.3301171	0.27	0.790	-.5592789 .7347563
bonds_tw					
L1.	.9206303	.1147032	8.03	0.000	.6958162 1.145444
L2.	-.259367	.0955876	-2.71	0.007	-.4467153 -.0720187
L3.	-.0001548	.0741834	-0.00	0.998	-.1455516 .1452421
L4.	-.0031859	.078252	-0.04	0.968	-.156557 .1501851
L5.	-.05464	.0698475	-0.78	0.434	-.1915385 .0822585
L6.	.0041234	.0834056	0.05	0.961	-.1593487 .1675954
L7.	-.029777	.066204	-0.45	0.653	-.1595344 .0999805

disaster						
F1.	.5417829	.3267642	1.66	0.097	-.0986632	1.182229
F2.	-.1343868	.1964717	-0.68	0.494	-.5194644	.2506907
F3.	-.3615061	.2127898	-1.70	0.089	-.7785664	.0555542
F4.	-.0921011	.2159915	-0.43	0.670	-.5154366	.3312344
F5.	-.0994345	.1897316	-0.52	0.600	-.4713016	.2724325
F6.	.2467235	.312916	0.79	0.430	-.3665807	.8600277
F7.	.1496582	.2439883	0.61	0.540	-.32855	.6278664
period						
2000	0	(empty)				
2001	.242507	.2697182	0.90	0.369	-.286131	.7711449
2002	.4970753	.5043881	0.99	0.324	-.4915073	1.485658
2003	-.8621624	.3636923	-2.37	0.018	-1.574986	-.1493385
2004	-.6359374	.2961593	-2.15	0.032	-1.216399	-.0554759
2005	-1.02544	.5304708	-1.93	0.053	-2.065144	.0142632
2006	-.8149908	.5824762	-1.40	0.162	-1.956623	.3266416
2007	-.9114185	.5736124	-1.59	0.112	-2.035678	.2128413
2008	-.4524012	.5975389	-0.76	0.449	-1.623556	.7187534
2009	-.4533506	.660707	-0.69	0.493	-1.748313	.8416113
2010	-2.740203	.6313269	-4.34	0.000	-3.977581	-1.502825
2011	-.8887916	.5915439	-1.50	0.133	-2.048196	.2706132
2012	-1.873361	.5891257	-3.18	0.001	-3.028026	-.7186953
2013	-1.648764	.6685774	-2.47	0.014	-2.959151	-.3383759
2014	-1.558357	.8360842	-1.86	0.062	-3.197052	.0803382
2015	-.7382119	.9120149	-0.81	0.418	-2.525728	1.049304
_cons	3.771924	1.372625	2.75	0.006	1.081629	6.462219

bonds_tw_h(1)						
	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
disaster						
--.	.664878	.2358687	2.82	0.005	.2025838	1.127172
L1.	.5416325	.3741333	1.45	0.148	-.1916553	1.27492
L2.	.9447173	.5513898	1.71	0.087	-.1359868	2.025421
L3.	.5907471	.5801162	1.02	0.309	-.5462598	1.727754
L4.	.0033319	.3899939	0.01	0.993	-.7610421	.767706
L5.	.1166112	.5079535	0.23	0.818	-.8789594	1.112182
L6.	.0887223	.476765	0.19	0.852	-.8457199	1.023164
bonds_tw						
L1.	.5871769	.1717617	3.42	0.001	.2505302	.9238236
L2.	-.2200306	.1302971	-1.69	0.091	-.4754082	.0353471
L3.	-.0013814	.0629769	-0.02	0.983	-.1248138	.1220511
L4.	-.1441243	.0910383	-1.58	0.113	-.3225561	.0343075
L5.	.0992741	.1154892	0.86	0.390	-.1270806	.3256288
L6.	-.0560062	.1052146	-0.53	0.595	-.262223	.1502106
L7.	-.0789211	.0743908	-1.06	0.289	-.2247245	.0668823
disaster						
F1.	.8853788	.4074187	2.17	0.030	.0868529	1.683905
F2.	.2306677	.3841449	0.60	0.548	-.5222425	.9835779
F3.	-.5586462	.3248076	-1.72	0.085	-1.195257	.077965
F4.	-.3937816	.3340495	-1.18	0.238	-1.048506	.2609433
F5.	-.2078307	.29164	-0.71	0.476	-.7794345	.3637731
F6.	.1385092	.3147698	0.44	0.660	-.4784283	.7554467
F7.	.4175896	.4048465	1.03	0.302	-.375895	1.211074
period						
2000	0	(empty)				
2001	.7395756	.7414079	1.00	0.319	-.7135572	2.192708
2002	-.1606702	.5683885	-0.28	0.777	-1.274691	.9533507
2003	-1.249291	.581592	-2.15	0.032	-2.389191	-.1093918
2004	-1.766599	.7455013	-2.37	0.018	-3.227754	-.3054429
2005	-1.82674	1.061812	-1.72	0.085	-3.907853	.2543729
2006	-1.377995	.9959236	-1.38	0.166	-3.329969	.5739796
2007	-1.285823	1.069961	-1.20	0.229	-3.382908	.8112625
2008	-.567467	1.167007	-0.49	0.627	-2.854758	1.719824
2009	-2.977706	1.052597	-2.83	0.005	-5.040758	-.9146545
2010	-3.320097	.9407467	-3.53	0.000	-5.163926	-1.476267

2011	-2.732736	.9617535	-2.84	0.004	-4.617738	-.8477336
2012	-3.307714	1.137529	-2.91	0.004	-5.537231	-1.078197
2013	-3.10437	1.373021	-2.26	0.024	-5.795442	-.4132973
2014	-2.423614	1.539478	-1.57	0.115	-5.440935	.5937072
2015	-2.229116	1.557852	-1.43	0.152	-5.282451	.8242182
_cons	7.313328	2.52503	2.90	0.004	2.364361	12.2623

bonds_tw_h(2)

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
disaster						
--.	.5163741	.2412013	2.14	0.032	.0436282	.9891201
L1.	.8238202	.5560152	1.48	0.138	-.2659495	1.91359
L2.	.7179247	.7562492	0.95	0.342	-.7642964	2.200146
L3.	.550254	.5461611	1.01	0.314	-.5202021	1.62071
L4.	.313258	.4687168	0.67	0.504	-.6054101	1.231926
L5.	.1486766	.5708947	0.26	0.795	-.9702564	1.26761
L6.	.2554554	.4069761	0.63	0.530	-.5422031	1.053114
bonds_tw						
L1.	.3414711	.1881894	1.81	0.070	-.0273732	.7103155
L2.	-.1322744	.0957118	-1.38	0.167	-.3198661	.0553172
L3.	-.1509848	.0693764	-2.18	0.030	-.2869601	-.0150095
L4.	.1104747	.1282156	0.86	0.389	-.1408232	.3617727
L5.	.0077668	.1290524	0.06	0.952	-.2451712	.2607049
L6.	-.0245099	.052795	-0.46	0.642	-.1279861	.0789663
L7.	-.1136618	.1011013	-1.12	0.261	-.3118167	.0844931
disaster						
F1.	.602224	.3600605	1.67	0.094	-.1034815	1.30793
F2.	.3272265	.3919986	0.83	0.404	-.4410766	1.09553
F3.	-.0823289	.3978368	-0.21	0.836	-.8620747	.6974169
F4.	-.5763729	.3360491	-1.72	0.086	-1.235017	.0822713
F5.	-.4606914	.3378859	-1.36	0.173	-1.122936	.2015527
F6.	.0826313	.2753119	0.30	0.764	-.4569701	.6222328
F7.	.3195025	.3521961	0.91	0.364	-.3707892	1.009794
period						
2000	0 (empty)					
2001	.2488316	.5894238	0.42	0.673	-.9064177	1.404081
2002	-.3466519	.4767646	-0.73	0.467	-1.281093	.5877896
2003	-1.798475	.7322318	-2.46	0.014	-3.233623	-.363327
2004	-2.235186	.8662259	-2.58	0.010	-3.932958	-.5374144
2005	-1.817034	.9841449	-1.85	0.065	-3.745923	.1118542
2006	-1.125837	1.033134	-1.09	0.276	-3.150742	.8990679
2007	-.5002384	1.184594	-0.42	0.673	-2.821999	1.821523
2008	-2.481456	1.152932	-2.15	0.031	-4.741161	-.2217512
2009	-2.94845	.983715	-3.00	0.003	-4.876496	-1.020404
2010	-3.629131	.8104609	-4.48	0.000	-5.217606	-2.040657
2011	-3.109901	1.035421	-3.00	0.003	-5.139289	-1.080514
2012	-3.567126	1.276448	-2.79	0.005	-6.068917	-1.065334
2013	-2.710985	1.512614	-1.79	0.073	-5.675653	.2536829
2014	-2.668723	1.560827	-1.71	0.087	-5.727888	.3904427
2015	-3.243778	1.601701	-2.03	0.043	-6.383055	-.1045005
_cons	8.315356	2.606618	3.19	0.001	3.206478	13.42423

bonds_tw_h(3)

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
disaster						
--.	.5734281	.3354318	1.71	0.087	-.0840061	1.230862
L1.	.6093694	.6447271	0.95	0.345	-.6542725	1.873011
L2.	.8279392	.7002155	1.18	0.237	-.544458	2.200336
L3.	.7114066	.5272921	1.35	0.177	-.3220669	1.74488
L4.	.3893881	.5039237	0.77	0.440	-.5982842	1.37706
L5.	.3180691	.4628902	0.69	0.492	-.5891791	1.225317
L6.	.5301314	.3877552	1.37	0.172	-.2298547	1.290118
bonds_tw						

L1.	.2020153	.1736103	1.16	0.245	-.1382546	.5422852
L2.	-.1736865	.0876139	-1.98	0.047	-.3454065	-.0019664
L3.	.0861763	.1221781	0.71	0.481	-.1532884	.325641
L4.	.0245465	.080806	0.30	0.761	-.1338304	.1829233
L5.	.0291054	.0815282	0.36	0.721	-.1306869	.1888977
L6.	-.0114805	.076646	-0.15	0.881	-.161704	.138743
L7.	-.1343401	.0978536	-1.37	0.170	-.3261296	.0574494
disaster						
F1.	.4824138	.435714	1.11	0.268	-.37157	1.336398
F2.	.1759221	.346008	0.51	0.611	-.5022412	.8540853
F3.	.1367751	.3739042	0.37	0.715	-.5960636	.8696138
F4.	-.0585222	.3948867	-0.15	0.882	-.8324859	.7154415
F5.	-.4608942	.3138065	-1.47	0.142	-1.075944	.1541553
F6.	-.145576	.2772887	-0.52	0.600	-.6890519	.3978999
F7.	.2682228	.2895088	0.93	0.354	-.2992041	.8356496
period						
2000	0	(empty)				
2001	.390643	.5895549	0.66	0.508	-.7648634	1.546149
2002	-.5838564	.577231	-1.01	0.312	-1.715208	.5474956
2003	-1.742756	.7768755	-2.24	0.025	-3.265405	-.2201084
2004	-1.769611	.6765238	-2.62	0.009	-3.095573	-.443649
2005	-.7985349	.8891738	-0.90	0.369	-2.541284	.9442138
2006	.0254775	1.003792	0.03	0.980	-1.941918	1.992873
2007	-1.541039	.9549381	-1.61	0.107	-3.412683	.3306053
2008	-1.872888	.9805359	-1.91	0.056	-3.794703	.0489272
2009	-2.422298	.7677958	-3.15	0.002	-3.92715	-.917446
2010	-2.873599	.8014929	-3.59	0.000	-4.444497	-1.302702
2011	-2.626875	1.125983	-2.33	0.020	-4.83376	-.4199896
2012	-2.037177	1.291753	-1.58	0.115	-4.568967	.4946126
2013	-1.983144	1.384424	-1.43	0.152	-4.696566	.7302773
2014	-2.542204	1.469502	-1.73	0.084	-5.422376	.3379681
2015	-1.934799	1.588728	-1.22	0.223	-5.04865	1.179051
_cons	7.454833	2.432962	3.06	0.002	2.686315	12.22335

bonds_tw_h(4)						

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	

disaster						
--.	.437436	.4175235	1.05	0.295	-.3808952	1.255767
L1.	.5327118	.5207373	1.02	0.306	-.4879146	1.553338
L2.	.8742864	.6609314	1.32	0.186	-.4211153	2.169688
L3.	.5507224	.5221065	1.05	0.292	-.4725874	1.574032
L4.	.3231409	.432174	0.75	0.455	-.5239047	1.170186
L5.	.4141981	.3956569	1.05	0.295	-.3612753	1.189671
L6.	.224525	.2987503	0.75	0.452	-.3610147	.8100648
bonds_tw						
L1.	.0310534	.1375961	0.23	0.821	-.2386301	.3007369
L2.	.1679183	.1409889	1.19	0.234	-.1084148	.4442514
L3.	-.0617542	.1182248	-0.52	0.601	-.2934706	.1699622
L4.	.0326804	.0564762	0.58	0.563	-.078011	.1433717
L5.	.0597757	.052715	1.13	0.257	-.0435437	.1630952
L6.	-.0201632	.1319925	-0.15	0.879	-.2788637	.2385373
L7.	-.1069656	.1049248	-1.02	0.308	-.3126144	.0986832
disaster						
F1.	.478806	.4019203	1.19	0.234	-.3089434	1.266555
F2.	.2034149	.4032076	0.50	0.614	-.5868575	.9936872
F3.	.0523825	.3235983	0.16	0.871	-.5818585	.6866234
F4.	.1198732	.3794272	0.32	0.752	-.6237904	.8635368
F5.	.1229087	.3632253	0.34	0.735	-.5889999	.8348173
F6.	-.2401188	.2634001	-0.91	0.362	-.7563735	.276136
F7.	-.0536587	.2536592	-0.21	0.832	-.5508217	.4435043
period						
2000	0	(empty)				
2001	-.1789896	.4945721	-0.36	0.717	-1.148333	.7903539
2002	-.9027328	.5583178	-1.62	0.106	-1.997016	.1915499
2003	-1.241336	.6416319	-1.93	0.053	-2.498911	.0162392

2004	-.8745523	.6128951	-1.43	0.154	-2.075805	.3266999
2005	.4988615	.797678	0.63	0.532	-1.064559	2.062282
2006	-.9438019	.6524422	-1.45	0.148	-2.222565	.3349614
2007	-1.028341	.7697843	-1.34	0.182	-2.537091	.4804081
2008	-1.284784	.7862551	-1.63	0.102	-2.825816	.2562477
2009	-1.590521	.8902297	-1.79	0.074	-3.335339	.1542973
2010	-2.046102	.9736115	-2.10	0.036	-3.954346	-.1378586
2011	-.936419	1.276933	-0.73	0.463	-3.439161	1.566323
2012	-.8784412	1.211148	-0.73	0.468	-3.252247	1.495365
2013	-1.59974	1.293485	-1.24	0.216	-4.134923	.9354431
2014	-.9728293	1.453057	-0.67	0.503	-3.820769	1.87511
2015	-.6858979	1.549137	-0.44	0.658	-3.722151	2.350355
_cons	5.888326	2.431077	2.42	0.015	1.123503	10.65315

bonds_tw_h(5)

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
disaster						
--	.3548635	.4036906	0.88	0.379	-.4363555	1.146083
L1.	.6619411	.7025569	0.94	0.346	-.7150451	2.038927
L2.	.6400513	.6301315	1.02	0.310	-.5949836	1.875086
L3.	.3402114	.4627977	0.74	0.462	-.5668555	1.247278
L4.	.4875784	.4225324	1.15	0.249	-.3405698	1.315727
L5.	.0592954	.2995104	0.20	0.843	-.5277343	.6463251
L6.	-.1082689	.2067939	-0.52	0.601	-.5135775	.2970396
bonds_tw						
L1.	.2096847	.0997543	2.10	0.036	.01417	.4051995
L2.	.0679932	.0928272	0.73	0.464	-.1139447	.2499312
L3.	-.111204	.0668836	-1.66	0.096	-.2422934	.0198853
L4.	.0909843	.046369	1.96	0.050	.0001028	.1818659
L5.	.0717699	.132106	0.54	0.587	-.1871531	.3306929
L6.	-.0809501	.082095	-0.99	0.324	-.2418533	.0799532
L7.	-.0379456	.091822	-0.41	0.679	-.2179134	.1420221
disaster						
F1.	.2664094	.3598141	0.74	0.459	-.4388133	.9716321
F2.	.3426481	.3691545	0.93	0.353	-.3808814	1.066178
F3.	.0816601	.3956298	0.21	0.836	-.69376	.8570802
F4.	.1185818	.3445794	0.34	0.731	-.5567815	.7939451
F5.	.1298608	.3367542	0.39	0.700	-.5301652	.7898868
F6.	.1533422	.3892398	0.39	0.694	-.6095538	.9162382
F7.	-.2772871	.2528567	-1.10	0.273	-.7728772	.218303
period						
2000	0 (empty)					
2001	-.5722429	.5536743	-1.03	0.301	-1.657425	.5129388
2002	-.6654303	.5844347	-1.14	0.255	-1.810901	.4800407
2003	-.4393007	.701541	-0.63	0.531	-1.814296	.9356943
2004	.4533036	.605258	0.75	0.454	-.7329802	1.639587
2005	-.3570037	.5545671	-0.64	0.520	-1.443935	.7299278
2006	-.5853128	.6598702	-0.89	0.375	-1.878635	.7080091
2007	-.6580231	.7683896	-0.86	0.392	-2.164039	.8479929
2008	-.5632397	.9394799	-0.60	0.549	-2.404587	1.278107
2009	-1.054217	.9944584	-1.06	0.289	-3.00332	.8948853
2010	-.1725116	1.143993	-0.15	0.880	-2.414697	2.069674
2011	.0328854	1.165684	0.03	0.977	-2.251814	2.317585
2012	-.4650102	1.156953	-0.40	0.688	-2.732596	1.802575
2013	-.1940881	1.262554	-0.15	0.878	-2.668648	2.280472
2014	.1909362	1.434497	0.13	0.894	-2.620627	3.002499
_cons	4.334436	2.4543	1.77	0.077	-.4759033	9.144775

Statistics by step

	N	R2	psR2	F	Chi2	Prob
0	453	0.677	.	381.50	.	0.000
1	450	0.429	.	133.65	.	0.000
2	448	0.306	.	89.84	.	0.000
3	446	0.230	.	30.65	.	0.000
4	444	0.185	.	13.46	.	0.000
5	399	0.183	.	35.95	.	0.000

Impulse Response Function

	IRF	Std.Err.	IRF LO~90	IRF UP 90	IRF LO~95	IRF UP 95
0	0.49190	0.18580	0.18015	0.80365	0.11813	0.86567
1	0.66488	0.23587	0.26911	1.06065	0.19037	1.13938
2	0.51637	0.24120	0.11166	0.92109	0.03114	1.00161
3	0.57343	0.33543	0.01060	1.13626	-0.10137	1.24823
4	0.43744	0.41752	-0.26314	1.13801	-0.40251	1.27738
5	0.35486	0.40369	-0.32311	1.03283	-0.45821	1.16794

```
.
. graph combine DTZ1 DTZ2, row(1) ///
> name(vul_bonds_combine, replace) ycommon

. graph export disasterbondsapr25.png, as(png) width(4000) replace
file disasterbondsapr25.png saved as PNG format

.
. *** Figure E1. Panel VAR - impact of natural disasters on bond yields
.
. xtvar bonds_tw disaster, lags(3) maxldep(8) ///
> fd collapse
note: 1 panel dropped due to no within-panel variation in bonds_tw.
note: 16 panels dropped due to no within-panel variation in disaster.
note: 2 panels dropped due to no within-panel variation in L2.disaster.
note: 1 panel dropped due to no within-panel variation in L3.disaster.
note: the estimation sample has gaps; see xtdescribe.
```

```
Panel-data vector autoregression          Number of obs   = 1,377
Group variable: imfcode                   Number of groups =    90
Time variable: period                    Obs per group:
                                         min =     1
                                         avg =   15.3
                                         max =    23
```

Fixed-effects transform: FD

Two-step results

(Std. err. adjusted for 90 clusters in imfcode)

		WC robust				
	Coefficient	std. err.	z	P> z	[95% conf. interval]	
bonds_tw						
bonds_tw						
L1.	1.017884	.0826652	12.31	0.000	.8558634	1.179905
L2.	-.233757	.0629318	-3.71	0.000	-.3571009	-.110413
L3.	.1418616	.0373733	3.80	0.000	.0686112	.215112
disaster						
L1.	.4448994	.1364639	3.26	0.001	.1774351	.7123637
L2.	.3841239	.1565362	2.45	0.014	.0773186	.6909292
L3.	.3521655	.1231233	2.86	0.004	.1108484	.5934827
disaster						
bonds_tw						
L1.	-.0013973	.0184169	-0.08	0.940	-.0374938	.0346991
L2.	-.0007851	.0112649	-0.07	0.944	-.0228638	.0212937
L3.	.00519	.007359	0.71	0.481	-.0092333	.0196133
disaster						
L1.	-.0093212	.0576149	-0.16	0.871	-.1222443	.103602
L2.	.0611907	.0561984	1.09	0.276	-.0489561	.1713375
L3.	.0143953	.0443547	0.32	0.746	-.0725384	.1013289

Hansen's test of overid. restrictions: $\chi^2(20) = 30.60$ Prob > $\chi^2 = 0.061$
 GMM-type instruments: L(2/9).(bonds_tw disaster)

```
.
. vargranger
```

Granger causality Wald tests

Equation	Excluded	chi2	df	Prob > chi2
bonds_tw	disaster	13.239	3	0.004
bonds_tw	ALL	13.239	3	0.004
disaster	bonds_tw	.53487	3	0.911
disaster	ALL	.53487	3	0.911

```
.
. irf create lags, set(example1) step(6) replace
(file example1.irf now active)
(file example1.irf updated)

.
. irf graph irf, irf(lags) name(irf2, replace)

.
. *irf cgraph (lags disaster bonds_tw irf) ///
> (lags bonds_tw disaster irf), ///
> ycommon name(oirf, replace)
.
. graph export bonds_var_irf_disaster.png, as(png) width(4000) replace
file bonds_var_irf_disaster.png saved as PNG format

.
. log close _all
      name: SLP_EE
      log: C:\Users\jame1\Dropbox\Latex\PROJECTS\24-03-emft-adb-private\manuscript\SI_EE\Replication_EE\SLP_EE.s
> mcl
      log type: text
      closed on: 22 Jun 2025, 21:21:20
-----
```